

Globally certified measurement selection with correlated errors

Abstract

When measurement errors are correlated, selecting observations changes the covariance matrix that must be inverted: the information of a schedule S is $J(S) = J_0 + F_S^\top R_{SS}^{-1} F_S$, generally not a sum of per-observation terms. We develop global certificates for this discrete design problem with Gaussian errors, known covariance and nominal mean sensitivities. First, a common surrogate built from entries of the full precision matrix overstates information by at most the Kantorovich factor of the covariance, which gives a sharp efficiency bound; in a public kinetics model it changes one of 33 exactly enumerated criterion–budget optima. Second, finite-history Gaussian conditionals satisfy explicit relative precision bounds, uniform over schedules, for noisy scalar chains, complete observed blocks, exponentially decaying covariances and fixed partial-observation packets. On a calendar-state graph they give a weighted-trace FPTAS with input-sized dimensions when the promise constants are fixed. For fixed information dimension and additive rational PSD atoms, polynomial sets of feasible paths in explicit acyclic graphs, or of bases of rationally represented matroids, approximate every feasible information matrix within $1 \pm \eta$ in PSD order with exact ranges, which yields D-, A-, E- and contrast approximation schemes. Third, we give rationally checked certificates: tangent support bounds, one witness that bounds every admissible diagonal virtual-noise relaxation from below, certificates for one schedule over finite scenarios with unknown optimal normalizers, and a nested latent-anchor hierarchy proved by Schur elimination. On ten stipulated rational instances, finite-history upper bounds lie strictly below the optimum of every admissible scalar virtual-noise relaxation, and on one instance below that of every diagonal one. Full-packet and fixed-physical-grid experiments show limits in bound strength and cost.

Keywords: optimal experimental design; correlated errors; discrete optimization; relative matrix approximation; global certificates; latent Gaussian models.

1 Introduction

Measurement selection combines a statistical decision with a discrete resource allocation problem. A kinetic experiment may choose which species to measure, when to sample and which instruments to install; a sensor schedule may acquire a whole multichannel packet whenever a device is activated. Installation costs, exact sample counts and minimum time gaps restrict the feasible experiments. Correlated errors couple these choices: the information from a new observation depends on which other observations are acquired. Correlated dynamic multiresponse design has a long history [47], and recent process-design formulations integrate measurement costs with model-based information criteria [57]. For optimization, the central question is how to bound the best attainable information over all feasible discrete experiments.

Write F_S for the nominal mean sensitivities of the selected observations, R_{SS} for their known positive definite error covariance, and J_0 for fixed prior information or a declared regularizer. The information matrix is $J(S) = J_0 + F_S^\top R_{SS}^{-1} F_S$. Even with a linear mean, $J(S)$ is generally not a sum of fixed per-observation terms. Taking selected entries of the full precision matrix R^{-1} instead

gives a different information model, as Liu et al. [34] already explained. A valid optimization bound must refer to the experiment that will actually be performed.

Even with the correct information formula, two further gaps must be kept separate from the numerical optimization gap. A continuous mixture of feasible schedules can have a larger log determinant than any single schedule, and closing the numerical gap does not close this integrality gap. Approximating the covariance adds an error that must be controlled uniformly over all schedules the optimizer may choose. In this paper a *certificate* is a feasible design with a checked lower bound on its objective, together with a global upper bound for the same discrete model. For p parameters, a log-determinant gap g certifies determinant-root efficiency at least $e^{-g/p}$, whether the design came from a heuristic, a convex relaxation or an exact combinatorial search.

1.1 Relationship to prior work

Correlated design and sensor selection. The selected-covariance information $J(S)$ is the established model in correlated design and sensor selection [34, 47, 48]. Liu et al. [34] also analyze the weak-correlation regime of gating. Virtual-noise designs and covariance-split sensor relaxations give concave continuous extensions [34, 48]; Hainy et al. [18, Proposition 3] prove that Liu’s scalar split and the virtual-noise design coincide, and optimize them by simplicial decomposition. Information-matrix hulls [19], subsystem criteria and mixed-integer second-order cone formulations for exact designs [51], and exact multiresponse designs [17] are also established. We use these results as building blocks and comparators; our comparisons concern the strength of specified relaxation families, not dense formulations in general.

Local Gaussian conditionals. Local conditional factorizations originate with Vecchia [55]. Their noisy and vector extensions, sparse inverse-factor interpretations and approximation theory are substantial existing work [20, 24, 25, 53], and Kozdoba et al. [28] use process-noise dissipation to obtain finite-memory Kalman forecasts. If the error process is fully observed and Markov, exact gap conditionals give an ordered path representation whose inverse-matrix hull is covered by the convexification results of Lee et al. [30], building on Wei et al. [59]. Independent measurement noise generally destroys this observed Markov property. We treat the noisy case through the actual local regressions, with explicit constants for chronological calendar windows; we do not introduce a new factorization or a general localization principle.

Approximation algorithms. Bounded-width Gaussian message schemes [36], approximate Pareto sets [39, 45, 54], nonlinear matroid optimization [7] and fixed-dimensional design algorithms [11] are the main algorithmic precedents. A scalar focused-target reduction of the Gaussian-message scheme already yields an approximation scheme (Appendix B), so we do not claim a first scalar FPTAS. Greedy guarantees for weakly supermodular objectives achieve sparse multiple-regression bounds with an enlarged support [32]; our schemes enforce the exact cardinality and do not rely on sub- or supermodularity. Appendix A.4 shows that a supermodularity step asserted for a pivot-set KL objective in Kaminitz [23] fails. Spectral spanners and coresets [21, 35] retain elements rather than returning one feasible object close to each target matrix. For A- and E-design with additive rank-one information under partition constraints, Bansal and Xu [5] rule out polynomial-factor approximation in a growing information dimension unless $P = NP$, consistent with our fixed-dimensional results.

Robustness, nuisance parameters and certification. Efficiency normalization over scenarios [12, 37], robust and scenario-based design [13, 16, 58], and nuisance or subsystem design [1, 31, 51] precede our robust and latent-anchor certificates. Log-determinant support conditions [19] and

convexity of $\log \det(I + KX^{-1})$ [26, 27] supply the tangent and dual inequalities we check in rational arithmetic. Disjunctive programming explains why grouping alternatives before convexification strengthens a relaxation [4, 46].

Table 1 lists what is inherited and what is developed here; detailed comparisons accompany the relevant proofs.

Table 1: Established antecedents and the developments in this paper. The new results combine established statistical and optimization tools under the hypotheses stated in the cited sections.

Component	Established antecedent	Development in this paper
Selected information	Selected covariance and gating distinction [34]; classical Kantorovich inequality [41]	Sharp design-efficiency consequences, equality/rank analysis, and exact public-model reanalysis (Sections 2, 6.2).
Finite history	Vecchia conditionals, inverse localization and covariance transport [28, 53, 55]	Explicit schedule-uniform relative bounds for four covariance classes, spacing refinements, and transfer to exact feasible support prices (Sections 3, 5.1).
Spectral approximation	Pareto/profile algorithms and guessed-vector normalization [7, 11, 54]	Deterministic all-target, two-sided PSD covers with polynomial accuracy dependence and exact ranges for explicit rational acyclic graphs and represented matroids (Section 4).
Global comparison	Virtual-noise concavity and equivalence [18, 34, 48]; inverse-logdet convexity [27]	Finite rational witnesses separating every scalar or diagonal split, including their stated affine-support intersections (Section 5.3).
Robustness and anchors	Efficiency normalization [12, 37]; nuisance and subsystem design [1, 51]	Certified uncertainty in optimal normalizers for one common schedule, and a proved ordering across nested changing-anchor representations (Sections 5.4, 5.5).

1.2 Main results

1. *Selected versus gated information.* Full-inverse gating never understates information. If $mI \preceq R \preceq MI$ and $\kappa = M/m$, then $J(S) \preceq J_{\text{gate}}(S) \preceq \alpha J(S)$ with the Kantorovich factor $\alpha = (\kappa + 1)^2/(4\kappa)$ (Propositions 2.1 and 2.3). A global gated D-optimum therefore has determinant-root efficiency at least $1/\alpha$, and this bound is sharp as an infimum, already for one parameter (Corollary 2.4, Example 2.5). In the public kinetics implementation of Wang et al. [57], exact evaluation of all 2347 schedules changes one of 33 criterion–budget optima: the trace-information choice at one budget, with a relative loss of about 3.11% (Section 6.2).
2. *Schedule-uniform locality bounds.* If the normalized residual covariance of the local conditionals is within δ of the identity, the local precision lies between $1 - \delta$ and $1 + \delta$ times the true precision (Lemma 3.1). We bound δ explicitly, for every selected schedule at once, for noisy scalar chains (Theorem 3.3), complete observed blocks with noncommuting transitions (Theorem 3.4), general covariances with exponential off-diagonal decay and a spectral floor (Theorem 3.6), and fixed partial-observation packets, including singular latent covariances (Theorem 3.8). The bounds decay geometrically in the window length and contain no dimension factor. For stationary scalar covariances, Proposition 3.5 refines them under minimum sampling gaps.

3. *Approximation with exact feasibility.* On a calendar-state graph, these bounds give a rational weighted-trace FPTAS with input-sized packet and parameter dimensions under fixed promises, including exact cardinality, mandatory and forbidden times, and minimum spacing (Theorem 4.1). Selecting individual channels within a packet admits no deterministic FPTAS for scalar information unless $P = NP$, even at a single time with a well-conditioned covariance (Proposition 4.2). For fixed information dimension, Theorems 4.3 and 4.6 return polynomial sets of actual paths in explicit acyclic graphs, or of bases of rationally represented matroids, with additive rational PSD atoms. For every feasible matrix, one returned matrix lies between its $1 - \eta$ and $1 + \eta$ multiples in PSD order and has the same range, even when singular. The proof combines rational normalization on exact ranges with bounded signed profiles and exact feasible-witness recovery. The covers give approximation schemes for D-, E- and conventional A-optimality and for estimable contrasts, with no condition-number bound. Under the fixed promises of Theorem 4.1, the DAG cover transfers to the true selected-covariance model (Corollary 4.5). Relative to Brown et al. [11], who give a randomized PTAS in the broader independence-oracle matroid model, our represented-matroid result is deterministic with polynomial dependence on $1/\eta$ and does not subsume their model. To the best of our knowledge, this combined guarantee is not stated in the inspected predecessors. The polynomial degree grows with the information dimension.
4. *Certificates and relaxation comparisons.* Any positive definite reference matrix and a globally priced linear support over the feasible graph give an upper bound on the original discrete objective (Theorem 5.1); rational arithmetic checks the reference, the full support and outward logarithm bounds. A fixed fractional point and a repaired PSD dual give one lower witness valid for every admissible diagonal virtual-noise split (Proposition 5.3). A strict separation from a valid integer upper bound therefore survives both complete continuous optimization and split tuning within that family. For one schedule shared by finitely many scenarios, we certify standardized D-efficiency while enclosing each scenario’s unknown optimal normalizer (Proposition 5.5). Under the fixed memory promises and feasibility conditions of Theorem 4.1, with fixed scenario count and information dimension, this problem has an FPTAS (Corollary 5.6).
5. *Latent separators.* When long calendar histories are expensive, latent anchors give an exact representation by conditional blocks, with its own support certificate (Theorem 5.7). For nested anchor sets, removing anchors never loosens the complete-schedule hull bound (Theorem 5.8), by a cross-representation Schur identity. Within established Schur and subsystem-design theory [51], the claimed novelty is this comparison across changing latent representations, which to the best of our knowledge is not stated in the inspected predecessors. Larger blocks cost more to enumerate, and even the empty-anchor hull can retain an integrality gap.
6. *Computational evidence* (Section 6). On ten rational instances (six synthetic, four kinetic), certified finite-history log gaps are at most 0.0039, and every admissible scalar virtual-noise relaxation has optimum at least 0.022 above the finite-history upper bound. On one kinetic instance a single witness separates every admissible diagonal split by more than 0.092. A three-scenario kinetic certificate guarantees at least 99.6% of the unknown optimal standardized robust efficiency. An exact eight-time study verifies strict tightening across four nested anchor sets, and a new far-pair argument improves the complete-block constant. The limits are reported as well: in full-packet examples the dense bound is tighter; on the finer fixed physical grids (96 and 192 points), the primary calendar runs complete no pricing step within the 30-second allowance; and the separator certificates miss the 0.01 log-gap target.

Scope. Throughout, the covariance is known and parameter independent, the sensitivities are local, and a certificate concerns the supplied finite experiment. PSD priors are allowed in the locality and spectral theory; the practical log-determinant and separator certificates assume a positive definite prior. The guarantees do not imply global nonlinear identifiability or robustness outside a declared scenario set.

Formal verification. Lean formalizes the explicit-DAG and represented-matroid covers (Theorems 4.3 and 4.6) from rational inputs, including exact singular ranges, criterion selection and bit-work bounds; the DAG core assumes verified topological vertex indices. The transfer in Corollary 4.5 is formalized conditional on its sandwich hypothesis; the stochastic locality estimates and the graph construction are not formalized (Sections 4.4 and 4.5).

Organization. Sections 2–6 follow the order of these results, and Section 7 discusses scope; a standalone supplement replays the certificates without a commercial solver.

2 Statistical model and the meaning of a certificate

A measurement schedule determines which responses enter the likelihood. With correlated errors, changing the schedule changes the covariance matrix that must be inverted. This section specifies that statistical target, compares it quantitatively with a commonly used surrogate, and defines what a global certificate establishes. The distinctions apply both to sensor installation and to sampling-time selection.

2.1 Selected observations and local information

There are n candidate measurement packets. Packet t contains d_t scalar responses, acquired together when that packet is selected. Set $d_{\text{tot}} = \sum_t d_t$; scalar candidates have $d_t = 1$. The candidate observation vector satisfies

$$Y = \mu(\theta) + \epsilon, \quad \epsilon \sim N(0, R), \quad R \in \mathbb{S}_{++}^{d_{\text{tot}}}, \quad (2.1)$$

where $\theta \in \mathbb{R}^p$, the mean is differentiable at a specified nominal parameter θ_0 , and the known covariance is independent of θ . A schedule is chosen before the data are observed. It determines which coordinates are observed and does not alter the process or covariance. Adaptive sampling and interventions on the process are outside this model.

Write $F = \partial\mu/\partial\theta|_{\theta_0} \in \mathbb{R}^{d_{\text{tot}} \times p}$. For a schedule $S \subseteq \{1, \dots, n\}$, E_S selects its scalar coordinates, $F_S = E_S F$, and $R_{SS} = E_S R E_S^\top$. Thus a matrix subscript S includes every coordinate of its selected packets. The feasible family $\mathcal{F}$ is finite and nonempty; it may encode cardinality, costs, installation, and spacing constraints. Complexity claims require the more specific families stated in each theorem; an arbitrary feasible family is not assumed to have an efficient oracle.

Marginalizing unobserved responses gives $Y_S \sim N(\mu_S(\theta), R_{SS})$. Its score at θ_0 and its Fisher information are

$$\nabla_\theta \log p_\theta(Y_S)|_{\theta_0} = F_S^\top R_{SS}^{-1}(Y_S - \mu_S(\theta_0)), \quad (2.2)$$

$$I(S) = F_S^\top R_{SS}^{-1} F_S. \quad (2.3)$$

The second identity is the second moment of the first. It is the exact Fisher information of the stated Gaussian model, also for a nonlinear mean evaluated at θ_0 . The approximation in local design lies in

using a nominal parameter for the unknown one, or local information to predict estimation accuracy; it does not justify interchanging selection and inversion. This selected-covariance formulation is established in correlated design and sensor selection [34, 47, 48].

We optimize the augmented matrix

$$J(S) = J_0 + I(S), \quad J_0 \in \mathbb{S}_+^p, \quad (2.4)$$

with $I(\emptyset) = 0$. A fixed J_0 can represent independent prior information or a declared regularization; these interpretations must be reported separately. In a linear Gaussian model, a Gaussian prior with precision $J_0 \succ 0$ makes $J(S)^{-1}$ the posterior covariance. For a nonlinear mean, adding a matrix at one nominal parameter does not generally compute exact posterior covariance or Bayesian expected information. A numerical diagonal regularization changes the design problem and does not establish information supplied by an actual prior experiment.

The fixed-covariance assumption is substantive. If $R = R(\theta)$, the (a, b) entry of Fisher information includes the additional term

$$\frac{1}{2} \text{tr} \left(R_{SS}^{-1} \frac{\partial R_{SS}}{\partial \theta_a} R_{SS}^{-1} \frac{\partial R_{SS}}{\partial \theta_b} \right). \quad (2.5)$$

Patan and Bogacka explicitly consider this dependence [47]. Our results assume a fixed covariance and do not apply to a model that requires this term. For matrices, $A \preceq B$ means $B - A$ is positive semidefinite, $\langle A, B \rangle = \text{tr}(A^\top B)$, and $\|A\|_2$ is the spectral norm. A singular prior is allowed; a singular observation covariance is not allowed in (2.1). A singular Gaussian likelihood requires analysis on its support, possibly with parameter-dependent deterministic constraints, and cannot generally be handled by replacing inverses with pseudoinverses.

2.2 Criteria, rank, and parameter coordinates

We use the following conventional criteria [18, 51]:

Criterion	Objective	Domain
D-optimality	maximize $\log \det J(S)$	$J(S) \succ 0$
Weighted trace information	maximize $\text{tr}(WJ(S))$, $W \succeq 0$	$J(S) \succeq 0$
A-optimality	minimize $\text{tr}(J(S)^{-1})$	$J(S) \succ 0$
E-optimality	maximize $\lambda_{\min}(J(S))$	$J(S) \succeq 0$

Trace information is distinct from trace inverse information; we use the descriptive name “trace information” to avoid ambiguity with other uses of the letter T in design theory. The criterion called A-optimality in Wang et al. [57] maximizes $\text{tr} J$, whereas our A-optimality minimizes $\text{tr}(J^{-1})$.

All logarithms are natural. We may extend $\log \det J = -\infty$ to singular positive semidefinite matrices. Every finite D-optimality certificate must include a positive definite feasible information matrix. If none exists, the optimum determinant is zero and there is no finite full-parameter log-determinant optimum. Inverse-based criteria are used only when their inverses exist. In a linear model with no prior, a contrast $c^\top \theta$ is estimable exactly when $c \in \text{range } I(S)$, and its minimum unbiased variance is $c^\top I(S)^\dagger c$. An analogous augmented-matrix expression is an algebraic or prior-regularized criterion and requires its declared interpretation. A restriction of all information matrices is valid for a declared reduced parameter model, or on a common supported invariant range expressed in an orthonormal basis. If complementary parameters remain unknown nuisance parameters, use the contrast covariance accounting for those parameters (in a nonsingular block model, by inverting the nuisance-adjusted Schur complement). Mere compression followed by

inversion need not give that covariance. A pseudodeterminant on design-dependent ranges would define a different objective.

For positive definite information, maximizing $\log \det J$, $\det J$, and $(\det J)^{1/p}$ gives the same schedules. The log-determinant and determinant root are concave in J ; the determinant itself is generally not concave for $p > 1$. Under an invertible change $\theta = T\eta$, sensitivities become FT and the prior must become $T^\top J_0 T$. The log-determinant changes by the schedule-independent constant $2 \log |\det T|$. D-optimal schedules and relative D-efficiencies are therefore invariant. Unweighted trace information, A-optimality, and E-optimality are generally coordinate-dependent. A weight or contrast can be transformed with the coordinates when its interpretation is to be preserved.

Positive definite local information does not imply global identifiability. Observing only the intermediate species in consecutive first-order reactions gives the mean

$$B(t; A_0, k_1, k_2) = \frac{A_0 k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t}), \quad A_0, k_1, k_2 > 0, \quad k_1 \neq k_2. \quad (2.6)$$

Substitution shows that (A_0, k_1, k_2) and $(A_0 k_1 / k_2, k_2, k_1)$ give the same entire trajectory. With fixed covariance, they give the same likelihood for every schedule. Nevertheless, at $(A_0, k_1, k_2) = (1, 1, 2)$, the sensitivity matrix with rows at times $\log 2$, $2 \log 2$, $3 \log 2$ and columns in the order (A_0, k_1, k_2) has determinant $-(\log 2)^2 / 2048$. It has full rank, so any positive definite observation covariance gives positive definite local data information for these three observations without a prior. A design certificate at one nominal parameter, including a regularized certificate, does not resolve this ambiguity. A physical restriction such as a known rate ordering changes the admissible parameter model and must be stated when used.

2.3 Selection before inversion and full-inverse gating

An alternative expression weights precomputed full-precision entries by pairs of selected coordinates. With $D_S = E_S^\top E_S$, it is

$$J_{\text{gate}}(S) = J_0 + F^\top D_S R^{-1} D_S F = J_0 + F_S^\top (R^{-1})_{SS} F_S. \quad (2.7)$$

Liu et al. distinguish this expression from marginal information and analyze its weak-correlation approximation [34, Sec. IV]. The distinction is theirs; we add proofs and quantitative consequences for interpreting optimization results.

Proposition 2.1 (Information inflation and equality). *For every schedule, $J(S) \preceq J_{\text{gate}}(S)$. For a nonempty proper selection, partition the scalar coordinates into selected and unselected sets and write*

$$R = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix}, \quad A = R_{SS}, \quad C = R_{S^c S^c}.$$

Then

$$J_{\text{gate}}(S) - J(S) = F_S^\top A^{-1} B (C - B^\top A^{-1} B)^{-1} B^\top A^{-1} F_S. \quad (2.8)$$

Equality holds exactly when $B^\top A^{-1} F_S = 0$.

Proof. Positive definiteness of R makes the diagonal blocks and both Schur complements positive definite. Block inversion gives $(R^{-1})_{SS} = A^{-1} + A^{-1} B (C - B^\top A^{-1} B)^{-1} B^\top A^{-1}$. Congruence by F_S proves the formula and positive semidefiniteness. The positive definite middle matrix makes the resulting quadratic form zero exactly under the stated condition. Empty and full selections give equality directly. $\square$

Equality covers independent errors, unions of complete independent covariance blocks, and special sensitivity directions even when $B \neq 0$. In particular, correlation within an all-or-nothing sensor block causes no mismatch when that block is independent of every other selection unit, so the mismatch does not affect every correlated instance.

Gating has a statistical interpretation under a different experiment. If the unselected raw noise values ϵ_{S^c} are supplied as parameter-independent side information, the conditional mean of Y_S is $\mu_S + BC^{-1}\epsilon_{S^c}$ and the covariance is $A - BC^{-1}B^\top$. Sensitivity remains F_S , giving (2.7). Equivalently, all channels can be observed under the altered model $Y = D_S\mu(\theta) + \epsilon$.

Conditioning on actual unselected responses is different. At fixed $Y_{S^c} = y_{S^c}$ the conditional mean is $\mu_S + BC^{-1}(y_{S^c} - \mu_{S^c})$, so its sensitivity is $F_S - BC^{-1}F_{S^c}$. Conditional information is consequently

$$(F_S - BC^{-1}F_{S^c})^\top (A - BC^{-1}B^\top)^{-1} (F_S - BC^{-1}F_{S^c}). \quad (2.9)$$

Adding $F_{S^c}^\top C^{-1}F_{S^c}$ gives full-response information. Describing gating as “conditioning on the other observations” without this sensitivity correction is therefore generally incorrect.

Example 2.2 (Exact information and decision discrepancies). Let $J_0 = 0$, $p = 1$, $R = \begin{pmatrix} 1 & 3/4 \\ 3/4 & 1 \end{pmatrix}$, and $F = (1, 1)^\top$. Observing only the first response gives marginal information 1 and gated information $16/7$. Observing both gives $8/7$; the first response conditional on the actual second has information $1/7$. The gated score therefore decreases when the second response is selected.

The discrepancy can change an optimum. Select exactly one of three equal-cost candidates with

$$R = \begin{pmatrix} 1 & 0 & 3/4 \\ 0 & 1 & 0 \\ 3/4 & 0 & 1 \end{pmatrix}, \quad F = (1, 6/5, 0)^\top, \quad J_0 = 0.$$

The marginal singleton values are $(1, 36/25, 0)$ and the gated values are $(16/7, 36/25, 0)$. Their optimizers select the second and first candidates, respectively. The same positive-information ranking applies to scalar log-determinant and inverse information; the third design is singular.

In contrast, marginal information is monotone under adding observations. For disjoint coordinate sets S, T , block inversion gives

$$\begin{aligned} I(S \cup T) = I(S) &+ (F_T - R_{TS}R_{SS}^{-1}F_S)^\top \\ &\cdot (R_{TT} - R_{TS}R_{SS}^{-1}R_{ST})^{-1} (F_T - R_{TS}R_{SS}^{-1}F_S). \end{aligned} \quad (2.10)$$

The increment is positive semidefinite; the empty-set case is interpreted directly. This identity does not assert submodularity of a scalar criterion.

2.4 Uniform quantitative bounds for a gated model

The next result specializes the classical inverse-compression form of the operator Kantorovich inequality [41, Eq. (1)]; it is not a new matrix inequality.

Proposition 2.3 (Condition-dependent comparison). *Suppose $0 < m \leq M$ and $mI \preceq R \preceq MI$. Set*

$$\kappa = M/m, \quad \alpha = \frac{(M+m)^2}{4Mm} = \frac{(\kappa+1)^2}{4\kappa}. \quad (2.11)$$

Every schedule satisfies

$$J(S) \preceq J_{\text{gate}}(S) \preceq \alpha J(S). \quad (2.12)$$

The two information matrices have the same kernel. If $J(S) \succ 0$ and $r_S = \text{rank}(R_{S,S^c})$, then

$$0 \leq \log \det J_{\text{gate}}(S) - \log \det J(S) \leq \min(p, r_S) \log \alpha. \quad (2.13)$$

For an empty or full selection set $r_S = 0$.

Proof. For $\lambda \in [m, M]$, $(\lambda - m)(\lambda - M) \leq 0$, hence $\lambda + Mm/\lambda \leq M + m$. Applying this inequality to an orthogonal eigendecomposition gives $R + MmR^{-1} \preceq (M + m)I$. Compress to selected coordinates and put $A = R_{SS}$, $K = (R^{-1})_{SS}$. Then

$$K \preceq \frac{(M + m)I - A}{Mm} \preceq \alpha A^{-1}.$$

The second inequality follows by diagonalizing A and using $t(M + m - t) \leq (M + m)^2/4$ for its positive eigenvalues. Congruence by F_S , and $J_0 \preceq \alpha J_0$, prove the upper comparison; the lower comparison is Proposition 2.1. The sandwich equates zero quadratic forms and hence kernels. For $J(S) \succ 0$, the eigenvalues of $J(S)^{-1/2} J_{\text{gate}}(S) J(S)^{-1/2}$ lie in $[1, \alpha]$. Equation (2.8) bounds the rank of its difference from the identity by r_S . At most $\min(p, r_S)$ eigenvalues differ from one; the logarithm of their product proves (2.13). $\square$

For positive diagonal D , replacing (R, F) by (DRD, DF) leaves both information matrices unchanged for every coordinate selection; so do invertible block-diagonal transformations when entire blocks are selected together. Scaling to unit marginal variances therefore avoids measuring conditioning in arbitrary physical units. Spectral enclosures for the scaled covariance used in a certificate must themselves be valid.

Corollary 2.4 (Consequences for design optimization). *Under the assumptions of Proposition 2.3, with α as in (2.11), suppose $\mathcal{F}$ contains a positive definite information design. Let S_G maximize $\log \det J_{\text{gate}}$ and let S_* maximize $\log \det J$ over $\mathcal{F}$. Then*

$$\log \det J(S_G) \geq \log \det J(S_*) - p \log \alpha, \quad \left(\frac{\det J(S_G)}{\det J(S_*)} \right)^{1/p} \geq \alpha^{-1}. \quad (2.14)$$

If $r_S \leq r$ uniformly over feasible schedules, the additive loss improves to $\min(p, r) \log \alpha$. A global gated optimizer of weighted trace information achieves at least $1/\alpha$ of the true optimum of that same criterion. A global gated A -optimal design has true trace-inverse objective at most α times the true A -optimum, on the common positive definite domain.

Proof. The D-optimality chain is

$$\log \det J(S_G) \geq \log \det J_{\text{gate}}(S_G) - p \log \alpha \geq \log \det J_{\text{gate}}(S_*) - p \log \alpha \geq \log \det J(S_*) - p \log \alpha.$$

Use (2.13) for the refinement. Taking the trace after multiplication by $W \succeq 0$ preserves the sandwich and proves the trace statement by the same optimality chain. Inversion reverses order: $\alpha^{-1} J(S)^{-1} \preceq J_{\text{gate}}(S)^{-1} \preceq J(S)^{-1}$. Take traces and compare the two minimizers to obtain the A -optimality claim. $\square$

These guarantees require global optimization of the gated criterion. An additive gated log-objective suboptimality $\eta \geq 0$ adds η to the loss in (2.14). A heuristic solution or a solver termination status alone does not establish global optimality.

Example 2.5 (Sharpness). For $0 < \rho < 1$, select exactly one observation from

$$R = \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad F = (1, 1/2, b)^\top, \quad J_0 = 0.$$

The eigenvalue bounds $m = 1 - \rho$, $M = 1 + \rho$ give $\alpha = (1 - \rho^2)^{-1}$. True singleton information values are $(1, 1/4, b^2)$, and gated values are $(\alpha, \alpha/4, b^2)$. For $1 < b^2 < \alpha$, gating selects the first candidate and the true objective selects the third. The true efficiency $1/b^2$ approaches $1/\alpha$ as $b^2 \uparrow \alpha$. Every singleton information matrix is positive. Rational ρ, b approximate the limit, so the efficiency lower bound is sharp as an infimum for rational data, one parameter, and three candidates; equivalently, the additive log-information loss has supremum $\log \alpha$. Letting $\rho \uparrow 1$ rules out a positive efficiency guarantee independent of covariance conditioning.

For a weak-correlation family $R = \Lambda + \varepsilon \Upsilon$, with fixed positive diagonal Λ and fixed symmetric zero-diagonal Υ , the cross block B is $O(\varepsilon)$. Equation (2.8) gives $O(\varepsilon^2)$ information error because the inverse blocks stay bounded near zero. This recovers the first-order agreement of Liu et al. [34, Proposition 2]. Weak correlation is sufficient asymptotically but not necessary for equality at a special selection.

2.5 A source comparison and independent time-block repair

Wang et al. [57] formulate measurement costs for all-or-nothing sensors and individually charged samples. We compare with arXiv:2406.09557v1 and the accepted manuscript OSTI 2447595 (the publisher’s typeset article was not retrieved). In the accepted manuscript, Eqs. (8)–(10) form pair contributions from full inverse-covariance entries, and Eq. (11) gates them by joint selection. The public implementation at commit `430090e610446aab88328ce495ffb15b684c56c4` computes the full covariance pseudoinverse before extracting entries or blocks [15]; for our positive definite examples this is the ordinary inverse in (2.7). The mismatch occurs with one measurement modality, so it does not depend on cross-modality counting conventions in the displayed formulation.

A local Fisher approximation does not remove a mismatch that is already present for an exactly linear Gaussian mean. Conversely, the argument does not invalidate the source’s independent-noise case, whose rotary-bed covariance is diagonal, and does not assert that every correlated optimizer changes. Agreement between optimization packages that reuse the same coefficients checks their stated objective, not the likelihood.

The source’s kinetics code uses eight independent time blocks. At each time, order three static-cost measurements (SCMs) before three dynamic-cost measurements (DCMs), with species A, B, C in each group. The code covariance is

$$R_6 = \begin{pmatrix} C_0 & C_0/2 \\ C_0/2 & C_0 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 1 & 1/10 & 1/10 \\ 1/10 & 4 & 1/2 \\ 1/10 & 1/2 & 8 \end{pmatrix}. \quad (2.15)$$

The leading principal minors of C_0 are 1, $399/100$, and $791/25$, so $C_0 \succ 0$. The other Kronecker factor has eigenvalues $1/2$ and $3/2$, so $R_6 \succ 0$. Same-species cross-modality correlation is $1/2$. The SCM–SCM block of R_6^{-1} is $(4/3)C_0^{-1}$, so selecting all SCMs and no DCMs gives gated information exactly $4/3$ times the actual data information. We do not assert that this pattern is feasible at every budget. In Table S-1 of the inspected accepted manuscript (printed p. S-2, PDF p. 46), the A-DCM row, C-DCM column is 0.01, whereas the transposed entry is 0.1. Equation (2.15) uses the

symmetric public-code covariance and does not treat that asymmetric table as a valid covariance model.

Independence across time allows a direct repair. For each time t and subset $P \subseteq \{1, \dots, 6\}$, precompute

$$Q_{t,P} = F_{t,P}^\top (R_6)_{PP}^{-1} F_{t,P}, \quad Q_{t,\emptyset} = 0. \quad (2.16)$$

Choose one pattern per time with binary $z_{t,P}$ and $\sum_P z_{t,P} = 1$. Then

$$J = J_0 + \sum_{t,P} z_{t,P} Q_{t,P} \quad (2.17)$$

is exactly the marginal information: reorder selected coordinates by time, invert the resulting block-diagonal covariance, and sum its sensitivity quadratic forms. SCM presence is tied across time, and installation, sampling and budget restrictions act on pattern membership. There are at most 64 patterns per time before pruning. This is ordinary finite pattern enumeration, not a new hull theorem. It does not justify independent temporal chains when same-time cross-channel covariance is present, or complete-state Markov formulas under arbitrary partial observation of a latent state.

2.6 What a global certificate establishes

For $f(S) = \log \det J(S)$ and a feasible positive definite design $\hat{S}$, a certificate supplies valid numbers ℓ, U such that

$$\ell \leq f(\hat{S}) \leq f_* := \max_{S \in \mathcal{F}} f(S) \leq U. \quad (2.18)$$

The achieved D-efficiency is at least $\exp(-(U - \ell)/p)$. Numerical primal values and support prices become rigorous certificates only after feasibility, arithmetic enclosures, and the relevant inequalities and model inputs have been justified. “Exact design” in design theory means a discrete design rather than a continuous weight measure; it does not itself mean globally certified optimality or exact numerical arithmetic.

By Proposition 2.1, a global upper bound U_G on the gated log optimum also bounds f_* . Re-evaluating a feasible schedule with its selected covariance supplies the lower side of (2.18). The gated solver’s own gap is not the marginal gap, but a valid bound from an existing model remains useful after its statistical objective is corrected.

Our rational certificates treat stored decimal model entries as exact rational numbers and enclose logarithms with outward bounds. They certify the optimization problem specified by those inputs, not that numerically generated sensitivities or estimated covariances equal physical quantities. Sensitivity verification, covariance assumptions, finite-scenario coverage and arithmetic certification answer different questions and are reported separately.

3 Local conditional information with explicit error bounds

Chronological Gaussian conditionals make information additive, but the exact conditional at a selected time generally depends on the entire selected past. We replace that past by a calendar window and bound the resulting error, uniformly over schedules, before optimizing. The construction is Vecchia’s approximation [55, Section 3.1], also a factorized sparse approximate inverse [20, 24, 53]. Noisy selected subsets and vector conditional blocks are already included in the general Vecchia framework [25, Section 2], and block Vecchia computation is developed by Pan et al. [44].

3.1 The exact complete-observation Markov case

Suppose first that the error itself is a completely observed Gaussian Markov chain, $\epsilon_t = X_t$, with $X_t = A_t X_{t-1} + w_t$, $P_1 = \text{Cov}(X_1) \succ 0$, and independent innovations $\text{Cov}(w_t) \succ 0$. All blocks have the same dimension in this subsection. Write $P_t = \text{Cov}(X_t)$ and $\Phi_{ji} = A_j \cdots A_{i+1}$ for $i < j$. The gap innovation covariance $\Omega_{ji} = P_j - \Phi_{ji} P_i \Phi_{ji}^\top$ is positive definite. For $S = \{t_1 < \cdots < t_k\}$ the likelihood factors into its first marginal and consecutive gap conditionals. Therefore

$$I(S) = M_{t_1} + \sum_{a=1}^{k-1} W_{t_a t_{a+1}}^+, \quad M_i = F_i^\top P_i^{-1} F_i, \quad W_{ij}^+ = (F_j - \Phi_{ji} F_i)^\top \Omega_{ji}^{-1} (F_j - \Phi_{ji} F_i). \quad (3.1)$$

Indeed, each gap residual is independent of the preceding selected errors; the triangular residual map diagonalizes R_{SS} , giving (3.1) by congruence with F_S .

The associated ordered graph has vertices $0, 1, \dots, n, n+1$, all forward arcs between candidate vertices, source arcs $(0, i)$ of weight M_i , and zero-weight sink arcs $(i, n+1)$ and $(0, n+1)$. Each source–sink path is one schedule. A nonnegative unit flow is a convex combination of paths: subtract the minimum flow on any positive source–sink path and repeat; acyclicity precludes a residual circulation. Thus its affine information image is exactly $\text{conv}\{J(S) : S \subseteq [n]\}$, together with visit indicators if these are retained. This is a direct sensitivity image of the known principal-inverse hull of Lee et al. [30], building on Wei et al. [59], not a new hull theorem.

We record the mapping and the innovation-direction equivalence explicitly. When all A_t are nonsingular, let $T_1 = I$ and $T_i = A_i \cdots A_2$. Then $R_{ij} = P_i T_i^{-\top} T_j^\top$ for $i \leq j$, which is the block-factorizable class of Lee et al. [30]. With $K_S = E_S^\top R_{SS}^{-1} E_S$, the affine map $(z, K) \mapsto (z, J_0 + F^\top K F)$ commutes with convex combinations. For $C_{ij} = R_{ij}$, the reverse gap weight is

$$W_{ij}^- = (F_i - C_{ij} P_j^{-1} F_j)^\top (P_i - C_{ij} P_j^{-1} C_{ij}^\top)^{-1} (F_i - C_{ij} P_j^{-1} F_j).$$

The two Schur-complement evaluations of the two-packet information give

$$M_i + W_{ij}^+ = W_{ij}^- + M_j. \quad (3.2)$$

Flow conservation cancels the vertex potentials M_i on every fractional flow, so forward source/gap weights and reverse gap/sink weights give the same relaxation. Singular transitions are covered by the displayed innovation formulas. To obtain the same conclusion from the factorized representation, perturb every A_t to nonsingular matrices converging to it, keeping positive initial and innovation covariances fixed. The finitely many principal inverses and arc coefficients converge. The common bounded flow polytope therefore has the limiting affine image asserted above. A transition inverse must not be used directly at the singular boundary.

Adding independent measurement noise usually destroys the observed Markov property. For example, the three-packet covariance with diagonal 2 and off-diagonal $R_{ij} = 2^{-|i-j|}$ has $\text{Cov}(\epsilon_1, \epsilon_3 \mid \epsilon_2) = 1/8 \neq 0$, so complete latent observation and noisy observation require different history lengths. Appendix A.1 records the established score-orthogonality extension of exact Markov paths beyond fixed-covariance Gaussian mean information; the noisy finite-history bounds do not use it.

3.2 Genuine local conditionals and a common error argument

Hereafter ϵ_t is a complete fixed packet of dimension d_t in (2.1). For an integer $L \geq 0$, set

$$\begin{aligned} \mathcal{H}_t(S) &= S \cap \{\max(1, t-L), \dots, t-1\}, \\ b_t &= R_{t\mathcal{H}_t} R_{\mathcal{H}_t\mathcal{H}_t}^{-1}, & D_t &= R_{tt} - R_{t\mathcal{H}_t} R_{\mathcal{H}_t\mathcal{H}_t}^{-1} R_{\mathcal{H}_t t}, \\ Z_t &= \epsilon_t - \sum_{j \in \mathcal{H}_t} b_{tj} \epsilon_j, & G_{t, \mathcal{H}_t} &= F_t - \sum_{j \in \mathcal{H}_t} b_{tj} F_j. \end{aligned} \quad (3.3)$$

An empty history has no regression term. Let A_L be the unit block lower-triangular residual map on S and $D_L = \text{diag}(D_t : t \in S)$. Define

$$\begin{aligned} C_L(S) &= D_L^{-1/2} A_L R_{SS} A_L^\top D_L^{-1/2}, & Q_L(S) &= A_L^\top D_L^{-1} A_L, \\ I_L(S) &= \sum_{t \in S} G_{t, \mathcal{H}_t}^\top D_t^{-1} G_{t, \mathcal{H}_t}, & J_L(S) &= J_0 + I_L(S). \end{aligned} \quad (3.4)$$

The product of the true densities $p(\epsilon_t | \epsilon_{\mathcal{H}_t})$ is a normalized Gaussian density with precision Q_L , as successive integration in reverse chronological order shows. Its mean-parameter information I_L is a working-model information matrix, not the true Fisher information or the covariance of an estimator under misspecification. Every regression is recomputed on its actual local history; deleting older coefficients from a full-history Kalman filter gives a different approximation.

Lemma 3.1 (Residual-to-information transfer). *If $\|C_L(S) - I\|_2 \leq \delta$, then*

$$(1 - \delta) R_{SS}^{-1} \preceq Q_L(S) \preceq (1 + \delta) R_{SS}^{-1}. \quad (3.5)$$

For $J_0 \succeq 0$ this implies $(1 - \delta)J(S) \preceq J_L(S) \preceq (1 + \delta)J(S)$. If $\delta < 1$, a sharper prior-aware upper matrix is

$$J(S) \preceq J_0 + \frac{I_L(S)}{1 - \delta}. \quad (3.6)$$

These statements require no sensitivity norm bound or positive prior eigenvalue.

Proof. Put $B = D_L^{-1/2} A_L R_{SS}^{1/2}$. The invertible square matrices $BB^\top = C_L$ and $B^\top B = R_{SS}^{1/2} Q_L R_{SS}^{1/2}$ have identical eigenvalues. Bounding these between $1 - \delta$ and $1 + \delta$ and applying congruence proves (3.5). Congruence with F_S gives the data-information inequalities. Adding J_0 proves the remaining statements, using $\delta J_0 \succeq 0$. $\square$

The next lemma contains the overlapping-history calculation shared by several models. Define, for $0 < x < 1$,

$$T_L(x) = \frac{x^{L+1}}{1 - x}, \quad N_L(x) = \sum_{h=1}^L \sum_{d=L+1-h}^L x^{h+2d} = \frac{x^{L+2}(1 - x^L)(1 - x^{L+1})}{(1 - x)(1 - x^2)}. \quad (3.7)$$

The empty near sum is zero. Summing the inner geometric series, and then $\sum_{h=1}^L (x^{2L+2-h} - x^{2L+2+h})/(1 - x^2)$, proves the identity.

Lemma 3.2 (Local regression row bound). *Suppose, uniformly in selected histories, $D_t \succeq d_* I$, $\|b_{tj}\|_2 \leq B_* x^{t-j}$ for $j \in \mathcal{H}_t$, and $\|\text{Cov}(Z_t, \epsilon_j)\|_2 \leq K_* x^{t-j}$ for $j < t - L$, where $d_* > 0$, $B_*, K_* \geq 0$, and $0 < x < 1$. Then*

$$\|C_L(S) - I\|_2 \leq \frac{2K_*}{d_*} \frac{x^{L+1}}{1 - x} \left[1 + B_* \frac{x(1 - x^L)}{1 - x} \right]. \quad (3.8)$$

If a sharper far-pair bound $\|\text{Cov}(Z_t, Z_s)\|_2 \leq K_* x^{t-s}$ is available for $t - s > L$, the right side can instead be $2K_*[T_L(x) + B_* N_L(x)]/d_*$.

Proof. For $s < t$, $h = t - s$, regression orthogonality and

$$\text{Cov}(Z_t, Z_s) = \text{Cov}(Z_t, \epsilon_s) - \sum_{j \in \mathcal{H}_s} \text{Cov}(Z_t, \epsilon_j) b_{sj}^\top \quad (3.9)$$

give the respective bounds

$$K_* x^h \left(1 + B_* \sum_{d=1}^L x^{2d} \right) \quad (h > L), \quad K_* B_* \sum_{d=L+1-h}^L x^{h+2d} \quad (1 \leq h \leq L).$$

For the near case, s and all $j \geq t - L$ in the sum belong to $\mathcal{H}_t$, so their covariances vanish; the older terms need not vanish. Normalization by $D_t^{-1/2}, D_s^{-1/2}$ costs at most $1/d_*$. There is at most one packet at each distance on each side. The row sum is at most $2K_*/d_*$ times $T_L(x)(1 + B_* \sum_{d=1}^L x^{2d}) + B_* N_L(x)$. The terms multiplying B_* sum to $T_L(x)x(1 - x^L)/(1 - x)$, proving (3.8). For any block matrix E , its scalar matrix of block norms majorizes its action on the vector of component norms. If E is symmetric, that scalar matrix is symmetric and its maximum row sum bounds $\|E\|_2$. Apply this to $E = C_L - I$, whose diagonal blocks vanish. Replacing the far majorant proves the final statement. $\square$

For empty selections, independent packets, or $L \geq n - 1$, the construction is exact and we use $\delta_L = 0$: with complete history, Z_t is orthogonal to every earlier selected observation and hence to every earlier residual, and with independent packets all regressions vanish. These exact cases are handled directly rather than through conservative infinite tails.

3.3 Noisy scalar chains and complete observed blocks

Consider scalar errors $\epsilon_t = X_t + v_t$, where $X_t = a_t X_{t-1} + w_t$. The centered Gaussian initial state, process noises and measurement noises are mutually independent. Assume

$$|a_t| \leq \rho, \quad 0 \leq \rho < 1, \quad \text{Var}(X_t) \leq \bar{P}, \quad r_t = \text{Var}(v_t) \geq r_{\min} > 0. \quad (3.10)$$

Transitions may be signed, marginal variances nonstationary, and process variances zero. Let $\kappa = \bar{P}/(\bar{P} + r_{\min})$ and let $d_* > 0$ be any valid lower bound on all local innovation variances. The choice $d_* = r_{\min}$ is always valid.

Theorem 3.3 (Scalar noisy Markov bound). *Under (3.10), for every selected subset,*

$$\|C_L(S) - I\|_2 \leq \delta_L^{\text{sc}} := \frac{2\bar{P}}{d_*} \{T_L(\rho) + \kappa N_L(\rho)\}. \quad (3.11)$$

If both latent marginal variance P and measurement variance r are constant, the near coefficient κ in this formula can be replaced by $\kappa(1 - \kappa)$, with $\bar{P} = P$ and $\kappa = P/(P + r)$. At $\rho = 0$ or $\bar{P} = 0$ the error is zero.

Proof. Compute the fresh filter for $\mathcal{H}_t$ from the unconditional marginal at its first observation. Its prediction variance is at most the unconditional one. Each update has gain $K_j = p_j^-/(p_j^- + r_j) \in [0, \kappa]$; the prediction coefficient on an observation at j is its gain multiplied by intervening transitions and later factors $1 - K_u$. Thus $|b_{tj}| \leq \kappa \rho^{t-j}$. For $j < t - L$, start the same fresh filter at time j with the unconditional state. New noises are independent of ϵ_j ; transitions multiply the cross covariance by a_u , and updates by $1 - K_u$. Consequently

$$\text{Cov}(Z_t, \epsilon_j) = \Phi_{tj} \text{Var}(X_j) \alpha_t, \quad \alpha_t = \prod_{u \in \mathcal{H}_t} (1 - K_u^{(t)}), \quad \Phi_{tj} = \prod_{u=j+1}^t a_u. \quad (3.12)$$

The gains $K_u^{(t)}$ are those of this particular local history. Their values coincide whether one starts at j without intervening updates or at the first history observation, because unconditional marginal propagation is exact. This proves the old-covariance bound with $K_* = \bar{P}$.

For $s < t - L$, put $e_s = X_s - \mathbb{E}(X_s \mid \epsilon_{\mathcal{H}_s})$ and $p_s^- = \text{Var}(e_s)$. Regression orthogonality gives $\text{Cov}(X_s, Z_s) = p_s^-$, since $Z_s = e_s + v_s$. All updates of the t -local filter occur strictly after s ; the same transport proves the exact identity

$$\text{Cov}(Z_t, Z_s) = \Phi_{ts} p_s^- \alpha_t, \quad |\text{Cov}(Z_t, Z_s)| \leq \bar{P} \rho^{t-s}. \quad (3.13)$$

Lemma 3.2 gives (3.11). For constant P, r , every nonempty fresh history starts with gain exactly κ . In a near pair $s \in \mathcal{H}_t$, so $\alpha_t \leq 1 - \kappa$. Every surviving old covariance in (3.9) receives this factor through (3.12). Its regression coefficient still contributes κ , proving the stated refinement. The far bound does not receive this factor because its local history can be empty. $\square$

The simpler gain-only constant, used for comparison with complete blocks, is

$$\delta_L^{\text{gain}} = \frac{2\bar{P}}{d_*} \frac{\rho^{L+1}}{1-\rho} \left[1 + \kappa \frac{\rho(1-\rho^L)}{1-\rho} \right]. \quad (3.14)$$

It follows directly from Lemma 3.2; the exact far identity makes (3.11) no larger. Replacing κ by one gives the more conservative constant $2\bar{P}\rho^{L+1}(1-\rho^{L+1})/[d_*(1-\rho)^2]$. Another improvement uses

$$d_* = \min_t \text{Var}(\epsilon_t \mid \epsilon_1, \dots, \epsilon_{t-1}) \geq r_{\min}. \quad (3.15)$$

Conditioning on more observations cannot increase conditional covariance, so one full-grid Kalman sweep gives a valid floor, while the gain constant continues to use $r_{\min}$. Any numerical floor used in a certificate must be bounded downward rigorously. For stationary $P = r = 1$ and $|a| = \rho$, the full-grid prediction recursion is $x \mapsto 1 - \rho^2 + \rho^2 x / (1 + x)$, started at $x = 1$. Its nonnegative fixed point is $\sqrt{1 - \rho^2}$; monotonicity of the map keeps every iterate above that point. Thus the explicit horizon-independent floor $d_* = 1 + \sqrt{1 - \rho^2}$ is valid in this special case.

Theorem 3.4 (Complete blocks in the observation-noise metric). *Let $\epsilon_t = X_t + v_t$, $X_t = A_t X_{t-1} + w_t$, with complete blocks of a common dimension and the same independence assumptions. Suppose $V_t = \text{Cov}(v_t) \succ 0$ and*

$$\|V_t^{-1/2} A_t V_{t-1}^{1/2}\|_2 \leq \rho, \quad 0 \leq \rho < 1, \quad V_t^{-1/2} P_t V_t^{-1/2} \preceq \bar{P} I. \quad (3.16)$$

Then (3.11) bounds $\|C_L(S) - I\|_2$ for every subset, with $\kappa = \bar{P}/(1 + \bar{P})$ and $d_ = 1$. One may replace 1 by any uniform lower eigenvalue bound on full-grid innovation covariances in these whitened coordinates. The weaker gain-only bound (3.14) also remains valid. Neither constant has a factor depending on block dimension. If $\rho = 0$ or $\bar{P} = 0$, use the exact bound zero.*

Proof. In coordinates $X'_t = V_t^{-1/2} X_t$, the measurement noise covariance is I and the transition norm is at most ρ . A fresh prediction covariance is at most $\bar{P}I$, so its gain $K = P^-(P^- + I)^{-1}$ is symmetric and satisfies $0 \preceq K \preceq \kappa I$ and $0 \preceq I - K \preceq I$. Expanding the predictor by successive transitions and updates yields $\|b_{tj}\|_2 \leq \kappa \rho^{t-j}$ by submultiplicativity, with no commutativity assumption. Propagating covariance with an excluded old observation starts from a covariance of norm at most $\bar{P}$; the same products give norm at most $\bar{P} \rho^{t-j}$. For a far pair $s < t - L$, full observation gives $\text{Cov}(X'_s, Z'_s) = \Pi_s^{-,s}$, the prediction covariance for the s -local history, of norm at most $\bar{P}$. All updates in the t -local history occur after s . Therefore $\text{Cov}(Z'_t, Z'_s) = \mathcal{T}_{ts}^{(t)} \Pi_s^{-,s}$, where $\mathcal{T}_{ts}^{(t)}$ is the ordered product of transitions and intervening $I - K_u^{(t)}$ updates. Each factor has the norm bound already proved, so this covariance has norm at most $\bar{P} \rho^{t-s}$. This far-pair improvement requires no scalar commutativity. Every innovation covariance is at least I . The sharp-far option of Lemma 3.2 proves

the stronger bound, and its original option gives (3.14). Full-grid conditioning gives the alternative floor. Whiten F_t together with each packet. Local regression residuals then transform by $V_t^{-1/2}$, so both true and local information are invariant. Normalized residual covariances before and after whitening are related by a block orthogonal congruence: each block $(V_t^{-1/2} D_t V_t^{-1/2})^{-1/2} V_t^{-1/2} D_t^{1/2}$ is orthogonal. The norm and precision conclusions therefore hold in the original coordinates. $\square$

The contraction in (3.16) must hold in the indicated noise metric; contraction of A_t in unrelated coordinates is insufficient. The theorem permits noncommuting transitions and time-varying correlated noise within packets. It does not allow selecting arbitrary channels from a packet or replacing full observation by an arbitrary observation matrix.

3.4 Spacing and finite-horizon row bounds

For a stationary scalar covariance $R_{ij} = Pa^{|i-j|} + r\mathbf{1}_{i=j}$, let $b = |a| < 1$, $P \geq 0$, $r > 0$, and require consecutive selected times to differ by at least $g \geq 1$. Set $\kappa = P/(P+r)$ and $m_L = \lfloor L/g \rfloor$. Define $T_g^{\circ 0}(P) = P$ and

$$T_g(x) = P(1 - b^{2g}) + b^{2g} \frac{rx}{r+x}, \quad d_*^{\text{sp}} = r + T_g^{\circ m_L}(P). \quad (3.17)$$

For distances $h \geq g$ put

$$\phi(h) = \begin{cases} Pb^h, & h > L, \\ P\kappa(1-\kappa)b^h \sum_{d=\ell_h, \ell_h+g, \dots, L} b^{2d}, & g \leq h \leq L, \end{cases} \quad \ell_h = \max(g, L+1-h). \quad (3.18)$$

For integers $M \geq 0$ define

$$B(M) = \begin{cases} 0, & M < g, \\ \max\{B(M-1), \phi(M) + B(M-g)\}, & M \geq g. \end{cases} \quad (3.19)$$

Proposition 3.5 (A finite-horizon bound respecting sampling gaps). *For every schedule satisfying the minimum gap,*

$$\|C_L(S) - I\|_2 \leq \delta_{n,L,g} := \max_{1 \leq t \leq n} \frac{B(t-1) + B(n-t)}{d_*^{\text{sp}}}. \quad (3.20)$$

The bound is zero directly if $P = 0$, $b = 0$, $L \geq n-1$, or $g \geq n$.

Proof. A fresh local history has at most m_L observations. An observation with prediction variance $x \leq P$, followed by $d \geq g$ calendar steps, has next prediction variance $T_d(x)$. This is increasing in x ; it is nondecreasing in d , because $rx/(r+x) \leq P$. Iteration of the increasing map T_g from P decreases and remains in $[0, P]$. Starting from the unconditional first prediction, replacing every gap by g and adding observations up to m_L can only decrease the target variance. This proves $D_t \geq d_*^{\text{sp}}$. The far bound is (3.13). For a near pair, surviving history distances $d = s - j$ lie in $[\ell_h, L]$ and are g -separated. If their ordered values are $d_1 < \dots < d_q$, then $d_i \geq \ell_h + (i-1)g$. Since b^{2d} decreases, summing these largest possible weights and applying the stationary near factor gives (3.18). The largest sum of these nonnegative pair weights on g -separated distances in $\{g, \dots, M\}$ is exactly $B(M)$: an optimal set either omits M or includes it and has all earlier distances at most $M-g$. Left and right distances from an anchor satisfy these restrictions separately. Bounding each side and dividing by the innovation floor proves the row bound, and the block row argument of Lemma 3.2 proves the norm bound. We do not assume that one schedule attains all pair majorants. $\square$

Using the unrefined near coefficient κ instead of $\kappa(1 - \kappa)$, or the far weight $Pb^h[1 + \kappa \sum_{j=1}^{m_L} b^{2gj}]$, gives valid but weaker bounds; Section 6.4 reports certificates with both the weaker and the refined versions. The stationary factor cannot be inferred from upper latent-variance and lower noise bounds alone; an exact three-time counterexample appears in Appendix A.2.

3.5 General covariance decay, including unequal packet sizes

The next result avoids a latent Markov assumption. Let d_t be arbitrary positive packet dimensions and suppose

$$R \succeq mI, \quad \|R_{ts}\|_2 \leq C\rho^{|t-s|} \quad (t \neq s), \quad m, C > 0, \quad 0 < \rho < 1. \quad (3.21)$$

No upper bound on diagonal blocks is imposed. Define

$$\begin{aligned} \vartheta &= \frac{4C\rho + m\rho(1 - \rho)}{4C\rho + m(1 - \rho)}, & B_* &= \frac{2C}{m} \frac{\rho}{\vartheta - \rho}, \\ K_* &= C \left(1 + B_* \frac{\rho}{\vartheta - \rho} \right), & K_{\text{dec}} &= \frac{2K_*}{m(1 - \vartheta)} \left(1 + B_* \frac{\vartheta}{1 - \vartheta} \right). \end{aligned} \quad (3.22)$$

Theorem 3.6 (Explicit relative precision from covariance decay). *Under (3.21), every selected subset satisfies*

$$\|C_L(S) - I\|_2 \leq \delta_L^{\text{dec}} := \frac{2K_*}{m} \frac{\vartheta^{L+1}}{1 - \vartheta} \left[1 + B_* \frac{\vartheta(1 - \vartheta^L)}{1 - \vartheta} \right] \leq K_{\text{dec}} \vartheta^{L+1}. \quad (3.23)$$

The error bound and its envelope constant K_{dec} depend only on C/m and ρ , not on the horizon or any packet dimension. For $C = 0$ or $\rho = 0$ the approximation is exact.

Proof. The choice in (3.22) lies in $(\rho, 1)$ and satisfies

$$2C \left(\frac{\rho}{\vartheta - \rho} - \frac{\rho}{1 - \rho} \right) = m/2. \quad (3.24)$$

Consider any principal block matrix R_{HH} and scalar block weights $W_i = w_i I$ such that both ratios obey $|w_i/w_j - 1| \leq \vartheta^{-|i-j|} - 1$. The diagonal blocks of $E = W^{-1}R_{HH}W - R_{HH}$ vanish. Every row and column sum of its block norm matrix is at most

$$2C \sum_{h \geq 1} \rho^h (\vartheta^{-h} - 1) = m/2.$$

The row-and-column Schur estimate gives $\|E\|_2 \leq m/2$: if a nonnegative scalar matrix has maximum row sum u and column sum v , Cauchy–Schwarz yields $\|M\|_2^2 \leq uv$, and application to component norms gives the block version. Since $R_{HH} \succeq mI$, $\|R_{HH}^{-1}E\|_2 \leq 1/2$. Factoring $R_{HH} + E = R_{HH}(I + R_{HH}^{-1}E)$ and summing its Neumann series proves

$$\|(W^{-1}R_{HH}W)^{-1}\|_2 \leq 2/m. \quad (3.25)$$

For example, $w_j = \vartheta^{-|j-i|}$ gives the usual inverse-block decay $\|(R_{HH}^{-1})_{ij}\|_2 \leq (2/m)\vartheta^{|i-j|}$. This is the classical weighted-conjugation localization argument; here it is used on each actual local regression.

On $\mathcal{H}_t$, take $w_j = \vartheta^{-(t-j)}$. These weights satisfy the ratio condition. The horizontal row operator obeys $\|R_{t\mathcal{H}_t}W\|_2 \leq C \sum_{h \geq 1} (\rho/\vartheta)^h = C\rho/(\vartheta - \rho)$, because its norm is at most the sum of the block norms. The identity

$$b_t W = (R_{t\mathcal{H}_t}W)(W^{-1}R_{\mathcal{H}_t\mathcal{H}_t}W)^{-1}$$

and (3.25) give $\|b_{tj}\|_2 \leq B_* \vartheta^{t-j}$. For $j < t - L$, all $u \in \mathcal{H}_t$ are later than j , whence

$$\begin{aligned} \|\text{Cov}(Z_t, \epsilon_j)\|_2 &\leq C\rho^{t-j} + B_*C \sum_{u \in \mathcal{H}_t} \vartheta^{t-u} \rho^{u-j} \\ &\leq C\vartheta^{t-j} \left[1 + B_* \sum_{r \geq 1} (\rho/\vartheta)^r \right] = K_* \vartheta^{t-j}. \end{aligned}$$

Finally, the Schur-complement variational formula gives $D_t \succeq mI$: for a vector v on the target packet, $v^\top D_t v$ is the minimum quadratic form of $R_{\{t\} \cup \mathcal{H}_t}$ over vectors with target block v , and is therefore at least $m\|v\|^2$. Lemma 3.2 proves (3.23); bounding $1 - \vartheta^L$ by one gives its last inequality. $\square$

An alternative $\vartheta \in (\rho, 1)$ may be used if the left side of (3.24) is at most $m/2$; the definitions of B_* and K_* then remain valid. The constants are conservative: at $m = C = 1$, $\rho = 2/5$, they give $\vartheta = 46/55$, $B_* = 11/6$ and $K_* = 193/72$. The first windows making the finite bound at most 0.05 and 0.01 are respectively 49 and 58, which imply enormous mask counts on a sufficiently long horizon. The theorem broadens the covariance scope; it is not a practical improvement over the Markov bounds.

Corollary 3.7 (Supplied block metrics). *Let $V = \text{diag}(V_t)$, with each $V_t \succ 0$. The same theorem holds if (3.21) is replaced by*

$$R \succeq mV, \quad \begin{pmatrix} C\rho^{|t-s|}V_t & R_{ts} \\ R_{st} & C\rho^{|t-s|}V_s \end{pmatrix} \succeq 0 \quad (t \neq s). \quad (3.26)$$

The metrics may have arbitrarily large Euclidean condition numbers.

Proof. Congruence by $\text{diag}(V_t^{-1/2}, V_s^{-1/2})$ identifies the second condition with $\|V_t^{-1/2} R_{ts} V_s^{-1/2}\|_2 \leq C\rho^{|t-s|}$: the symmetric block matrix with diagonal cI and off-diagonal $B, B^\top$ is positive semidefinite exactly when $\|B\|_2 \leq c$, as follows by singular-value decomposition. Apply Theorem 3.6 to $V^{-1/2} R V^{-1/2}$. The block-orthogonal residual transformation in Theorem 3.4 preserves the norm bound and congruence returns the precision bound to the original coordinates. $\square$

For rational input and rational m, C, ρ, V_t , the promises are rational positive-semidefinite tests. All local inverses and information can be computed in original coordinates; no irrational whitening is required. Here “metric” means a supplied positive-definite matrix on each packet; the calendar distance remains $|t - s|$.

A direct application is $\epsilon_t = H_t X_t + v_t$ with $\|A_t\|_2 \leq \rho$, $P_t \preceq \bar{P}I$, $\|H_t\|_2 \leq \bar{H}$, and $V_t \succeq mI$. The independent-noise component gives $R \succeq mI$ and $R_{ts} = H_t A_t \cdots A_{s+1} P_s H_s^\top$ for $t > s$ gives $C = \bar{H}^2 \bar{P}$. This covers fixed partial packets without any process-noise floor, at the cost of the slower generic rate.

3.6 Partial observations using covariance contraction

For a more direct partial-observation bound, take $X_t = A_t X_{t-1} + w_t$ and $\epsilon_t = H_t X_t + v_t$ with independent Gaussian noises, $P_t = \text{Cov}(X_t)$, process covariance $\Xi_t = \text{Cov}(w_t)$, and $V_t = \text{Cov}(v_t) \succ 0$. Latent and packet dimensions may be part of the input. The matrices H_t are fixed; selecting a packet acquires all its responses. Assume, for constants $0 \leq \gamma < 1$ and $s \geq 0$,

$$\Xi_t \succeq (1 - \gamma^2)P_t \quad (t \geq 2), \quad H_t P_t H_t^\top \preceq s V_t \quad (t \geq 1). \quad (3.27)$$

The initial covariance may be singular. These promises are invariant under invertible changes of state and observation coordinates and do not assume that a partial-observation update is a Euclidean contraction.

Theorem 3.8 (Intrinsic partial-packet bound). *Put $\kappa = s/(1+s)$. Under (3.27), every selected subset satisfies*

$$\|C_L(S) - I\|_2 \leq \delta_L^{\text{part}} := 2\kappa\{T_L(\gamma) + \sqrt{\kappa s} N_L(\gamma)\}. \quad (3.28)$$

Use zero for $s = 0$, $\gamma = 0$, or complete history. No dimension appears in the constant.

Proof. We give the covariance transport details, including singular covariances. For any fresh filter and any chosen update pattern, let Π_t^- and Π_t^+ denote its prediction and posterior state covariances. Conditional covariance monotonicity gives $\Pi_t^+ \preceq \Pi_t^- \preceq P_t$. At an update,

$$D = H\Pi^-H^\top + V, \quad K = \Pi^-H^\top D^{-1}, \quad E = I - KH, \quad \Pi^+ = E\Pi^-E^\top + KVK^\top.$$

At a skipped time use $E = I$ and $\Pi^+ = \Pi^-$. For every transition,

$$A_t\Pi_{t-1}^+A_t^\top = \Pi_t^- - \Xi_t \preceq \gamma^2\Pi_t^-. \quad (3.29)$$

Indeed $\Xi_t \succeq (1 - \gamma^2)P_t \succeq (1 - \gamma^2)\Pi_t^-$. Multiplying (3.29) by later update and transition matrices and using $E\Pi^-E^\top \preceq \Pi^+$ proves

$$\mathcal{T}_{tj}\Pi_j^+\mathcal{T}_{tj}^\top \preceq \gamma^{2(t-j)}\Pi_t^-. \quad (3.30)$$

Here $\mathcal{T}_{tj}$ transports a state error just after j to the prediction at t , including intervening updates and no update at t . If there is no update at j , its starting covariance is the prediction there; in particular the formula holds when a fresh filter starts with unconditional covariance P_j .

Two endpoint estimates retain the signal-to-noise ratio. For any $0 \preceq \Pi \preceq P_t$, put $D = H_t\Pi H_t^\top + V_t$ and $M = \Pi^{1/2}H_t^\top V_t^{-1}H_t\Pi^{1/2}$. The nonzero eigenvalues of M are those of $V_t^{-1/2}H_t\Pi H_t^\top V_t^{-1/2}$, so $0 \preceq M \preceq sI$. The singular-value decomposition and the scalar inequality $u/(1+u) \leq \kappa$ for $0 \leq u \leq s$ give

$$\|D^{-1/2}H_t\Pi^{1/2}\|_2 \leq \sqrt{\kappa}, \quad (3.31)$$

$$KV_tK^\top \preceq \kappa\Pi^+. \quad (3.32)$$

For the second inequality use the identities, valid also for singular Π , $\Pi^+ = \Pi^{1/2}(I + M)^{-1}\Pi^{1/2}$ and $KV_tK^\top = \Pi^{1/2}M(I + M)^{-2}\Pi^{1/2}$. For use below, if $UU^\top \preceq c\Pi$ then $U = \sqrt{c}\Pi^{1/2}B$ for a contraction B (and $U = 0$ if $c = 0$). To prove this without assuming Π invertible, diagonalize Π : the inequality makes the rows of U on its nullspace zero; divide the remaining rows by positive square roots and use $BB^\top \preceq I$. Together with (3.30) and (3.31), this factorization converts covariance order into the following norm estimates.

For a far pair $s < t - L$, let $\Pi_s^{-,s}$ be the prediction covariance from $\mathcal{H}_s$. Orthogonality yields $U = \text{Cov}(X_s, Z_s) = \Pi_s^{-,s}H_s^\top$ and

$$UD_s^{-1}U^\top \preceq \kappa\Pi_s^{-,s} \preceq \kappa P_s.$$

The t -local filter starts at s with P_s and updates only later. Its cross covariance with Z_s is $H_t\mathcal{T}_{ts}U$. Transport, factorization, and the target endpoint bound therefore give

$$\|D_t^{-1/2}\text{Cov}(Z_t, Z_s)D_s^{-1/2}\|_2 \leq \kappa\gamma^{t-s}. \quad (3.33)$$

For $j < t - L$, $\text{Cov}(X_j, \epsilon_j) = P_j H_j^\top$, and the signal promise implies $P_j H_j^\top V_j^{-1} H_j P_j \preceq s P_j$. The same transport now gives

$$\|D_t^{-1/2} \text{Cov}(Z_t, \epsilon_j) V_j^{-1/2}\|_2 \leq \sqrt{\kappa s} \gamma^{t-j}. \quad (3.34)$$

Finally, for $j \in \mathcal{H}_s$, its local regression coefficient is $b_{sj} = H_s \mathcal{T}_{sj} K_j$ with gains from the s -local filter. Using (3.32) before transport and (3.31) after it yields

$$\|D_s^{-1/2} b_{sj} V_j^{1/2}\|_2 \leq \kappa \gamma^{s-j}. \quad (3.35)$$

For a near pair, insert $V_j^{-1/2} V_j^{1/2}$ in each surviving term of (3.9) and use (3.34)–(3.35). Its normalized block norm is at most $\kappa \sqrt{\kappa s} \sum_{d=L+1-h}^L \gamma^{h+2d}$. Summing these near weights and the far weights (3.33) on both sides proves (3.28). If $s = 0$ the observed state signal vanishes. If $\gamma = 0$, $A_t P_{t-1} A_t^\top = P_t - \Xi_t = 0$, so every cross-time observation covariance vanishes. These cases are exact even with singular state covariances. $\square$

Corollary 3.9 (Euclidean process-noise floor). *Suppose $P_t \preceq \bar{P}I$, $\Xi_t \succeq qI$ for $t \geq 2$, $V_t \succeq rI$, and $\|H_t\|_2 \leq \bar{H}$, with $0 < q \leq \bar{P}$ and $r > 0$. Then Theorem 3.8 applies with $\gamma = \sqrt{1 - q/\bar{P}}$ and $s = \bar{H}^2 \bar{P}/r$. The weaker unnormalized bound is also valid:*

$$\delta_L^{\text{Euc}} = \frac{2\bar{H}^2 \bar{P}}{r} \left\{ T_L(\gamma) + \bar{H} \sqrt{\bar{P}/r} N_L(\gamma) \right\}. \quad (3.36)$$

Proof. $qI \succeq (q/\bar{P})P_t$ and $H_t P_t H_t^\top \preceq \bar{H}^2 \bar{P}I \preceq s V_t$ give the intrinsic promises. For the weaker constants, use $K V K^\top \preceq \Pi^+$ in place of (3.32); transport gives $\|b_{tj}\|_2 \leq \bar{H} \sqrt{\bar{P}/r} \gamma^{t-j}$. For an old observation, $P_j H_j^\top H_j P_j \preceq \bar{H}^2 \bar{P} P_j$, so transport gives $\|\text{Cov}(Z_t, \epsilon_j)\|_2 \leq \bar{H}^2 \bar{P} \gamma^{t-j}$. For a far residual use $U U^\top \preceq \bar{H}^2 \bar{P} \Pi_s^{-,s} \preceq \bar{H}^2 \bar{P} P_s$ in the same way. Lemma 3.2 with $d_* = r$ proves (3.36). The intrinsic bound is no larger because $\kappa \leq s$ and $\kappa \sqrt{\kappa s} \leq s \sqrt{s}$. $\square$

Covariance contraction and finite-memory filtering have substantial prior history. In particular, Kozdoba et al. [28, Theorems 1–2] use process-noise dissipation in a covariance norm to approximate forecasts after burn-in for observable time-invariant systems. Here the sufficient covariance inequalities are imposed on the supplied finite, possibly time-varying horizon, transport is proved separately for every fresh local selection history, and the bounded quantity is the relative precision of the resulting local conditionals. Theorem 3.8 relies on the relative process-noise floor $\Xi_t \succeq (1 - \gamma^2)P_t$, whereas Theorem 3.4 allows zero process noise.

3.7 A finite graph for local information and its scope

Before deciding at time t , a state records the last L selection bits. A skip arc shifts in zero and has information weight zero. A choose arc shifts in one and has positive-semidefinite weight

$$W_{t,H} = G_{t,H}^\top D_{t,H}^{-1} G_{t,H}, \quad H = \{j \in [t - L, t - 1] : \text{the state selects } j\}. \quad (3.37)$$

Start from the all-zero history and connect all terminal states to a sink. There are $O(n2^L)$ states and arcs. Each schedule yields one path and its arc sum is exactly $I_L(S)$, because the state determines the entire local conditional in (3.3). Conversely, every path specifies that schedule. Path decomposition therefore gives the exact hull of the surrogate information and calendar visit vectors.

If visit marginals are binary, every path in a flow decomposition has those same decisions, and the deterministic state transitions make the path unique. An exact cardinality adds a count layer, giving $O(n(k+1)2^L)$ states and arcs. Mandatory or forbidden times remove arcs. General linking constraints preserve integer exactness, but intersection with such constraints need not preserve the unconstrained continuous hull.

For minimum gap g , a valid L -bit mask contains no two ones within distance $g-1$. The number of such masks is

$$M(L, g) = 1 + \sum_{q=1}^{\lceil L/g \rceil} \binom{L - (g-1)(q-1)}{q}. \quad (3.38)$$

To count masks with q ones, subtract $(g-1)(i-1)$ from the i th ordered one-position; this bijects to q unconstrained positions in a line of length $L - (g-1)(q-1)$. If $L < g-1$, the information mask forgets a last observation before the spacing restriction expires. A correct graph also stores a cooldown counter, or uses $\max(L, g-1)$ bits for feasibility while still using only L bits for information. This covers $L = 0$ and $g > L$. Feasible exact counts satisfy $k \leq \lceil n/g \rceil$. Any support objective $\sum \langle N, W_{t,H} \rangle$, with fixed symmetric N , is optimized exactly by the ordinary longest-path recurrence on this acyclic graph; scalar weights need not be nonnegative for that recurrence. Complexity and approximation guarantees for particular objectives are developed separately.

The construction uses a *calendar* window. Keeping the last m *selected* observations instead is a valid conditional approximation, and the scalar contraction bound applies after reindexing the selected sequence (each selected gap contracts by at most ρ). A direct graph for it remembers an ordered tuple of up to m calendar indices and may require $O(n^{m+1})$ states and arcs, a different state representation. Refining a physical-time grid can make the per-index contraction arbitrarily close to one. The window and graph bounds are conditional on the stated decay constants and do not establish uniform scalability under grid refinement. Simultaneous scalar responses should be treated as one packet, not as successive strictly contracting calendar steps.

3.8 Relationship to existing locality guarantees

The preceding arguments give explicit sufficient error bounds for the specified local histories; their ingredients have extensive antecedents. Inverse-factor localization is studied by Benzi and Tuma [6], Krishtal et al. [29], and Rubensson et al. [49]; finite-predictor tail inequalities include Meyer et al. [38] and Inoue et al. [22]. The former works concern banded inverse factors, inverse-closed decay algebras, or localized iterative factorizations; the latter use stationary spectral or multivariate long-memory assumptions. We prove the needed local-regression statement directly rather than presume that a theorem for an arbitrary localized inverse factor identifies the factor in (3.4). In particular, the fixed exponential rate does not follow merely from the admissible-weight algebra in Krishtal et al. [29, Corollary 5.1]: its admissible weights satisfy the Gelfand–Raikov–Shilov condition, which a fixed exponential weight fails. We also do not use an unproved direct-sum or factorization-inheritance argument.

Schäfer et al. [53, Theorem 2.1] characterize fixed-pattern KL-optimal inverse-Cholesky factors. Their Theorem 3.4 gives rigorous error and complexity bounds for Green’s covariances with geometric ordering; the neighborhood radius has a $\log(n/\epsilon)$ dependence. Their added-noise construction in Section 4.1 uses a latent precision approximation and incomplete Cholesky refinement. Huan et al. [20] study conditional selection for sparse inverse factors, including its experimental-design interpretation. Kaminetz and Webber [24] identify a partial-Cholesky-plus-Vecchia model with an augmented Vecchia pattern and prove fixed-pattern Kaporin optimality. These are direct antecedents for the approximation and its analysis. Our comparisons concern model hypotheses, factor and

ordering, constants and horizon dependence, not whether KL can control Fisher information: Appendix A.3 proves that a global Gaussian KL bound does imply a relative precision and mean-information sandwich.

Two reductions further limit the novelty of Theorem 3.6. First, pad a selected covariance at its original calendar positions with independent mI dummy blocks at deleted times. The padded covariance preserves (3.21); its selected local conditionals are unchanged, and its normalized residual covariance is, after permutation, $C_L(S) \oplus I$. Thus a suitable *class-uniform full-calendar* theorem already implies subset uniformity. Second, unbounded original diagonal blocks do not preclude reduction to a bounded-spectrum class. Put $D = \text{diag}(R_{tt})$, $E = R - D$, and $b_0 = 2C\rho/(1 - \rho)$. The block row bound gives $\|E\|_2 \leq b_0$, while $D \succeq mI$. Hence

$$D \preceq R + b_0I \preceq (1 + b_0/m)R, \quad R \preceq D + b_0I \preceq (1 + b_0/m)D,$$

and therefore

$$\frac{m}{m + b_0}I \preceq D^{-1/2}RD^{-1/2} \preceq (1 + b_0/m)I. \quad (3.39)$$

Its off-diagonal blocks decay with amplitude C/m and rate ρ . Consequently, neither subset uniformity nor the absence of an upper bound on the original diagonal is by itself a new localization principle, and we make no first-localization claim. The contribution is the fully explicit local-regression bounds (3.11), (3.20), (3.23), and (3.28), and their use with a fixed-calendar information graph to certify the original selected-covariance design objective.

4 Approximation with exact discrete feasibility

Relative information control has two algorithmic consequences. A linear criterion, weighted trace, can be optimized on the local graph without fixing the parameter dimension (Section 4.1). When that dimension is fixed, several nonlinear criteria can instead share one finite set of feasible designs (Sections 4.3 to 4.5). The two results use different computational models.

4.1 A rational weighted-trace scheme

All inputs in this subsection are rational and explicitly encoded in binary, including the sensitivity matrices, $J_0 \succeq 0$, and a common weight $W \succeq 0$. We maximize

$$\mathcal{I}(S) = \text{tr}(WJ(S)).$$

This is a weighted trace of information, not conventional A-optimality. An FPTAS is an algorithm, polynomial in input length and $1/\varepsilon$, that returns a feasible value at least $1 - \varepsilon$ times the maximum.

Theorem 4.1 (Weighted trace with input-sized dimensions). *Suppose the covariance belongs to one of the following promised classes:*

1. *The scalar model (3.10), with fixed rational upper bounds $0 \leq \rho \leq \rho_0 < 1$ and $\bar{P}/r_{\min} \leq B_0$.*
2. *The full-packet model of Theorem 3.4, supplied in rational coordinates with $P_t \preceq B_0V_t$ and $A_tV_{t-1}A_t^\top \preceq \rho_0^2V_t$, for fixed rational $B_0 \geq 0$, $0 \leq \rho_0 < 1$.*
3. *The general-decay model (3.21), or its supplied metric version (3.26), with fixed rational upper bounds on C/m and $\rho < 1$.*
4. *The partial-packet model (3.27), with fixed rational upper bounds $s \leq s_0$ and $\gamma \leq \gamma_0 < 1$.*

Packet, latent-state, and parameter dimensions may be part of the input. For rational $0 < \varepsilon < 1$, an exact cardinality k , mandatory and forbidden times, and an optional integer minimum calendar gap $g \geq 1$, there is a deterministic rational FPTAS, and it detects infeasibility exactly. The polynomial depends on the fixed promise constants, but its exponent does not grow with these dimensions.

Proof. Each class has a rational envelope

$$\delta_L \leq C_0 \beta^{L+1}, \quad 0 < \beta < 1, \quad (4.1)$$

with C_0, β bounded by fixed constants. For the scalar and full-packet classes one may use $C_0 = 2B_0/(1 - \rho_0)^2$, $\beta = \rho_0$, from the weaker gain bound (3.14); the stronger bound is not needed for complexity. For general decay use $K_{\text{dec}}, \vartheta$ in (3.22), formed with the fixed upper promises in place of C/m and ρ (equivalently, apply the theorem with those larger decay/floor-ratio bounds). For partial packets put $\kappa_0 = s_0/(1 + s_0)$ and use

$$C_0 = 2\kappa_0 \left\{ \frac{1}{1 - \gamma_0} + \frac{s_0 \gamma_0}{(1 - \gamma_0)(1 - \gamma_0^2)} \right\}, \quad \beta = \gamma_0.$$

This follows from $\sqrt{\kappa_0 s_0} \leq s_0$ in (3.28). Zero signal variance or zero contraction makes the relevant conditional model exact with $L = 0$; no logarithm or expression involving $1/\log(1/\beta)$ is needed in that case.

For $n \geq 1$ choose the first $L \geq 0$ for which $C_0 \beta^{L+1} \leq \varepsilon/2$, stopping at $L = n - 1$ if necessary. Use rational powers and comparisons. At the cap all histories are complete and their error is zero. Compute (3.3) in the original rational coordinates and assign arc score $\text{tr}(WW_{t,H}) \geq 0$ to (3.37). Add $\text{tr}(WJ_0)$ once. Exact longest-path optimization then maximizes $\mathcal{I}_L(S) = \text{tr}(WJ_L(S))$ over the graph.

The following polynomial extension of the graph enforces feasibility. Clamp g to $\min(g, n + 1)$, which leaves every feasible schedule unchanged. Before time t record a counter $c \in \{0, \dots, g - 1\}$ for the number of still-forbidden times. A skip replaces c by $\max(c - 1, 0)$; a choose is permitted only at $c = 0$ and sets the next counter to $g - 1$. The count starts at zero, increases by one on a choose, and must end at k . Initialize the counter and history at zero. Remove skips at mandatory times and chooses at forbidden times. The graph has $O(n(k + 1)g2^L)$ states and arcs and represents exactly the required family, including $L = 0$ and $g > L$. Reachability detects conflicting restrictions. If $k \notin \{0, \dots, n\}$ the family is infeasible. For $k = 0$, or $n = 0$, return the empty schedule only when all restrictions permit it.

Let S^* maximize the true criterion and $\hat{S}$ maximize its surrogate. Lemma 3.1 and $W, J_0 \succeq 0$ imply

$$\mathcal{I}(\hat{S}) \geq \frac{\mathcal{I}_L(\hat{S})}{1 + \delta_L} \geq \frac{1 - \delta_L}{1 + \delta_L} \mathcal{I}(S^*) \geq (1 - \varepsilon) \mathcal{I}(S^*). \quad (4.2)$$

No division by the optimum is used, so a zero optimum is included.

For the size bound, put $a = \log 2 / \log(1/\beta)$. Minimality before the full-history cap gives

$$2^L \leq \max\{1, (2C_0/\varepsilon)^a\}.$$

If the cap is encountered earlier this bound remains valid: when $L > 0$ the previous window failed the accuracy test. Thus the graph has polynomial size. A simple algorithm first forms the whole covariance by rational state recursions, or reads the supplied dense covariance. A local solve uses at most the scalar coordinates of $L + 1$ packets; the number of histories is $O(n2^L)$. These matrix operations are polynomial in the explicitly supplied dimensions. In the metric cases whitening is used only in the proof; no irrational square root is computed by the algorithm.

It remains to bound bit complexity, not only arithmetic counts. State recursions expand into sums of products along at most n transitions. A common denominator formed from input denominators to polynomial powers has polynomial bit length; the logarithm of the number of expanded coordinate paths is also polynomial. Hence the covariance entries have polynomial bit length. Rational principal solves have polynomial-size numerators and denominators by determinant bounds and can be computed by fraction-free elimination. Each path value sums at most n rational arc scores and one prior score. Its denominator bit length is bounded by the sum of those polynomial bit lengths. Rational comparison is therefore polynomial even when many paths merge into a state. This proves the Turing bound. $\square$

The theorem takes model sensitivities as input; it does not assert that an arbitrary nonlinear differential-equation model has polynomial-time sensitivity evaluation. Promise verification is rational linear algebra: for example, $BB^\top \preceq a^2 I$ tests $\|B\|_2 \leq a$ without computing eigenvalues. For supplied block metrics use the corresponding Schur-complement matrix inequality. Such tests certify the promises of one finite input; the fixed-constant assumption restricts a sequence of instances. If contraction approaches one under grid refinement, the exponent a is no longer uniform. Arbitrary binary resource budgets need not admit a polynomial-size graph.

4.2 Scope, hardness, and the scalar predecessor

Theorem 4.1 selects complete packets, and this restriction is substantive. The following conditioning calculation shows why; we do not claim to introduce hardness of correlated selection.

Proposition 4.2 (Individual-channel boundary). *Unless $P = NP$, there is no deterministic FPTAS for scalar information under exact individual-channel cardinality, even at a single time with covariance condition number at most $5/3$, independent isotropic observation noise, and the fixed full-packet variance promise $B_0 = 1$.*

Proof. Independent Set is NP-hard on cubic graphs [40, Theorem 4.1(a)]. For a graph on m vertices with adjacency matrix A , take $R = I + A/12$, scalar sensitivity $\mathbf{1}$, latent covariance $P = I/3 + A/12$, and observation noise $V = 2I/3$. Since $\|A\|_2 \leq 3$, $P \succ 0$, $P \preceq (7/8)V$, and $3I/4 \preceq R \preceq 5I/4$. The transition promise is vacuous. For a k -subset S let $x = R_{SS}^{-1}\mathbf{1}$. A Neumann-series bound from $\|A_{SS}/12\|_\infty \leq 1/4$ gives $\|x\|_\infty \leq 4/3$. Each equation for x_i therefore gives $x_i \geq 1 - 3(4/3)/12 = 2/3$. Summing those equations yields

$$\mathbf{1}^\top R_{SS}^{-1} \mathbf{1} = k - \frac{1}{12} \sum_{i \in S} \deg_S(i) x_i.$$

This is k for an independent set and at most $k - 1/9$ otherwise. With optional fixed prior $J_0 = 1$, an FPTAS at $\varepsilon = 1/[18(m+1)]$ must return an independent k -set whenever one exists, since its permitted loss is at most $1/18$. Checking that returned set would decide the NP-hard problem in polynomial time. Trivial impossible cardinalities can be checked first. The reduction chooses separate channels within one packet, which Theorem 4.1 does not allow. $\square$

For scalar information, strong positive results already exist. The bounded-treewidth Gaussian message scheme of Mahalanabis and Štefankovič [36, Section 3.2, Theorem 43] can be specialized to one unobservable target and designated observable candidates. Appendix B supplies the recurrence adaptation, a rational normalization and regularization, and the conversion from posterior variance to information. This yields the scalar approximation guarantee in the predecessor’s spectral-rounding

model, so we do not claim a first scalar FPTAS. The explicit rational finite-history realization in Theorem 4.1 avoids importing that numerical model.

For a growing latent block or a growing-rank target, the corresponding Gaussian separator contains growing-dimensional precision messages, so the bounded-width runtime does not by itself give the dimension dependence of Theorem 4.1. Nor can independent scalar target problems be optimized separately when they must share a schedule. This is a scope distinction, not a lower bound on every older method. Weighted information and common-support multiple-response approximation are established: Liu et al. [34] treats correlated Fisher criteria, while Liberty and Sviridenko [32, Section 5] gives sparse multiple-regression guarantees with enlarged support. The exact cardinality in our statement matters.

4.3 A relative matrix cover of feasible paths

We now fix the information dimension $p \geq 1$. Let $G = (V, E)$ be an explicitly encoded directed acyclic graph, with source s , sink t , rational $Q_e \succeq 0$ on each edge and rational $Q_0 \succeq 0$. For a path P write $J(P) = Q_0 + \sum_{e \in P} Q_e$. This notation is local to the additive graph; Q_e is an information atom, not an approximate observation precision.

Theorem 4.3 (Feasible spectral approximation set). *For rational $0 < \eta < 1$, a deterministic algorithm returns a set $\mathcal{C}$ of actual s - t paths such that for every feasible path P some $\hat{P} \in \mathcal{C}$ satisfies*

$$(1 - \eta)J(P) \preceq J(\hat{P}) \preceq (1 + \eta)J(P). \quad (4.3)$$

The set size and Turing bit complexity are polynomial in the entire input length and $1/\eta$ for fixed p . No positive lower eigenvalue, positive definite prior, or condition-number bound is assumed. The two matrices in (4.3) have exactly the same kernel. Infeasible graphs return the empty set; if $s = t$ the sole path is empty and is returned exactly.

The following normalization lemma serves both the graph algorithm and the represented-matroid extension below.

Lemma 4.4 (Rational normalization with exact ranges). *Consider m rational PSD atoms of order p , a common PSD prior, and feasible objects selecting each atom at most once. Decompose each matrix into at most p labeled terms $w_j v_j v_j^\top$, where $w_j > 0$ and v_j are rational; retain the atom or prior owning each label. There are $M \leq p(m + 1)$ labels. There is a rational trial for each independent set of r labels, $1 \leq r \leq p$, with the following properties. Retained matrices have ranges contained in a common subspace $\mathcal{S}$, a reversible rectangular congruence to $r \times r$ matrices whose entries have absolute value at most $4p$, and at most r forced atom owners. Every accepted object has transformed information at least I_r and range exactly $\mathcal{S}$. Every feasible object of positive information rank survives at least one trial. The algebraic normalization, filters, and owner requirements have polynomial bit complexity for fixed p ; feasibility is handled by the later family-specific algorithms.*

Proof. Rational LDL factorization supplies the decomposition. The reference implementation inspects the leading diagonal of each residual. If that entry is zero, its row and column vanish because $|a_{ij}|^2 \leq a_{ii}a_{jj}$, and the coordinate is skipped. Otherwise the pivot is positive and leaves a rational PSD Schur complement. Each step removes one coordinate, proving termination even for singular matrices, without square roots. Fix an independent label set with column matrix $V_b = [v_{b_1}, \dots, v_{b_r}]$, and define

$$\begin{aligned} L_b &= (V_b^\top V_b)^{-1} V_b^\top, & \Pi &= V_b L_b, \\ T &= \text{diag}(\tau_i) L_b, & K &= V_b \text{diag}(\tau_i^{-1}), \quad 1 \leq \tau_i^2 w_{b_i} < 4. \end{aligned} \quad (4.4)$$

Here each τ_i is a power of two, found by rational comparisons. We have $TK = I_r$, $KT = \Pi$, and Π is the orthogonal projector onto $\mathcal{S} = \text{range } V_b$. Reject the trial if $\Pi Q_0 \neq Q_0$; discard an atom if $\Pi Q_e \neq Q_e$. For any retained symmetric matrix,

$$Q = K(TQT^\top)K^\top. \quad (4.5)$$

Indeed $\Pi Q = Q$ implies $Q\Pi = Q$, so the right side is $\Pi Q\Pi$. Put $A_e = TQ_eT^\top$ and $A_0 = TQ_0T^\top$. Reject the trial if some diagonal of A_0 exceeds $4p$, and discard atoms with that property. Positive semidefiniteness bounds all remaining entries by $4p$ in magnitude. Require all distinct nonprior owners of the basis labels. Their factors are present with their original multiplicities, so every accepted object's information A obeys

$$A \succeq \text{diag}(\tau_i^2 w_{b_i}) \succeq I_r. \quad (4.6)$$

Together with the range tests this gives exact range $\mathcal{S}$.

To prove coverage, fix an object whose information has rank $r > 0$. The factors in its atoms and prior span its information range: a vector is in the kernel of a sum of PSD matrices exactly when it is in every summand's kernel. Introduce $u_j = \sqrt{w_j}v_j$ for this proof only, and choose r such vectors of maximum r -dimensional volume. For their matrix B , every other selected factor has coordinates $u = Bx$ with $|x_i| \leq 1$. Replacing column i by u multiplies volume by $|x_i|$, which proves the assertion. The formal proof carries out this determinant replacement in coordinates of the factor span, including when it is a proper subspace of the ambient space. The algorithm enumerates this basis's rational labels. Its range tests preserve the object, and $TB = \text{diag}(\tau_i \sqrt{w_{b_i}})$ has diagonal in $[1, 2)$. Every transformed selected factor therefore has coordinates of magnitude less than two. Each matrix has at most p factors, so its transformed diagonal is at most $4p$ and it survives every filter. Its forced owners were selected by construction.

Exact LDL, rational inverses, projectors and filters have polynomial bit length by determinant bounds. A very small rational pivot changes the dyadic exponent by at most its encoding length; the dyadic rational itself still has polynomial encoding length. Thus no square-root or maximum-volume oracle is used, and all actual trial operations are polynomial in bit complexity. $\square$

Proof of Theorem 4.3. After the stated reachability and $s = t$ cases, let $N = \max(1, |V| - 1)$ bound path length. Apply Lemma 4.4 with edge owners. For a rank- r trial use $h = \eta/(rN)$ and the signed integer labels

$$z_{e,ij} = \lfloor A_{e,ij}/h \rfloor, \quad 1 \leq i \leq j \leq r. \quad (4.7)$$

Do not round the prior. In topological order keep one actual path at every state consisting of a vertex, a bit mask of forced owners already visited, and the sum of upper-triangular integer labels. Transitions add labels and update masks. Every continuation depends only on this state. Strict topological increase prevents a suffix from revisiting either prefix, so substituting a representative preserves feasible continuations as well as their labels. At t return all representatives with complete masks, across all trials.

Fix a feasible rank- r path and its surviving trial from the lemma. Its terminal state has a representative $\hat{P}$. For each entry a path's rounding residual lies in $[0, Nh)$; this holds for negative entries too because the operation is floor. Equal integer sums and the common prior give $|A(\hat{P})_{ij} - A(P)_{ij}| < Nh$. The spectral norm of their symmetric difference is at most the maximum absolute row sum, hence at most $rNh = \eta$. Since $A(P) \succeq I_r$, this proves the two-sided relative inequality in normalized coordinates. Congruence by K and (4.5) prove (4.3). For rank zero, if $Q_0 = 0$, search the subgraph of exactly zero-matrix edges and include one path when possible. A zero PSD sum has only zero summands, so this covers every zero-information path.

For an explicit size bound, put $d_r = r(r+1)/2$ and

$$C_r = \left\lceil 8prN^2/\eta + N \right\rceil + 2.$$

Every signed accumulated coordinate has at most C_r values, because $|A_{e,ij}| \leq 4p$ and at most N labels are added. There are at most $|V|2^r C_r^{d_r}$ states and $|E|2^r C_r^{d_r}$ edge extensions per trial, up to dictionary and fixed-dimensional arithmetic costs. In particular,

$$|\mathcal{C}| \leq 1 + \sum_{r=1}^p \binom{M}{r} 2^r C_r^{d_r}. \quad (4.8)$$

Normalization, floors and integer sums have polynomial bit length; stored path reconstruction has length at most N . For the reference list implementation, write $B_r = 2^r C_r^{d_r}$. The retained table has at most $|V|B_r$ entries, while a transient candidate list can contain $1 + |E|B_r$ paths. A full-scan bound on key comparisons is $|V|(1 + |E|B_r)^2$. Key evaluation, edge scans, rational arithmetic, and copying whole paths contribute additional polynomial factors; state and extension counts alone are not total memory or bit-work bounds. This gives a polynomial realization without assuming an optimized dictionary. Exact ranges and (4.6) prove the kernel claim. $\square$

Exact ranges cannot be omitted. For $a = (1, 0)^\top$ and $b = (1, t)^\top$ with nonzero rational t , the matrices $aa^\top$ and $bb^\top$ approach each other entrywise as $t \rightarrow 0$ but have different kernels. Neither can dominate a positive multiple of the other. Also, $\mathcal{C}$ need not be a Pareto-optimal set: deleting a dominated matrix may destroy the upper half of (4.3). The exponent in (4.8) grows with p . This is a complexity theorem, not a practical running-time claim or a fixed-parameter tractability statement with a dimension-independent exponent.

4.4 Criteria, singular information, and finite-memory transfer

One computed set supports objectives chosen afterwards, provided their values can be compared. If $\Phi \geq 0$ is Loewner-nondecreasing and homogeneous of degree $a > 0$, the best representative has value at least $(1 - \eta)^a \max_P \Phi(J(P))$. This follows by applying Φ to the lower sandwich; concavity is unnecessary. Monotonicity alone supplies no such multiplicative guarantee.

For D-optimality, $\eta = \varepsilon/p$ gives determinant ratio $(1 - \eta)^p \geq 1 - \varepsilon$ by Bernoulli's inequality. Rational determinants can be compared exactly. Determinant roots have ratio $1 - \eta$; these are not multiplicative log-determinant statements. For E-optimality the minimum eigenvalue has ratio $1 - \eta$. The reference comparator computes rational characteristic coefficients of the Kronecker difference $A \otimes I - I \otimes B$. A coefficient bound separates every nonzero eigenvalue difference from zero. Exact rational PSD threshold tests and a prescribed finite dyadic bisection then distinguish strict order and equality, including repeated eigenvalues. The degree is fixed when p is fixed; no exact-real comparison oracle is supplied. If all paths are singular the E optimum is zero.

For conventional A-optimality give singular information infinite cost. Inversion of the lower sandwich gives $\text{tr}(J(\hat{P})^{-1}) \leq (1 - \eta)^{-1} \text{tr}(J(P)^{-1})$ on positive definite matrices. The choice $\eta = \varepsilon/(1 + \varepsilon)$ gives ratio $1 + \varepsilon$; inverse traces are rational. Existence of a positive definite feasible matrix is detected by the set itself. For a specified contrast c , give $c^\top J^\dagger c$ infinite cost unless $c \in \text{range } J$. The common-kernel property permits inversion on the common range and gives

$$\frac{c^\top J(P)^\dagger c}{1 + \eta} \leq c^\top J(\hat{P})^\dagger c \leq \frac{c^\top J(P)^\dagger c}{1 - \eta}.$$

This is not an assertion that pseudoinverses reverse order across different kernels. Multiple fixed estimable contrasts with nonnegative weights follow by summation. Adding any common $H \succeq 0$

preserves (4.3), as does any common congruence. The same set can therefore be reused after either operation, with the stated objective-evaluation qualifications.

Corollary 4.5 (Cover for the true finite-memory design model). *Suppose an explicit rational PSD-edge graph satisfies, for every feasible schedule, $(1 - \delta)J(S) \preceq J_L(S) \preceq (1 + \delta)J(S)$, $0 \leq \delta < 1$. The set from Theorem 4.3 satisfies*

$$\frac{(1 - \eta)(1 - \delta)}{1 + \delta} J(S) \preceq J(\hat{S}) \preceq \frac{(1 + \eta)(1 + \delta)}{1 - \delta} J(S) \quad (4.9)$$

for a representative of every feasible S . Under any fixed-promise class of Theorem 4.1, fixed p gives a deterministic polynomial relative cover, and hence the stated D , E , A and contrast schemes, for the true selected covariance.

Proof. Apply the local sandwich once to each of $S, \hat{S}$ and the graph cover between them to obtain (4.9). Given a desired $0 < \xi < 1$, take $\eta = \xi/4$ and a window with $\delta \leq \xi/8$. The lower factor's deficit is at most $\eta + 2\delta \leq \xi/2$. The upper factor's excess is $(\eta + 2\delta + \eta\delta)/(1 - \delta) \leq 17\xi/28 < \xi$. The graph construction and cover are polynomial under the fixed promises; the finite-history cap is handled as in Theorem 4.1. Choose ξ according to the desired scalar criterion ratio. All these sandwiches preserve kernels, including singular priors. $\square$

Lean verification boundary. The accompanying Lean formalization covers the explicit-DAG construction from the original rational prior and atoms on a graph with verified topological vertex indices: the two-sided cover and exact ranges, finite output counts, and the criterion and conditional transfer arguments above. It contains executable rational normalization and a representative-merging path DP; it is not a theorem conditional on a supplied successful trial or cover certificate. The complete bit-work bound starts from the original matrix and accuracy encodings and includes factor initialization, rejected trials, exact arithmetic, stored-data access, dictionaries and path recovery. Exact D , E , A and rational contrast selection also have proved execution bounds. The cost model uses schoolbook arithmetic and explicit storage and scan charges; it gives no wall-clock bound for Lean's runtime. The upstream stochastic locality estimates and finite-history graph construction required by the full Corollary 4.5 are not part of this verification. The represented-matroid theorem below is formalized separately.

4.5 Rationally represented matroids and contribution boundary

A different feasible family admits the same cover, through an established exact profile algorithm. Let $A \in \mathbb{Q}^{a \times m}$ represent a matroid: a set is independent precisely when its columns of A are independent. Its rank $q = \text{rank } A$ is input-sized. Give its elements rational PSD atoms Q_e of fixed order p , and define $J(B) = Q_0 + \sum_{e \in B} Q_e$ for bases.

Theorem 4.6 (Represented-matroid spectral cover). *For the preceding rational representation, one can deterministically return actual bases satisfying (4.3) for every target base, with polynomial output size and Turing bit complexity in input length and $1/\eta$ for fixed p . Singular information and an input-sized matroid rank are allowed. The same criterion and reuse consequences apply.*

The complete proof in Appendix C combines Lemma 4.4 with the squared-determinant profile polynomial, interpolation and deletion recovery of Bernstein et al. [7]; those algebraic algorithms are prior work. Our construction uses an extra owner-count coordinate instead of contraction, cached polynomial-time determinant evaluation, and tensor Lagrange interpolation. The Lean formalization

proves the assembled construction from the supplied rectangular rational representation, including row reduction, exact singular ranges, polynomial output size, polynomial charged bit work at fixed p , and exact criterion selectors with their work bounds. Its cost model uses schoolbook arithmetic and finite storage and scans, and excludes the construction of proof and observer transcripts. Rational Vandermonde representations include uniform matroids and their direct sums (partition quotas); oriented incidence matrices include graphic matroids and spanning forests, including labelled parallel edges and loops. The result requires the supplied characteristic-zero representation. It does not construct one from an independence oracle or finite-field representation, and it does not cover matroid intersections. Most importantly, additive matroid atoms do not encode arbitrary correlated history or an additional matroid restriction on the calendar graph without a further reduction.

Approximate Pareto sets and additive profile schemes are established [39, 45, 54]. Their nonnegative coordinate formulations do not directly turn relative entrywise errors into relative PSD errors, because cancellation and singular ranges matter. Brown et al. [11] already normalize, filter large vectors, force a guessed subset, and handle several monotone homogeneous criteria in a randomized fixed-dimensional PTAS for the broader independence-oracle matroid model. Our rationally represented specialization has deterministic polynomial accuracy dependence; it does not subsume their oracle model. Generic decomposition of multiresponse PSD atoms into tied rank-one copies is also established [17, Theorem 1]; decomposition itself is not a contribution here.

Covariance-state pruning predates these design schemes [3, 56]. Their additive-error results concern dynamic Riccati trajectories and process-noise/covariance bounds. Multiplicative relaxed dynamic programming is older still [2, 33]. A quadratic lower-envelope guarantee has a different quantifier: a retained matrix may depend on the tested direction. For example,

$$\min\{x^\top \text{diag}(0, 2)x, x^\top \text{diag}(2, 0)x\} \leq x^\top Ix \quad \text{for every } x,$$

yet neither retained singular matrix is a relative PSD approximation to I . Objective-independent spectral spanners and coresets are also prior art [21, 35]; they retain elements and use fractional replacements or criterion-specific rounding, rather than returning one feasible integral object within $1 \pm \eta$ of each target matrix. The projected-vertex method of Onn and Rothblum [42] addresses convex maximization. Its vertices are insufficient here: among the three feasible matrices $\text{diag}(1, 3)$, $2I$, $\text{diag}(3, 1)$, the middle projected profile uniquely maximizes both determinant and minimum eigenvalue. Fixed-decision-dimensional polynomial optimization [14] does not supply attainability of these fixed-dimensional matrix profiles: the number of path or element decisions is still input-sized. Recent one-exact Pareto frameworks [10] likewise require a suitable positive-objective restriction oracle; they do not automatically supply the signed-profile to PSD-order reduction used here.

To the best of our knowledge, the specific combined guarantee in Theorems 4.3 and 4.6—a polynomial set of feasible integral objects giving an all-target, two-sided relative PSD cover, exact singular ranges, and deterministic polynomial accuracy dependence under the stated explicit representations—is not stated in these predecessors. The contribution is the normalization-to-bounded-profiles reduction and its complete guarantees, not dynamic programming, profile interpolation, PSD pruning, or objective independence separately. The large polynomial exponents rule out any inference of practical superiority; Section 6 uses support pricing and certificates instead of these spectral sets.

5 Certificates, relaxation comparisons, and latent separators

A useful design bound does not require an optimization run to terminate successfully: a feasible schedule and a separately checked upper witness suffice. This section gives such witnesses, compares

them with established virtual-noise relaxations, and constructs an alternative exact representation for cases where a long calendar window is expensive. Throughout this section $\mathcal{F}$ is nonempty and finite, and $J_0 \succ 0$ unless stated otherwise; this stronger prior assumption ensures that every logarithm and inverse used in the certificates is defined.

5.1 Prior-aware support certificates

Suppose a graph represents exactly $\mathcal{F}$, a path P_S represents schedule S , and the local increments $Q_e \succeq 0$ satisfy

$$J_L(S) = J_0 + \sum_{e \in P_S} Q_e, \quad J(S) \preceq J_0 + \frac{1}{1-\delta} \sum_{e \in P_S} Q_e, \quad 0 \leq \delta < 1.$$

The second inequality is the data-information consequence of Lemma 3.1; it retains the prior without inflation. For scalar, complete-block, general-decay, and partial-packet models, Section 3 supplies the corresponding sufficient δ .

Theorem 5.1 (Support certificate). *For any $N \succ 0$, let*

$$\begin{aligned} c_N &= \log \det N - p + \text{tr}(N^{-1} J_0), \\ a_e &= \frac{\text{tr}(N^{-1} Q_e)}{1-\delta}, \quad U_N = c_N + \max_{S \in \mathcal{F}} \sum_{e \in P_S} a_e. \end{aligned} \tag{5.1}$$

Then $\max_{S \in \mathcal{F}} \log \det J(S) \leq U_N$. If $S_0 \in \mathcal{F}$ and $\ell \leq \log \det J(S_0)$, an upper enclosure $U \geq U_N$ certifies D -efficiency at least $\exp\{(\ell - U)/p\}$ relative to the true optimum.

Proof. For any $X \succ 0$, concavity gives $\log \det X \leq \log \det N - p + \text{tr}(N^{-1} X)$. Apply monotonicity of log determinant to the prior-aware upper matrix, then this tangent inequality, and maximize the additive term. The efficiency statement follows by exponentiating the difference between the incumbent lower bound and the optimum upper bound. $\square$

For a weighted-trace objective with $W \succeq 0$, the same matrix bound gives

$$\max_{S \in \mathcal{F}} \text{tr}(W J(S)) \leq \text{tr}(W J_0) + \max_{S \in \mathcal{F}} \sum_{e \in P_S} \frac{\text{tr}(W Q_e)}{1-\delta}.$$

This requires no logarithm or tangent reference; the partial-packet trace certificates use it. Dividing an exact feasible lower value by a positive upper value certifies relative performance; a zero upper value proves that every feasible value is zero.

Neither N nor $J_0 + (N - J_0)(1 - \delta)$ needs to be the information of a feasible schedule. For example, optimizing log determinant over a convex mixture of path matrices can propose N , and an exchange heuristic can propose S_0 ; the mixture supplies no feasible-experiment lower bound. The support maximum in (5.1) must range over all feasible paths, not only over paths discovered by an optimizer. Exact cardinality, mandatory/forbidden choices, and minimum gap are enforced by the count, mask, and cooldown graph of Section 3.7; a spacing constraint remains in the graph even if its required cooldown exceeds the information window.

The theorem permits real witnesses. For the exact rational checker, choose a rational symmetric reference N (or its rational inverse), for example by rounding a numerical proposal, and verify $N \succ 0$ by exact positive pivots or leading principal minors; a failed check rejects the witness. With rational model data, the information of S_0 , every Q_e , and N^{-1} can then be recomputed by rational

elimination. The weighted-trace checker likewise requires its prescribed weight W to be rational and verifies $W \succeq 0$. Because rational model data need not make the analytic locality bound rational, every score uses a certified rational enclosure $\delta \leq \hat{\delta} < 1$ in place of δ . This enlargement preserves the upper matrix bound because the increments are PSD. Rational envelopes or outward interval evaluation of the locality formulas give such enclosures; square-root endpoints can be verified by squaring. If the enclosure is not below one, refine it, increase the memory window and recompute the bound, or reject that certificate. No numerical reference, solver status, or reported gradient is used as a proof. Appendix D gives complete outward score and logarithm formulas. If a_e is replaced by $\lceil G_s a_e \rceil / G_s$ and every path has at most k nonzero selection arcs, the support increase is at most k/G_s (and is zero when $k = 0$). More general integer interval scores remain valid, with their actual excess explicitly charged. The rational certificate proves a statement about the encoded inputs. Exact arithmetic does not remove uncertainty in the physical covariance or the sensitivity construction; that requires additional assumptions.

Logdet support conditions for finite information hulls are classical [19], and correlated-design simplicial decomposition is already developed by Hainy et al. [18, Section 4.4]. The role of Theorem 5.1 is to attach an explicit uniform selected-covariance error bound to this established support machinery.

5.2 The virtual-noise comparator and exact continuous bounds

Here we index scalar candidate responses by $i = 1, \dots, n$; packet selection ties their coordinates. Let

$$D = \text{diag}(a_i) \succ 0, \quad B = R - D \succeq 0, \quad Z(z) = \text{diag}(z_i/a_i), \quad z \in [0, 1]^n.$$

The established covariance-split extension is

$$V(z) = (I + BZ(z))^{-1}F, \quad J_D(z) = J_0 + F^T Z(z)V(z), \quad \varphi_D(z) = \log \det J_D(z). \quad (5.2)$$

These formulas do not invert $Z(z)$ and therefore apply at zero weights. The matrix $I + BZ$ is invertible: Sylvester's determinant identity reduces its determinant to that of $I + B^{1/2}ZB^{1/2} \succ 0$.

Proposition 5.2 (Established virtual-noise identities). *For $T = \{i : z_i > 0\}$,*

$$J_D(z) = J_0 + F_T^T \left[R_{TT} + \text{diag} \left(a_i \frac{1 - z_i}{z_i} : i \in T \right) \right]^{-1} F_T. \quad (5.3)$$

The empty-support value is J_0 . The map J_D is matrix concave on the cube, φ_D is concave and differentiable there by extension to a neighborhood, and

$$\frac{\partial J_D}{\partial z_i} = a_i^{-1} V_i^T V_i, \quad g_i(z) := \frac{\partial \varphi_D}{\partial z_i} = a_i^{-1} V_i J_D(z)^{-1} V_i^T. \quad (5.4)$$

For binary $z = \mathbf{1}_S$, $J_D(z) = J(S)$, independently of D .

Proof. The nonzero rows of ZV solve $(B_{TT} + Z_{TT}^{-1})(ZV)_T = F_T$, proving (5.3). For $B \succ 0$,

$$Z(I + BZ)^{-1} = B^{-1} - B^{-1}(B^{-1} + Z)^{-1}B^{-1}$$

is matrix concave because matrix inversion is convex. Approximate $B \succeq 0$ by $B + \epsilon I$ and pass to the continuous resolvent limit; this proves concavity also on the split boundary. Monotone concave log determinant preserves concavity. Differentiating the resolvent gives $d\{Z(I + BZ)^{-1}\} = (I + BZ)^{-1}(dZ)(I + BZ)^{-1}$, which proves (5.4), including the cube boundary. Binary substitution into (5.3) proves exactness. $\square$

The scalar choice $D = aI$ is the Liu relaxation [34, Section II.B]. Its equality with the convex virtual-noise design of Pázman et al. [48], under the measure change $\xi_i = z_i/k$ for $k > 0$, was already proved by Hainy et al. [18, Proposition 3], including zero weights. Their modified variance uses $D = \tilde{\kappa} \text{diag}(R_{ii})$ with $\tilde{\kappa} \leq \lambda_{\min}(\text{diag}(R_{ii})^{-1/2} R \text{diag}(R_{ii})^{-1/2})$. This is a subfamily of the diagonal splits above; for constant $R_{ii} = v$, it is the scalar family $a = v\tilde{\kappa}$. The equivalence and the concavity are prior results, not contributions of this paper.

Let $\mathcal{Z}$ be a specified compact convex relaxation satisfying

$$\text{conv}\{\mathbf{1}_S : S \in \mathcal{F}\} \subseteq \mathcal{Z} \subseteq [0, 1]^n.$$

A rational query z in the cube gives

$$\max_{x \in \mathcal{Z}} \varphi_D(x) \leq \overline{\log \det J_D(z)} - g(z)^T z + \max_{x \in \mathcal{Z}} g(z)^T x. \quad (5.5)$$

Here the overline is a rigorous logarithm upper enclosure, and the last quantity must be computed or bounded globally. For $\mathcal{Z}_k = \{x \in [0, 1]^n : \sum_i x_i = k\}$ it is the sum of the k largest rational gradient coordinates. If $\mathcal{Z}$ is the true indicator hull with spacing constraints, linear optimization uses its feasible graph instead. At a verified feasible $z \in \mathcal{Z}$, $\log \det J_D(z)$ is a lower bound on the continuous optimum. An infeasible query still supplies a tangent upper bound but no such lower bound. Thus continuous optimization error and the discrete relaxation gap can be measured separately. A strict comparison between a continuous lower bound and a competing integer-objective upper bound does not require knowing the integer optimum.

Exact dense elimination suffices to check all these expressions. For scalar AR(1)-plus-nugget covariance, Appendix E derives covariance-filter and reverse-adjoint formulas that are linear in n , and exact tridiagonal split checks; these implement the same relaxation. Floating-point precision-space solves near unit latent correlation, and a first implementation based on subtracting RTS smoothed means, can produce invalid tangents; the appendix explains the two failure mechanisms. Only recomputed rational witnesses make (5.5) a rigorous bound.

5.3 A certificate against every diagonal split

A virtual-noise relaxation can appear weak merely because it was optimized incompletely. The following comparison rules out both incomplete continuous solution and tuning within the diagonal-split family. It does not compare complete integer branch-and-bound methods.

Fix $z^0 \in \mathcal{Z}$ and set $T = \{i : z_i^0 > 0\}$, $h_i = (1 - z_i^0)/z_i^0$ for $i \in T$. For every positive vector a , define

$$S(a) = R_{TT} + \text{diag}(h_i a_i : i \in T), \quad \phi(a) = \log \det \{J_0 + F_T^T S(a)^{-1} F_T\}.$$

This fixed-point value exists even if $R - \text{diag}(a)$ is not PSD. For $T = \emptyset$, use $J = J_0$, $g = 0$, and omit all empty matrix terms.

Proposition 5.3 (All-split lower witness). *Choose any $a^0 > 0$, put $V = S(a^0)^{-1} F_T$, $J = J_0 + F_T^T V$, and let*

$$g_i = -h_i V_i J^{-1} V_i^T \quad (i \in T), \quad g_i = 0 \quad (i \notin T).$$

For every $Y \succeq 0$ with $\text{diag}(Y) \geq -g$, the number

$$L_{\text{split}} = \phi(a^0) - g^T a^0 - \text{tr}(RY) \quad (5.6)$$

satisfies $\varphi_D(z^0) \geq L_{\text{split}}$ for every admissible $D = \text{diag}(a) \succ 0$, $R - D \succeq 0$. Consequently it lower-bounds every such continuous optimum and

$$\sup_{z \in \mathcal{Z}} \inf_D \varphi_D(z) \geq L_{\text{split}}. \quad (5.7)$$

Every relaxation consisting of $z \in \mathcal{Z}$ and any collection of globally valid affine upper supports $t \leq l_D(z)$ from these splits also has optimum at least L_{split} .

Proof. Put $C = F_T J_0^{-1} F_T^T \succeq 0$. The determinant lemma gives $\phi(a) = \log \det J_0 + \log \det(S(a) + C) - \log \det S(a)$. This is a classical convex function of S [26, 27]. For completeness, with $A = S^{-1} \succeq B_1 = (S+C)^{-1}$, its Hessian in a symmetric direction H is $\text{tr}(AH AH) - \text{tr}(B_1 H B_1 H) \geq 0$: $A \otimes A - B_1 \otimes B_1 = (A - B_1) \otimes A + B_1 \otimes (A - B_1) \succeq 0$. Affine substitution proves convexity in a , with gradient g as displayed. For $w = -g \geq 0$, admissibility and the dual conditions imply

$$w^T a \leq \text{tr}(DY) \leq \text{tr}(RY).$$

The convex lower tangent $\phi(a) \geq \phi(a^0) + g^T(a - a^0)$ therefore proves (5.6). The same point z^0 works for all D , which proves (5.7) without a minimax interchange. For any globally valid affine upper support, $l_D(z^0) \geq \varphi_D(z^0) \geq L_{\text{split}}$. Hence (z^0, L_{split}) remains feasible in their intersection. $\square$

The proof uses established logdet convexity and weak semidefinite duality; its useful output is a finite checkable witness against an entire family. For rational a^0 , all entries except the logarithm are rational. Any rational factor B_0 gives a valid dual matrix by construction:

$$Y = B_0 B_0^T + \text{diag} \left(\max\{0, -g_i - \sum_j (B_0)_{ij}^2\} \right). \quad (5.8)$$

A floating-point SDP solution can propose B_0 but is not assumed feasible. The lower logarithm enclosure is used in (5.6). The reference a^0 need not be admissible, and neither strong duality nor optimality of this reference or dual factor is required.

There is a simpler scalar witness. For fixed z^0 , increasing a increases $R_{TT} + a \text{diag}(h)$, so $\varphi_{aI}(z^0)$ is nonincreasing. Therefore

$$\bar{a} \geq \lambda_{\min}(R) \implies \max_{z \in \mathcal{Z}} \varphi_{aI}(z) \geq \varphi_{\bar{a}I}(z^0) \quad (0 < a \leq \lambda_{\min}(R)). \quad (5.9)$$

The right side is evaluated using (5.3), not by an inadmissible concavity assertion. Any nonzero rational v supplies $\bar{a} = v^T R v / (v^T v) \geq \lambda_{\min}(R)$; alternatively an exact nonpositive leading pivot of $R - \bar{a}I$ proves the same inequality. Appendix E gives the smaller tridiagonal witness for a stationary latent chain.

Example 5.4 (A persistent gap). Let $R = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $F = (1, -1)^T$, $J_0 = 1$, and $k = 1$. Every feasible schedule has information $3/2$. At $z^0 = (1/2, 1/2)$ and $a^0 = (1, 1)$, $J = 2$, $g = (-1/8, -1/8)$, and $Y = \frac{1}{8} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ has $\text{diag } Y = -g$ and $\text{tr}(RY) = -g^T a^0 = 1/4$. Thus $L_{\text{split}} = \log 2$, leaving a gap at least $\log(4/3)$ over the integer optimum for every diagonal split, including strict splits and pointwise split optimization. All affine upper cuts in the stated family retain this witness. Branching, additional restrictions excluding z^0 , or cuts outside this family can remove it. This elementary example is not a novelty claim for the relaxation or the convexity argument.

5.4 One shared schedule over finite scenarios

Let $s = 1, \dots, q$ index a declared finite set of known covariance and sensitivity models, all with information dimension p , positive definite priors, and the same nonempty feasible schedule family $\mathcal{F}$. Define

$$\begin{aligned} f_s(S) &= \log \det J_s(S), & b_s^* &= \max_{S \in \mathcal{F}} f_s(S), \\ g_c(S) &= \min_s (f_s(S) - c_s), & g_*(S) &= \min_s (f_s(S) - b_s^*), & E_*(S) &= \exp(g_*(S)/p). \end{aligned} \quad (5.10)$$

Raw maximin sets $c_s = 0$. Standardized D-efficiency compares each scenario with its own best attainable information. Under a scenario-dependent invertible congruence $J_s \mapsto T_s J_s T_s^T$, both f_s and b_s^* gain $2 \log |\det T_s|$, so g_* is invariant if priors transform too. The maximizing schedules of raw maximin are invariant under a common transformation, but need not be under unrelated transformations.

Efficiency normalization and its support-based optimization are established: Burclová and Pázman [12, Section 3.2 of the preprint] computes individual criterion optima before maximizing minimum efficiency by affine supports, and Maus et al. [37] studies maximin relative efficiency under uncertain AR(1) correlation. Robust kinetic design and scenario exchange also have direct precedents [16, 58]; the latter preprint uses independent-trial sampling information. Chowdhary et al. [13] develops budget-constrained robust correlated-noise sensor design through stochastic policy optimization. Our contribution is an explicit certificate for one common discrete schedule that rigorously propagates the uncertainty in its optimal standardizers.

Suppose the common path graph supplies $Q_{s,e} \succeq 0$ and $\delta_s < 1$ with $J_s(S) \preceq J_{0,s} + \sum_{e \in P_S} Q_{s,e} / (1 - \delta_s)$. Choose any $N_s \succ 0$ and exactly nonnegative λ_s with $\sum_s \lambda_s = 1$. Then

$$\max_{S \in \mathcal{F}} g_c(S) \leq \sum_s \lambda_s [\log \det N_s - p + \text{tr}(N_s^{-1} J_{0,s}) - c_s] + \max_{S \in \mathcal{F}} \sum_{e \in P_S} \sum_s \frac{\lambda_s \text{tr}(N_s^{-1} Q_{s,e})}{1 - \delta_s}. \quad (5.11)$$

Indeed, the minimum is at most a convex average, each scenario obeys Theorem 5.1, and one common path prices the combined scores. Separate scenario support maxima also give a valid bound but may be strictly weaker. The rational checker uses rational SPD references N_s and certified rational enclosures $\delta_s \leq \hat{\delta}_s < 1$ in all scenario scores, as above. It also uses rational simplex weights: round a numerical proposal to rationals, clip negative entries, and normalize exactly if positive mass remains; otherwise use a specified rational simplex point or reject. Approximately normalized or negative weights do not satisfy the proof.

Proposition 5.5 (Unknown standardizers). *Suppose verified bounds $\ell_s \leq b_s^* \leq U_s$ are available. For a common feasible S_0 , let*

$$L = \min_s [f_s(S_0) - U_s]. \quad (5.12)$$

In (5.11), replace c_s by ℓ_s and round every term outward to obtain an upper bound U' . Then

$$L \leq g_*(S_0) \leq \max_{S \in \mathcal{F}} g_*(S) \leq U := \min(U', 0), \quad E_*(S_0) \geq e^{L/p}, \quad \frac{E_*(S_0)}{\max_{S \in \mathcal{F}} E_*(S)} \geq e^{(L-U)/p}.$$

Proof. Pointwise, $\min_s (f_s(S) - U_s) \leq g_*(S) \leq \min_s (f_s(S) - \ell_s)$. The left inequality gives the incumbent lower bound; the right and (5.11) give the upper bound. Since $f_s(S) \leq b_s^*$, $g_*(S) \leq 0$. Exponentiation gives both efficiency conclusions. $\square$

The individual lower bounds ℓ_s can come from different schedules, but each must be feasible for the same family defining b_s^* . A reference optimum over a larger family gives an upper bound, not automatically a valid lower bound. The rational log interval is the certificate; floating-point exponentials are only display summaries. These statements certify exactly the listed scenarios and do not cover an intervening continuous parameter region.

Corollary 5.6 (Fixed-scenario approximation). *Under the fixed rational memory promises and feasibility conditions of Theorem 4.1, with q and p fixed, standardized D-efficiency has an FPTAS. This is a direct consequence of the spectral approximation set, not a runtime guarantee for the practical mixture solver.*

Proof. Take one graph with the largest required memory. Apply Theorem 4.3 to the block-diagonal additive matrix $\text{diag}(J_{L,1}(S), \dots, J_{L,q}(S))$ of fixed order qp . If every memory error is at most δ , every target schedule has a single returned representative satisfying, in every scenario,

$$J_s(\hat{S}) \succeq a J_s(S), \quad a = \frac{(1-\eta)(1-\delta)}{1+\delta}. \quad (5.13)$$

This follows by transferring true information to local information, using the common spectral cover, and transferring back. For the returned set $\mathcal{C}$, compute the positive rational numbers $d_s = \max_{S \in \mathcal{C}} \det J_s(S)$ and choose $\hat{S} \in \mathcal{C}$ maximizing $\min_s \det J_s(S)/d_s$. If $D_s^* = \max_{S \in \mathcal{F}} \det J_s(S)$, then $a^p D_s^* \leq d_s \leq D_s^*$. Consequently the efficiency $E_{\mathcal{C}}$ normalized by d_s satisfies $E_*(S) \leq E_{\mathcal{C}}(S) \leq E_*(S)/a$. A representative S' of a robust optimum has $E_*(S') \geq a \max_S E_*(S)$, whence

$$E_*(\hat{S}) \geq a E_{\mathcal{C}}(\hat{S}) \geq a E_{\mathcal{C}}(S') \geq a^2 \max_S E_*(S).$$

Choose $\eta = \epsilon/4$ and $\delta \leq \epsilon/8$, or full history with $\delta = 0$. Then $a \geq 1 - \eta - 2\delta \geq 1 - \epsilon/2$ and $a^2 \geq 1 - \epsilon$. The graph and output set have polynomial size under the stated fixed promises; exact true determinants and the displayed rational comparisons have polynomial bit complexity. No logarithmic or root comparison oracle is needed. The polynomial degree depends on qp . $\square$

With known positive rational determinant normalizers, maximizing the corresponding rational determinant ratios over $\mathcal{C}$ loses only the single factor a in determinant-root efficiency. Unknown normalizers cost a second factor in the same-set procedure above. Appendix F gives a sharp tie example and a common-mixture integrality gap.

5.5 Exact latent-separator representations

When a fixed physical covariance is sampled on finer grids, the per-step contraction can approach one and the sufficient calendar window can become large. An alternative keeps selected latent states as nuisance variables and conditions short blocks on them exactly, without a truncation parameter. For this construction let

$$Y = F\theta + X + v, \quad X \sim N(0, K), \quad v \sim N(0, rI), \quad K \succ 0, \quad r > 0, \quad (5.14)$$

where X is a scalar Gaussian Markov chain and X, v are independent. Unlike the singular-state locality theorem, the inverse formulas here require $K \succ 0$; independent or zero-latent cases can be evaluated directly. For an anchor set A , write $U = X_A$, $K_A = K_{AA}$, and

$$H_A = K_{:A} K_A^{-1}, \quad D_A = K - K_{:A} K_A^{-1} K_A + rI \succ 0. \quad (5.15)$$

With $A = \emptyset$, omit the anchor terms and put $D_A = R = K + rI$. Conditioning on anchors separates the intervening Markov intervals. Each anchor-time observation is assigned to exactly one neighboring block; its conditional latent residual is zero and its observation noise remains independent. Thus D_A is block diagonal over a disjoint partition $I_1, \dots, I_m$. For example, partition into consecutive blocks of at most b times and anchor every block's right endpoint except the final time. Each block then has at most two adjacent anchors.

For any schedule define the augmented information

$$M_A(S) = \text{diag}(J_0, K_A^{-1}) + [F_S, H_{A,S}]^T (D_A)_{SS}^{-1} [F_S, H_{A,S}]. \quad (5.16)$$

It is the fixed prior plus one PSD atom for each within-block pattern $S \cap I_j$. Both its full matrix and its nuisance principal block are SPD. Woodbury's identity gives exactly

$$\mathcal{S}_\theta(M_A(S)) = J_0 + F_S^T R_{SS}^{-1} F_S = J(S), \quad (5.17)$$

where $\mathcal{S}_\theta$ eliminates all nuisance coordinates. The anchors are latent nuisance variables with prior precision K_A^{-1} ; they are not additional observations available at no cost.

For a generic SPD matrix $M = \begin{pmatrix} A_0 & B_0 \\ B_0^T & C_0 \end{pmatrix}$ with parameter block of order p , put

$$J = A_0 - B_0 C_0^{-1} B_0^T, \quad G_* = -C_0^{-1} B_0^T, \quad E_* = [I_p; G_*], \quad f(M) = \log \det J.$$

Completing the square gives, for every nuisance matrix G ,

$$[I_p; G]^T M [I_p; G] = J + (G - G_*)^T C_0 (G - G_*) \succeq J. \quad (5.18)$$

This identity proves matrix concavity and monotonicity of the Schur map: its quadratic form in any parameter vector is the minimum over nuisance vectors of a jointly linear-in- M quadratic form. Thus f is concave. Differentiation at the minimizing G_* gives

$$\nabla f(M) = E_* J^{-1} E_*^T \succeq 0, \quad \text{rank } \nabla f(M) = p, \quad \text{tr}(\nabla f(M) M) = p. \quad (5.19)$$

The rank remains p although the nuisance dimension grows.

Theorem 5.7 (Separator support certificate). *For any $W \succ 0$ of order p and any $|A| \times p$ matrix G ,*

$$\begin{aligned} \max_{S \in \mathcal{F}} \log \det J(S) &\leq -\log \det W - p + \text{tr}(W J_0) + \text{tr}(W G^T K_A^{-1} G) \\ &\quad + \max_{S \in \mathcal{F}} \sum_{j=1}^m \text{tr} \left[W (F_{S_j} + H_{A, S_j} G)^T (D_A)_{S_j S_j}^{-1} (F_{S_j} + H_{A, S_j} G) \right], \end{aligned} \quad (5.20)$$

where $S_j = S \cap I_j$. Empty patterns contribute zero and empty anchor terms are omitted. If $\mathcal{F} = \{S : |S| = k\}$, the support term can be priced by enumerating at most 2^b patterns per block and an exact count dynamic program, using $O(m(k+1)2^b)$ transitions after pattern scores are available.

Proof. Apply (5.18) to (5.16), then use $\log \det X \leq -\log \det W - p + \text{tr}(WX)$. Block diagonality of D_A separates the pattern scores and the fixed anchor/prior term is added once. For block j , let $v_j(c)$ be the largest score among its patterns of size c (or $-\infty$ if none). Starting with $V_0(0) = 0$ and other states $-\infty$, use $V_j(t) = \max_c \{V_{j-1}(t-c) + v_j(c)\}$. Its final state $V_m(k)$ is the support maximum. Pattern enumeration plus these transitions satisfies the stated bound. $\square$

Local forbidden or mandatory observations can be enforced while enumerating patterns. Cross-block constraints such as minimum spacing require additional boundary states; expected-count constraints on independent block mixtures are not a substitute for the hull of complete feasible schedules. Any rationally rounded G is valid in (5.20); only $W \succ 0$ needs a definiteness check. Rational conditional covariances, integer interval pattern scores, and the logarithm bounds of Appendix D make this an exact certificate. At a mixture matrix M , choosing $W = J^{-1}$ and $G = G_*$ recovers its ordinary upper tangent. Arbitrary witnesses need not solve these stationarity equations.

Theorem 5.8 (Nested-anchor hierarchy). *For the same model and feasible family, define the mathematical hull optimum*

$$U_A = \max_{\alpha \in \Delta(\mathcal{F})} \log \det \mathcal{S}_\theta \left(\sum_{S \in \mathcal{F}} \alpha_S M_A(S) \right).$$

For nested anchor sets $A \subseteq B$,

$$\max_{S \in \mathcal{F}} \log \det J(S) \leq U_A \leq U_B. \quad (5.21)$$

Thus removing anchors and merging conditional blocks tightens these hull bounds. The statement compares complete hull optima, not unfinished numerical upper witnesses or nonnested partitions.

Proof. Order M_B by $(\theta, X_A, X_{B \setminus A})$ and let $\mathcal{S}_{\theta, A}$ eliminate only $X_{B \setminus A}$. The exact identity

$$M_A(S) = \mathcal{S}_{\theta, A}(M_B(S)) \quad (5.22)$$

is proved with all Gaussian prior cross terms in Appendix G. It is not an identification of the smaller matrix with a principal block. Concavity gives

$$\sum_S \alpha_S M_A(S) \preceq \mathcal{S}_{\theta, A} \left(\sum_S \alpha_S M_B(S) \right).$$

Apply monotonicity of the remaining parameter Schur map and of log determinant. Successive Schur elimination equals joint elimination, so this bounds the objective for A by that for B at the same α . Maximize over the common simplex. A point mass on any feasible schedule and (5.17) prove the first inequality. $\square$

The empty-anchor level is the hull of the true information matrices. It can still exceed the best single schedule: with $R = I_2$, $F = 2I_2$, $J_0 = I_2$, and $k = 1$, the two information matrices are $\text{diag}(5, 1)$ and $\text{diag}(1, 5)$, whose equal mixture has determinant 9 whereas each feasible determinant is 5. This gap does not arise from covariance truncation.

When every block is a singleton, mixture matrices depend only on the marginal selection weights z . Their parameter Schur complement equals (5.2) for the diagonal residual split D_A ; a direct factor proof is in Appendix G. With all times anchored, $D_A = rI$, recovering the physical-nugget Liu split. Under the right-endpoint convention and $b = 1$, the final time is unanchored: for stationary variance P and $n \geq 2$ the actual split is $\text{diag}(r, \dots, r, r + P(1 - \rho^2))$. For $n = 1$ it is $[r + P]$. This distinction matters when choosing a baseline. Improving on the physical split does not prove dominance over every admissible virtual-noise split.

The statistical and optimization ingredients have direct predecessors. Focused Bayesian design already uses marginalized nuisance precision [1]; Levine and How [31, Section 7] uses latent augmentation to bound focused sensor information through an augmented-target criterion. General PSD experiment atoms and their design hulls are established [17, 19]. In particular, set $E = [I_p, 0]^T$ in the subsystem criterion of Sagnol and Harman [51, Theorem 4.3 and Corollaries 4.4–4.5]: $(E^T M^{-1} E)^{-1} = \mathcal{S}_\theta(M)$. Treating separator patterns and the fixed prior as atoms embeds this exact integer design in their general conic framework. Our specialized result is the comparison across the changing latent representations in (5.22)–(5.21), together with short-block support certificates for the original correlated experiment. Grouping discrete alternatives before convexification has a separate disjunctive-programming precedent [4, 46]. That principle explains a strength–enumeration tradeoff, but does not by itself compare statistical representations whose latent variables change; the cross-representation Schur identity supplies that step here. To the best of our knowledge, this particular nested Markov-anchor hierarchy is not stated in the inspected predecessors. This qualified claim does not assert a new Schur complement, generalized D criterion, generic conic formulation, or design-mixture algorithm. Its practical cost includes pattern construction and the growing nuisance system, in addition to pricing.

6 Computational investigation

The strongest empirical result is a certified separation between bounds for *the same discrete problem*. On ten rational input instances, a feasible point in every admissible scalar virtual-noise relaxation has a value larger than a finite-history upper bound on the integer optimum. One kinetic instance also separates every admissible diagonal split. These comparisons do not depend on convergence of a competing numerical solver. Other tests show the limits of the approach: full-packet examples favor the tested dense bound, and fine physical grids defeat the present calendar implementation. All observation-error models in the temporal experiments are stipulated; none is fitted to experimental residuals.

6.1 Evidence and reproducibility

We distinguish four kinds of evidence: exact rational identities and rankings; rational global certificates with outward logarithm bounds; floating-point solver results; and numerical checks of sensitivity generation. A certificate treats each stored JSON decimal as its exact rational value. It therefore certifies the supplied design instance, without asserting that its decimal sensitivities equal exact real derivatives of an exponential mean model. Reported log gaps are $U - \ell$; relative information gaps are $(U - \ell)/\ell$ when $\ell > 0$. A log gap g gives determinant-root efficiency at least $\exp(-g/p)$. Decimal bounds in the tables are rounded outward when presented as certified inequalities; ordinary numerical comparisons are labeled as such.

The portable supplement contains model arrays, feasible selections, tangent and dual witnesses, exact checkers, numerical producers, and an explicit file-hash manifest. Its validation command uses no commercial solver and recomputes the mathematical certificate quantities rather than trusting saved solver objectives. It also independently checks all source-model rankings, 864 temporal sensitivity entries against an augmented mass-balance ODE, and 16 sensitivity rows against 70-digit matrix exponentials. The largest observed absolute sensitivity differences in the latter two checks were $1.65 \cdot 10^{-13}$ and $1.48 \cdot 10^{-16}$, respectively. These are numerical consistency tests, not validated ODE enclosures. Tiny exact tests exercise the locality, approximation-set, Schur, robust, and interval-score arguments. The spectral-set algorithms of Section 4 are theoretical schemes, not the practical solvers used below.

Historical runs were made on a shared machine with Python 3.13.11, NumPy 2.5.3, SciPy 1.18.1, SymPy 1.14.0 and, for the mixed-integer outer-approximation comparisons, Gurobi 13.0.3. BLAS and Gurobi used one thread per invocation. Several certificate batches ran concurrently, so individual timings are reported as observed, not as controlled runtime comparisons. The new exhaustive studies use the same Python and scientific-library versions, with no mixed-integer solver. We report setup, incumbent generation, numerical witness search, and exact certification times separately; a sum of historical stage times is an accounted cost, not a newly timed pipeline. The archive retains time limits, iteration caps, unsuccessful status messages, input hashes, and earlier results superseded by stronger certificates.

6.2 Reanalysis of the inspected public kinetics implementation

The source comparison of Section 2.5 uses the public `measurement-opt` revision identified there and the supplied 24×4 sensitivity file `kinetics_Q_drop0.csv`. The eight times are 7.5, 15, $\dots$, 60, using the source’s dropped-zero sensitivity labels. Its internal spacing map is shifted by -7.5 ; this leaves every pairwise time difference and feasible schedule unchanged (see the supplement’s source-data provenance note). Three species may each use either a static-cost measurement (SCM), acquiring

all eight times for cost 2000, or a dynamic-cost measurement (DCM), with installation cost 200 and cost 400 per sample. The two modalities cannot be combined on one species. There is at most one DCM at a time, no DCM at adjacent times, and at most four DCM samples. A DCM installation without any sample is omitted as a redundant description of the same acquired data. The prior is $10^{-4}I_4$.

At each time the six modality/species responses have covariance

$$R_6 = \begin{pmatrix} B & B/2 \\ B/2 & B \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1/10 & 1/10 \\ 1/10 & 4 & 1/2 \\ 1/10 & 1/2 & 8 \end{pmatrix};$$

different times are independent. The same species sensitivity is used for both modalities, as in the inspected code. Thus this example concerns *within-time channel selection*; it is separate from the complete-packet temporal locality experiments.

Two independent enumerations produce the same 2347 nonempty acquisition schedules. We evaluate budgets 1000, 1400, $\dots$, 5000 for trace information, D-optimality, and conventional A-optimality. Exact rational arithmetic gives all 66 formula-criterion-budget optima, each unique, under selected-covariance information and full-inverse gating. A separate implementation constructs the full 48×48 covariance in a different row order, directly inverts each selected covariance, and verifies all 14,082 objective values. Comparing rational determinants suffices for D-optimality; conventional A-optimality is ranked by the rational trace of the inverse. No floating tie decision remains.

Exactly one of the 33 criterion-budget choices changes: trace information at budget 3000. The correct choice is SCM for species B and DCM for species C at times 45 and 60. Gating instead chooses SCM for B and DCM for A at times 7.5 and 22.5. Their true information traces are approximately 83.03029588 and 80.44572850, respectively, giving a certified relative loss of 3.112800396788% to the displayed precision. The gated traces are approximately 108.08074141 and 108.30708633, which explains the reversal. The D-optimal and conventional A-optimal choices agree between the two formulas at every tested budget; agreement of the optimizers does not make their information values equal. The supplement retains every objective and runner-up margin. This is a reanalysis of the pinned code’s feasible family and supplied data. It does not reconstruct uninspected stored published solutions or validate how the sensitivity CSV was originally generated.

6.3 Scalar temporal designs and relaxation separation

The six synthetic cases use $p = 3$, $k = n/3$, $J_0 = 0.1I_3$, $R_{ij} = (2/5)^{|i-j|} + \mathbf{1}_{i=j}$, and $n \in \{48, 96\}$. Each sensitivity array has independent standard-normal entries generated by NumPy’s default generator with seed 0, 1 or 2. The 48-row array is the prefix of the 96-row array with the same seed; the six cases are not six independent replications. The saved decimals specify the certified instances. The numerical finite-history method optimizes a mixture of complete feasible paths and evaluates its proposed integer selections under the true covariance. Its $L = 8$ runs close the surrogate numerical gap; the rational certificates in Table 2 still leave a positive gap to the true discrete optimum.

The kinetic mean is the consecutive first-order model

$$B(t) = \frac{A_0 k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t}), \quad A_0 = 1, \quad (6.1)$$

with parameters $(\log A_0, \log k_1, \log k_2)$ and prior $0.01I_3$. The early and late regimes have $(k_1, k_2) = (0.7, 0.2)$ and $(0.18, 0.045)$. Times are $12j/n$, $j = 1, \dots, n$, $k = n/3$, and

$$R_{ij} = \frac{1}{800} ((2/5)^{|i-j|} + \mathbf{1}_{i=j}). \quad (6.2)$$

The total pointwise noise standard deviation is 0.05 in normalized concentration units; the observed first-lag correlation is 0.2. Holding the per-grid correlation fixed while changing n changes the physical correlation scale, so these four cases are separate stipulated instances. The analytic derivatives are given in Appendix H and checked in the supplement. The rate-swap ambiguity of (2.6) remains; these are local-information designs, not global parameter-recovery guarantees.

Table 2: Certified scalar-temporal log bounds. g_{mem} is an upward rounded finite-history true-design gap. g_{cont} bounds the remaining fixed-split continuous optimization gap. $s_{\text{all}} > 0$ is a downward rounded lower bound on every scalar-split relaxation optimum minus the memory upper bound. The fixed split is verified rationally near $0.99\lambda_{\min}(R)$.

Family	n/case	$g_{\text{mem}} \leq$	$g_{\text{cont}} \leq$	$s_{\text{all}} \geq$	Memory cert. s
Synthetic	48/0	.00270073	$1.12 \cdot 10^{-7}$	.04721614	1.000
	48/1	.00187230	$6.48 \cdot 10^{-8}$	.03593564	.872
	48/2	.00387283	$8.39 \cdot 10^{-8}$	.04639027	.989
	96/0	.00183482	$1.62 \cdot 10^{-7}$	.03913455	2.336
	96/1	.00186234	$6.63 \cdot 10^{-8}$	.02276202	2.444
	96/2	.00178161	$3.54 \cdot 10^{-7}$	.02635127	2.446
Kinetics	48/early	.00190758	$5.19 \cdot 10^{-7}$	.10923746	.952
	48/late	.00182972	$6.27 \cdot 10^{-7}$	.08378101	.958
	96/early	.00193602	$6.64 \cdot 10^{-7}$	.10436604	2.299
	96/late	.00194271	$1.30 \cdot 10^{-6}$	.08771349	2.314

Within each row, all bounds refer to the same exact input model. In each case, a rational feasible fractional point and a rational upper witness for $\lambda_{\min}(R)$ establish the all-scalar-split lower bound by (5.9). The ten positive separations do not assume that the incumbents are integer optimal. All four kinetic memory gaps are below 0.001943, implying at least 99.935% determinant-root efficiency. Numerical memory solve times were 0.62–2.64 seconds for the synthetic cases and 0.63–3.23 seconds for the kinetics cases; the table gives the additional certificate times. Exact fixed-split continuous certificates, including their new numerical point proposals and incumbent checks, took 0.056–0.249 seconds and 0.191–1.203 seconds, respectively. The all-scalar-split extensions add separate value-only verification. None of these are complete integer solver timings.

The initial dense outer-approximation and branching runs received the best same-case memory incumbent as a disclosed start. Their five-second synthetic and ten-second kinetic limits left numerical gaps of approximately 0.018–0.035 and 0.118–0.240. These gaps combine an incomplete continuous outer approximation with a fractional relaxation gap. We retain the runs in the archive, but the table uses the tightly certified continuous comparison to separate the two effects. The structured covariance filtering/adjoint oracle of Appendix E repairs the earlier near-unit-correlation precision-coordinate instability; the performance of the failed numerical route is not evidence of a stronger relaxation.

For the 48-candidate early kinetic instance, the all-diagonal witness of Proposition 5.3 gives

$$\inf_{\text{admissible diagonal splits}} \phi_{z^0}(a) > 15.043116838, \quad \max_{S \in \mathcal{F}} \log \det J(S) < 14.950572423, \quad (6.3)$$

and its exact rational difference exceeds 0.0925444163328848. As in Example 5.4, the common fractional point survives pointwise split optimization and every globally valid affine upper cut from this family; branching or other cuts can exclude it. The numerical split proposal failed its SLSQP subproblem stopping test and had a small negative split eigenvalue; the proposed dual also had a small negative eigenvalue. Neither is trusted: rational factor repair and exact point recomputation

prove the displayed lower bound. The archived proposal and certificate times are 1.56 and 0.226 seconds. In contrast, the analogous synthetic seed-0 fixed-point probe gives a numerical lower estimate 7.68927318, below the memory upper bound 7.69935038. That witness fails to separate all diagonal splits; it does not establish the optimum of the diagonal-split family.

6.4 Spacing, full packets, and partial observations

A spacing-constrained kinetic case uses $n = 96$, $k = 32$, $g = 2$, $L = 13$ and the exact decimal correlation 0.6324555320336759, chosen near $\sqrt{0.4}$. Its covariance approximately preserves the earlier 48-point physical decay, but its observation budget is doubled, so it is not the fixed-budget grid study below. The compact producer uses 610 feasible masks instead of 8192 unrestricted masks and takes 3.238 seconds. Direct rational arc evaluation certifies gap < 0.01155765 in 36.609 seconds. The integer-interval mode increases the bound by exactly 10^{-8} and takes 1.087 seconds, an observed arithmetic tradeoff for this instance and reference, not a generic speedup factor. Across its 31,900 reachable arcs, 418 upper scores increase by one integer unit, none decrease, and the maximum path price increases by one unit. Recomputing with the refined stationary pair bound reduces δ from approximately .00379560 to .00318692 and gives a new certificate gap < 0.00972559 in 1.088 seconds, with the same incumbent. Reference and coefficient grids are 10^8 and 10^{12} ; the sensitivity and score grids are 10^{18} and 10^8 . A nonpositive rounded innovation floor causes rejection.

The complete-packet examples acquire all $d = 4$ or $d = 16$ spectral channels at a selected time. They have $n = 24$, $k = 8$, $L = 6$, one unknown parameter $\log k_1$, and scalar prior .01. Their stylized Beer-Lambert mean combines the three concentrations of (6.1) with fixed Gaussian channel profiles specified in Appendix H. The latent and observation-noise covariances are both I_d , and the transition is $A = (1/5)(C + \text{diag}(1, -1, \dots))$, where C is the cyclic shift. Thus $\|A\| \leq 2/5$, while $AA^T \neq A^T A$ in the tested dimensions. Increasing d adds observations at the same per-channel noise level, so information values across d do not have a common normalization.

All three tested methods—finite-history scalar-information pricing, greedy with single exchanges, and dense continuous rounding—select the first eight times in both examples. The archived numerical memory relative gaps are .70469% and .72802%, whereas the dense tangent gaps are tighter, .24301% and .20574%. Memory pricing takes about .022 and .026 seconds; greedy/exchange takes .009 and .032 seconds; dense optimization takes .006 and .061 seconds, excluding generation of its supplied common incumbent. On these examples, the floating-point results show no advantage for memory pricing in either selected designs or upper bounds.

The complete-block far-pair argument in Theorem 3.4 reduces the dimension-free $L = 6$ constant from .007274322 to .006754203 (upward displays) for the same transition/noise promises. This compares theoretical constants; it does not relabel earlier certificates. We also perform a new exact experiment with $n = 8$, $d = 3$, $k = 3$, $L = 3$, the same transition construction, prior .01, and rational sensitivity $F_{t,j} = ((7t + 3j) \bmod 13 - 6)/10$, using zero-based t, j . All 56 schedules and their local regressions are enumerated exactly. The old and sharp-far errors are .111957334 and .103863638 (upward displays), and the same local-optimal schedule has true information exceeding .90391527. The respective true-information upper bounds are below 1.01644959 and 1.00735957; relative gaps decrease from 12.4497% to 11.4441% (upward rounding). The direct selected-inverse checks verify every sandwich and the exhaustive optimum lies below both bounds. The single fresh run takes .011 seconds for model construction and .095 seconds for exact enumeration and certification. The constant improvement is useful but modest; we do not claim that the uniform bound is sharp for these sensitivities.

For partial observation, two independent latent modes have transition $\text{diag}(2/5, 1/5)$, stationary covariance $I_2/800$, observation row $(3/5, 4/5)$, and independent measurement variance $1/800$.

Consequently

$$R_{ij} = \frac{1}{800} \left\{ \frac{9}{25}(2/5)^{|i-j|} + \frac{16}{25}(1/5)^{|i-j|} + \mathbf{1}_{i=j} \right\}. \quad (6.4)$$

This is not a single-mode scalar covariance. Using the four kinetic sensitivity arrays above, $W = I_3$, $J_0 = .01I_3$, and $L = 8$, one exact integer price per case gives relative trace-information gaps bounded by .059924%, .059732%, .060361%, and .059925%, in order 48/early, 48/late, 96/early, 96/late. The normalized error bound uses $\gamma = 2/5$, $s = 1$, $\kappa = 1/2$, and the rational upper enclosure $\sqrt{1/2} \leq 707106781187/10^{12}$; the resulting δ is below .000583886. Certificate times are .617, .777, 1.314 and 1.262 seconds, additional to numerical design generation. The numerical memory schedules improve on completed greedy/exchange search in all four cases. The dense numerical comparator is faster, .046–.176 seconds versus .224–.685 for the $L = 8$ numerical price, but its matched-incumbent relative gaps are .9438–1.8741%. Its 96/late solve reaches the iteration cap; all reported dense upper bounds come from evaluated tangents, not a success flag. The normalized $L = 8$ certificates supersede the two preliminary $n = 48$, $L = 6$ prices and the weaker original partial bound, which remain in the archive.

6.5 One schedule for three kinetic scenarios

The finite scenario set consists of the early and late regimes and a central regime $(k_1, k_2) = (.4, .1)$, with $A_0 = 1$ in every case. Covariance, log coordinates, prior, cardinality and time grids are those of (6.2). All scenarios share one selected schedule. We propagate exact bounds on the unknown individual scenario optima as in Proposition 5.5; no feasible nominal reference is treated as an exact optimum.

Table 3: Exact standardized robust log intervals. The 96-point polished row uses a new feasible schedule with the original shared support witness. Certificate time includes all three individual-reference certificates and excludes numerical reference/design generation and polishing.

n	Incumbent	ℓ	U	$U - \ell$	Cert. s
48	Greedy/exchange	-.091701377259	-.083176142020	.008525235239	2.97
96	Original hull	-.091162249685	-.074519785108	.016642464577	8.25
96	Hull plus exchanges	-.086007784227	-.074519785108	.011487999119	8.18

The polished 96-point schedule guarantees worst-scenario determinant-root efficiency exceeding 97.1737%, and at least 99.6177% of the unknown optimal standardized robust efficiency. The 48-point guarantee relative to its unknown robust optimum exceeds 99.7162%. These guarantees apply only to the three specified scenarios, not to a continuous region of rate constants.

The numerical comparator study uses the same fixed offsets for all methods. The original hull scores at $n = 48, 96$ are approximately -.0942782 and -.0892262; completed single-exchange polishing improves them to -.0897938 and -.0840718. Independently rounded dense visits start at -2.3402441 and -2.6497962, improving after their own exchange searches to -.1173976 and -.0896783. The dense branch does not receive a hull schedule as its polishing start. The original winners are mixed: at 48 points a separate greedy/exchange run improves the hull proposal; at 96 points the hull proposal is better than that greedy run. The final reported local-optimum status refers to a checked neighborhood of the returned schedule, not the preceding iterate.

The dense continuous fixed-offset upper bounds are approximately .03204293 and .03506056, with continuous residual gaps below $8 \cdot 10^{-6}$. Their positive values reflect fractional information; the unknown-optimum standardized discrete objective is always at most zero. The individual reference

bounds already improve the raw dense discrete upper bounds, but the negative shared-price upper bounds improve those reference-only caps further. Scenario weights are normalized and the combined support is recomputed; a floating LP objective is not accepted as the support price.

Individual numerical reference generation costs 2.076 and 9.980 seconds. Including these references, the original shared solver and its polishing cost 3.311 and 15.507 seconds; the dense solve and its independent rounding/polishing cost 2.732 and 20.075 seconds. Exact certification is additional. These accounted totals are sums of the original single-run times. Supplying the best common incumbent to the dense *bound* comparison incurs its separate generation cost and does not attribute that selection to dense rounding. Earlier exact comparisons against the saved central nominal schedules guarantee at least 3.33% and .80% better worst-scenario efficiency before the 96-point polishing; they do not compare against every possible nominal-design algorithm. Robust maximin and scenario-exchange strategies have established predecessors (Section 5.4); the evidence here concerns the selected-covariance certificates and their uncertainty-normalized guarantees.

6.6 Fixed physical grids and latent separators

A fixed physical process reveals a limitation concealed by fixed per-step correlation. Use the early kinetic mean, prior $.01I_3$, horizon 12, exactly 16 observations on every grid, and

$$R(t, u) = \frac{1}{800}e^{-\lambda|t-u|} + \frac{1}{800}\mathbf{1}_{t=u}, \quad \lambda = -16 \log(4/5). \quad (6.5)$$

For $n = 48, 96, 192$, the exact adjacent latent correlations are $256/625$, $16/25$, and $4/5$. Covariances are nested exactly and saved sensitivity rows agree at shared times. A shared greedy/exchange incumbent is charged to each pipeline; setup plus heuristic costs are .170, 1.193 and 3.027 seconds. Every alternative then receives the remainder of a nominal 30-second pipeline allowance. Cooperative checks permit small overruns. The 256-MiB workspace estimate is not a process resident-memory bound.

Table 4: Fixed physical-grid numerical outcomes; none of these rows is an exact certificate. Refined transfers use completed saved surrogate bounds and leave the original solve records unchanged.

n	Method	Pipeline s	True log gap	Outcome
48	Memory $L = 8$, refined	.772	.005971	Target .01 reached
48	Dense OA/branching	30.048	.161219	Time limit
96	Memory $L = 14$	30.002	2.368082	No completed price
96	Dense OA/branching	30.028	.415473	Time limit
96	Memory $L = 12$, refined	28.047	.037097	Separate diagnostic
192	Memory $L = 13$	30.002	2.688686	No completed price
192	Dense OA/branching	30.024	.672499	Time limit

The finer primary memory runs finish construction and their initial path but complete no pricing step, so their upper bounds are full-selection bounds. They are first-pricing timeouts, not proved construction failures. The 96-point $L = 12$ run is a separately budgeted diagnostic, not a replacement for the failed $L = 14$ run. Its surrogate hull closes, but even the refined transfer misses .01. Sufficient refined one-sided windows for a .01 three-parameter transfer component are 7, 15 and 35; the latter two exceed the workspace estimate cap. These windows suffice for a conservative bound; they are not lower bounds on necessary memory or a proof that exponential cost is unavoidable. The dense prototype also leaves incomplete root outer approximations; the table ranks neither all dense methods nor all possible finite-history implementations.

Latent separators avoid a finite-history error term. Table 5 reports the rational block-pattern certificate with consecutive blocks of at most b observations and an anchor at the last point of every block except the last. Its lower bounds use the separator incumbent at $n = 48$ and the better shared heuristic incumbent at $n = 96, 192$. The certificate verifies arbitrary rational nuisance witnesses and prices all count-feasible block patterns, so its upper bounds do not depend on convergence of the numerical mixture search.

Table 5: Exact separator log bounds on the fixed physical model. Decimal endpoints are the stored terminating rational bounds. Generation and certification are separate historical measurements.

n	b	ℓ	U	Gap	Generate s	Certify s
48	8	14.925507606504	15.005243683880	.079736077376	.090	.047
96	12	14.945662679078	15.051419395187	.105756716109	.655	.872
96	16	14.945662679078	15.025970711361	.080308032283	6.956	16.147
192	12	14.953046919008	15.139311234872	.186264315864	1.247	1.238
192	16	14.953046919008	15.110331327215	.157284408207	14.409	20.587

Adding charged shared work gives accounted totals 2.720 and 24.295 seconds for the two 96-point rows, and 5.511 and 38.023 seconds for the 192-point rows. In particular, the 192-point $b = 16$ generation fits 30 seconds while generation plus certification exceeds it. All separator rows miss the .01 target. The 192-point certificates improve the recorded dense numerical upper bound, but at 96 points the refined calendar diagnostic is tighter. Block sizes 12 and 16 define nonnested anchor sets here, so their comparison is empirical; Theorem 5.8 does not order arbitrary block sizes. Enumerating all configurations costs more than any preselected row.

6.7 A matched exact study of nested anchor sets

To test the hierarchy itself, we construct a new rational instance with $n = 8$, $k = 3$, $p = 2$, $K_{ij} = (3/5)^{|i-j|}$, $R = K + I$, $J_0 = \text{diag}(1/10, 1/5)$ and $F_t = (1, ((3t + 2) \bmod 7 - 3)/3)$, $t = 0, \dots, 7$. Every relaxation uses the *same 56 complete size-three schedules* and the same true optimum incumbent $S = \{1, 4, 6\}$; its log determinant lies in $[\cdot 77290072518411, \cdot 77290072518413]$. We enumerate each augmented information atom, optimize a mixture numerically, normalize rational mixture weights, and recompute the Schur information and its support price over all 56 atoms in exact arithmetic. Thus the endpoints in Table 6 enclose the mathematical mixture optimum. All 224 individual-schedule Schur identities are checked independently using joint-covariance congruences. The lower endpoint is a feasible *mixture* value, not a feasible integer objective.

Table 6: Fresh matched nested-anchor mixture intervals, rounded outward. Times are one observed run in seconds. “Setup” constructs exact atoms; “search” includes all numerical gradient pricing; “certify” includes exact support over the complete feasible family. Common model/optimum setup adds .028 seconds to each alternative. Indices are zero-based.

Anchor set	Mixture lower	Mixture upper	Setup	Search	Certify
$\emptyset$	.7747129913	.7747130145	.019	.010	.014
$\{3\}$	.7837750653	.7837754218	.030	.008	.017
$\{1, 3, 5\}$	.8352816893	.8352816952	.040	.006	.019
$\{0, \dots, 7\}$	.9008412717	.9008414860	.059	.007	.037

The disjoint exact intervals establish strict tightening as anchors are removed. Even the no-anchor information hull retains an integrality gap above the common discrete optimum. The

same-instance dense scalar split, chosen and verified near $0.99\lambda_{\min}(R)$, has continuous upper bound below .839905825 and residual continuous gap below $1.14 \cdot 10^{-7}$. It is tighter than the all-anchor hierarchy member and weaker than the three-anchor member; there is no ordering of every separator representation against every scalar split. Complete-history information at $L = 7$ reproduces the no-anchor hull. At $L = 4$ the conservative transferred hull upper bound is 2.212989488, and at $L = 2$ the error bound exceeds one, so no finite-history transfer is issued. All numerical proposal searches in this study terminate successfully; exact witness checks, not that status, supply the reported bounds.

This small experiment isolates the mathematical hierarchy on a common feasible family. The larger fixed-physical study shows the practical cost: fewer anchors require larger pattern enumerations, while finer anchor sets increase nuisance dimension and can weaken the bound. Together, the evidence supports choosing a representation for the certificate it attains; a closed surrogate gap or a fast numerical solve does not by itself give a useful guarantee for the original discrete design.

7 Discussion and conclusion

7.1 What improves a usable design certificate

What can be certified depends on keeping the statistical information model separate from the optimization relaxation. The public-source enumeration shows that full-inverse gating can change the selected experiment even in a small, exactly enumerable model, and that in this model the effect depends on the criterion: D-optimal and conventional A-optimal choices agree across all eleven tested budgets, while one trace-information choice changes. Because gating never understates information (Proposition 2.1), a gated global upper bound can still be reused once the incumbent is evaluated under the selected covariance, and the condition-dependent comparison bounds the possible efficiency loss.

For temporal designs, finite histories are useful when their conditional structure admits a tractable price over the full feasible family and their uniform error is small enough for the desired certificate. The all-split comparisons are stronger evidence of relaxation strength than a timed solver contest: their common fractional witnesses prove that better continuous optimization or scalar split selection cannot remove the measured separations. The kinetic all-diagonal witness extends this conclusion to the diagonal family and its specified affine upper supports; it says nothing about branching or other valid cuts. Conversely, in the complete-packet examples the tested dense bound is tighter and all methods select the same design. No universal dominance is demonstrated.

The results suggest allocating work according to the part of the bound that remains loose. A large continuous support residual calls for a better reference or further numerical optimization. A large locality correction calls for a stronger model-specific bound, a longer history, or a different representation. A remaining mixture integrality gap requires restrictions that distinguish one feasible experiment from a fractional mixture. More accurate arithmetic can reduce enclosure loss, as the spacing experiment illustrates, but cannot remove a statistical truncation bound or a mixture gap. Because the parts are separate, inexpensive heuristics can supply incumbents while independently chosen witnesses certify them.

Latent separators expose a second tradeoff. For nested anchors and the same complete feasible family, removing anchors improves the mathematical hull bound; the exact eight-time study verifies strict improvements for nested sets, not for unrelated partitions or unfinished solver values. Fewer anchors, however, require larger conditional-block patterns, and more anchors increase the nuisance dimension. The larger fixed-physical experiments show these costs: the best reported 192-point separator certificate still misses the .01 log-gap target and exceeds the nominal 30-second allowance

once certification is included. The hierarchy orders the bounds; it does not identify a uniformly efficient representation.

7.2 Scope of the statistical and complexity conclusions

Fixed covariance and nominal sensitivities make the information objective precise but limit its interpretation. The kinetic examples are local designs for stipulated error models. Independent balance-equation and high-precision checks support their sensitivity implementation, and the exact certificates apply to the stored rational arrays; neither validates a physical noise law.

Finite-scenario standardization answers a different question from raw maximin information. It measures each scenario against its own attainable optimum, and its certificate must therefore include uncertainty in those optima. The shared support bound prices one design over all scenarios and improves on separate scenario caps in the reported tests.

The approximation theorems identify tractable promised classes; they do not predict the runtime of the numerical prototypes. Weighted trace permits growing packet and parameter dimensions because its local support is linear. Nonlinear criteria use a fixed-dimensional relative PSD cover with large polynomial exponents. The correlated temporal schemes require complete packets and the stated fixed covariance promises, while the abstract additive cover requires fixed information dimension and an explicit rational feasibility representation. This restriction is consistent with the hardness result of Bansal and Xu [5] for growing information dimension (Section 1.1). A finer grid for one physical process makes the per-step correlation approach one, and the measured calendar-price failures are consistent with this loss of uniform promises. The conservative sufficient windows do not prove a lower bound for every possible algorithm. The represented-matroid result concerns additive information atoms and gives no automatic way to impose arbitrary matroid constraints on correlated histories.

7.3 Conclusion

The paper gives a route from a specified correlated experiment to a checkable global design bound. Explicit, schedule-uniform errors of local conditionals support finite-state certificates and approximation guarantees; normalization on exact ranges extends the approximation to several matrix criteria; and Schur elimination gives a nested alternative when calendar histories are costly. The computations establish strict separations from virtual-noise relaxation families and useful finite-scenario guarantees, and also report cases with weaker bounds or unfavorable cost. Every guarantee consists of a feasible experiment and a verified upper witness for the same selected-covariance objective, so it can be checked independently of the numerical method that proposed the design.

Data and code availability

The standalone computational supplement contains the model arrays, selected schedules, rational witnesses, exact checkers, numerical producers, a dependency lock and file-hash manifests. Its README maps each reported study and table to the saved artifacts and gives separate commands for certificate validation and numerical reproduction. Full validation replays 46 principal saved certificates, independently checks every public-source schedule, checks the kinetic sensitivities and verifies the fresh exhaustive studies. Once dependencies are installed, validation needs no network access or commercial solver. Optional reproduction of the historical mixed-integer proposals uses Gurobi under its license. The public sensitivity CSV keeps its immutable source reference [15] and hash, with no new licensing assertion about that third-party data; third-party literature and uninspected stored author selections are not included. The Lean formalizations of Theorems 4.3

and 4.6 are distributed with the accompanying source repository, not with the computational supplement.

A Locality scope and exact counterexamples

A.1 Regular Markov likelihoods beyond fixed covariance

Exact gap-information additivity is not restricted to Gaussian mean parameters. Suppose a fully observed Markov family is regular in a parameter neighborhood: its marginal and transition densities admit differentiation under their integrals, their scores have finite second moments, and the sampled process is not changed by selecting observations. These assumptions, including any support conditions needed for differentiating normalizing integrals, are additional to the finite-history model. For a fixed schedule $t_1 < \dots < t_k$, its log likelihood is a first-marginal term plus gap terms. The gap score u_a satisfies $\mathbb{E}(u_a \mid Y_{t_1}, \dots, Y_{t_a}) = 0$ by differentiating the conditional density's normalizing integral. Any earlier score is measurable with respect to that past, so the tower property gives zero cross second moments. Expected Fisher information is consequently the sum of the first-marginal and expected gap Fisher matrices. A gap's expectation uses the unconditional marginal at its left endpoint, which is independent of the earlier sampling choices. This is the familiar score-orthogonality argument; for example, Pagendam and Pollett [43, Section 2] use gap Fisher information in designing observations of a simple death process.

For an explicit Gaussian version, let a gap satisfy

$$Y_j \mid Y_i \sim N\{\mu_j + A(Y_i - \mu_i), \Omega\}, \quad \text{Cov}(Y_i) = P_i,$$

where all matrices and means may depend differentiably on the parameter. For coordinate a define $c_a = \mu_{j,a} - A\mu_{i,a}$, $A_a = \partial_a A$, and $\Omega_a = \partial_a \Omega$. The expected gap Fisher entry is

$$(I_{ij}^{\text{gap}})_{ab} = c_a^\top \Omega^{-1} c_b + \text{tr}(\Omega^{-1} A_b P_i A_a^\top) + \frac{1}{2} \text{tr}(\Omega^{-1} \Omega_a \Omega^{-1} \Omega_b). \quad (\text{A.1})$$

To verify it, hold observed Y_i fixed while differentiating the conditional mean: its derivative is $c_a + A_a(Y_i - \mu_i)$. With the centered gap error e , its score is

$$[c_a + A_a(Y_i - \mu_i)]^\top \Omega^{-1} e + \frac{1}{2} \{e^\top \Omega^{-1} \Omega_a \Omega^{-1} e - \text{tr}(\Omega^{-1} \Omega_a)\}.$$

The conditional Gaussian error is independent of Y_i and has covariance Ω . Its third moments vanish. Its fourth moments satisfy $\mathbb{E}(e_i e_j e_k e_l) = \Omega_{ij} \Omega_{kl} + \Omega_{ik} \Omega_{jl} + \Omega_{il} \Omega_{jk}$, obtained by differentiating $\mathbb{E}e^{t^\top e} = e^{t^\top \Omega t/2}$ four times at zero. Taking score second moments gives the covariance trace term and the mean quadratic term; averaging the latter over Y_i gives (A.1). The first marginal has the corresponding mean term and $\frac{1}{2} \text{tr}(P_i^{-1} P_{i,a} P_i^{-1} P_{i,b})$. These positive-semidefinite expected gap matrices can be used as exact path weights when the hypotheses hold. They do not extend the noisy finite-history relative bounds to covariance parameters: derivatives of the working precision introduce additional terms that those bounds do not control.

A.2 Stationarity and contraction cannot be dropped

The stationary near-pair improvement in Theorem 3.3 needs a first-gain lower bound, not just a gain upper bound. Take three times, all selected, $L = 1$, latent initial variance 1, both transitions $1/2$, zero process noise, and measurement variances all 1. Then

$$R = \begin{pmatrix} 2 & 1/2 & 1/4 \\ 1/2 & 5/4 & 1/8 \\ 1/4 & 1/8 & 17/16 \end{pmatrix}, \quad Z_2 = \epsilon_2 - \epsilon_1/4, \quad Z_3 = \epsilon_3 - \epsilon_2/10.$$

Direct multiplication gives $\text{Cov}(Z_3, Z_2) = -1/20$. The general scalar promises hold with $\bar{P} = r_{\min} = 1$ and $\rho = 1/2$, but using their $\kappa = 1/2$ in the stationary improved near formula would give only $1/32 < 1/20$. The fresh history at time three starts from latent prediction variance $1/4$, with gain $1/5$, not $1/2$. Even with stationarity, the far term cannot receive a universal $1 - \kappa$ factor: at $L = 0$, $P = r = 1$, $a = 1/2$, the far covariance is $1/2$, whereas that proposed bound would be $1/4$.

Strict temporal decay is also substantive. For a common random intercept, $\epsilon_t = X + v_t$, with $\text{Var}(X) = \text{Var}(v_t) = 1$ and constant scalar sensitivity $F_t = 1$, the full selected covariance for n observations is $I + \mathbf{1}\mathbf{1}^\top$. Its information is $n/(n+1)$ by the rank-one inverse formula. Conditioning on m previous observations gives regression coefficients $1/(m+1)$, residual variance $(m+2)/(m+1)$, and information increment $1/[(m+1)(m+2)]$. Hence, for fixed window L and $n \geq L$,

$$I_L([n]) = \frac{L}{L+1} + \frac{n-L}{(L+1)(L+2)}. \quad (\text{A.2})$$

The first L increments telescope. The remaining increments are constant, so the local information grows unboundedly with horizon while the true information stays below one. No fixed-window uniform relative-information bound holds for this class without strict decay. Adding a fixed positive prior does not remove this divergence or its unbounded log-determinant error.

A.3 Gaussian KL, relative spectral error, and the Kaporin metric

Let $R, Q \succ 0$ have dimension d and let λ_i be the eigenvalues of $R^{1/2}QR^{1/2}$. Evaluation of the Gaussian densities and expectation of their quadratic forms gives

$$2 \text{KL}\{N(0, R) \parallel N(0, Q^{-1})\} = \text{tr}(QR) - \log \det(QR) - d = \sum_{i=1}^d (\lambda_i - \log \lambda_i - 1). \quad (\text{A.3})$$

The function $f(x) = x - \log x - 1$ is nonnegative, strictly decreasing on $(0, 1)$, strictly increasing on $(1, \infty)$, and diverges at both ends. Thus a KL bound $\epsilon > 0$ places each eigenvalue between the two positive roots $a_\epsilon < 1 < b_\epsilon$ of $f(x) = 2\epsilon$. At $\epsilon = 0$ both roots equal one. Congruence gives

$$a_\epsilon R^{-1} \preceq Q \preceq b_\epsilon R^{-1}, \quad a_\epsilon (J_0 + F^\top R^{-1} F) \preceq J_0 + F^\top Q F \preceq b_\epsilon (J_0 + F^\top R^{-1} F)$$

for any F and $J_0 \succeq 0$. A global KL guarantee therefore already implies relative mean-Fisher control without a sensitivity-norm assumption. The converse with a horizon-independent total KL constant is false: a spectral sandwich permits an error in each eigenvalue, and the divergence adds over dimensions. For example, if $|\lambda_i - 1| \leq \delta < 1$, Taylor's theorem gives $f(\lambda_i) \leq \delta^2/[2(1-\delta)^2]$ and KL is at most $d\delta^2/[4(1-\delta)^2]$, which contains the dimension.

An exact noisy-chain example separates these scalings. Let $n = 2m$, $P_t = r_t = 1$, $a_t = \rho$ at even t , and $a_t = 0$ at odd $t \geq 3$, with process variance $1 - a_t^2$, for $0 < \rho < 1$. Then

$$R = \text{diag}(B, \dots, B), \quad B = \begin{pmatrix} 2 & \rho \\ \rho & 2 \end{pmatrix}, \quad Q_0 = I/2.$$

Each pair contributes relative-precision eigenvalues $1 \pm \rho/2$. Deleting observations leaves complete pairs and independent singletons, so the maximum relative spectral error over all subsets is exactly $\rho/2$, independently of m . For the full-rank Kaporin definition in Kaminetz and Webber [24, Definition 1.1],

$$\kappa_{\text{Kap}}(R, Q_0^{-1}) = \frac{\{\text{tr}(RQ_0)/(2m)\}^{2m}}{\det(RQ_0)} = (1 - \rho^2/4)^{-m}. \quad (\text{A.4})$$

Thus $\log \kappa_{\text{Kap}} = -m \log(1 - \rho^2/4)$ grows linearly, and KL is one half of this value by (A.3). The dimension-normalized Kaporin root would instead be $(1 - \rho^2/4)^{-1/2}$, a different normalization. This is a comparison of error measures, not a counterexample to the approximation theorems of Kaminetz and Webber [24].

A.4 A precise boundary for pivot-set supermodularity

A separate issue in the literature matters if one attempts to replace exact graph pricing by a general greedy pivot guarantee. Theorem 4.3 and equation (4.10) of the thesis [23] assert supermodularity of its pivot-set KL objective for arbitrary positive-definite covariance. The following rational example refutes that step. The assertion does not appear in the distinct March 2026 paper [24], so the example does not concern that paper.

Set

$$R = \begin{pmatrix} 1 & 0 & 3/5 \\ 0 & 1 & 3/5 \\ 3/5 & 3/5 & 1 \end{pmatrix}, \quad \det R = 7/25 > 0. \quad (\text{A.5})$$

Its first two leading principal minors are one, so $R \succ 0$. With fixed order $1 < 2 < 3$ and pivot set A , condition row j on $\{i \in A : i < j\}$ and let Q_A be its Vecchia precision. Direct regression gives

$$Q_\emptyset = I, \quad Q_{\{1\}} = \begin{pmatrix} 25/16 & 0 & -15/16 \\ 0 & 1 & 0 \\ -15/16 & 0 & 25/16 \end{pmatrix}, \quad Q_{\{2\}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 25/16 & -15/16 \\ 0 & -15/16 & 25/16 \end{pmatrix},$$

$$Q_{\{1,2\}} = R^{-1} = \begin{pmatrix} 16/7 & 9/7 & -15/7 \\ 9/7 & 16/7 & -15/7 \\ -15/7 & -15/7 & 25/7 \end{pmatrix}.$$

For all four patterns $\text{tr}(Q_A R) = 3$, since each standardized residual has variance one. If $g(A) = \text{KL}\{N(0, R) \| N(0, Q_A^{-1})\}$, (A.3) gives

A	$\emptyset$	$\{1\}$	$\{2\}$	$\{1, 2\}$
$\exp(2g(A))$	$25/7$	$16/7$	$16/7$	1

and hence

$$g(\emptyset) + g(\{1, 2\}) - g(\{1\}) - g(\{2\}) = \frac{1}{2} \log(175/256) < 0. \quad (\text{A.6})$$

Supermodularity requires the opposite inequality. The thesis uses an additional positive objective scaling, which does not change the sign. The same four precisions arise if selected pivots are moved first and the remaining variables made conditionally independent given the pivots: the first two variables are independent, so their reordered regressions change no nonzero coefficients. The supermodularity failure thus holds under both conventions.

The invalid marginal identity can be identified explicitly. For the pivot-first convention, the determinant of the working covariance is $\det R_{AA} \prod_{j \notin A} \text{Var}(Y_j \mid Y_A)$, with empty determinant one. The conditional means remain true, so $\text{tr}(Q_A R) = 3$ here (or the full dimension generally). Cancellation of the newly selected pivot's variance term yields the correct marginal formula

$$2\{g(A \cup \{\ell\}) - g(A)\} = \sum_{j \notin A \cup \{\ell\}} \log \frac{\text{Var}(Y_j \mid Y_A, Y_\ell)}{\text{Var}(Y_j \mid Y_A)}.$$

It is a sum of individual conditional-variance logarithms, not a joint residual log determinant. Adding pivot one to the empty set in (A.5) gives $\log(16/25)$, while substituting the joint conditional-variance ratio used in the thesis’s equation (4.9) gives $\log(7/16)$.

Under the literal fixed-order interpretation there is also a failure of the displayed exponential greedy bound: pivot three changes no conditioning set, greedy first selects one or two, and a two-pivot optimum is zero. At one step the claimed bound would require $g(\{1\}) \leq e^{-1/2}g(\emptyset)$. But $16^8 > 25^5 7^3$ implies $\log(16/7)/\log(25/7) > 5/8$, while $e^{1/2} > 1 + 1/2 + 1/8 > 8/5$ gives $e^{-1/2} < 5/8$. This last conclusion is limited to fixed order. Under pivot-first ordering, pivot three gives $\exp(2g(\{3\})) = 256/175$ and greedy selects it; the same one-step contradiction then does not apply. The convention-independent findings are the failed supermodularity step and its marginal identity. None of the local-information guarantees proved in this paper uses that step or a generic greedy-pivot approximation ratio.

B The scalar Gaussian-message predecessor

This appendix reduces the scalar problem to prior work; it makes no additional novelty claim. We state exactly which interpretation of Mahalanabis and Štefankovič [36] is used. Their Section 3.2, Theorem 43 approximates total conditional variance on a Gaussian precision graph of treewidth w , with runtime

$$\left(\frac{Nw \text{cond}(\Lambda)}{\alpha}\right)^{O(w^2)} k^2 + N2^{O(w^3)} \quad (\text{B.1})$$

for relative variance error $1 + \alpha$. Here N counts Gaussian vertices, not candidate times. The theorem as printed sums all unobserved variances and permits all nodes as observations. We need only a single unobservable target, and only designated observation candidates.

B.1 The focused recurrence and a local-cost issue

Their Definition 18, equation (37), assigns a conditional-variance cost to the private variables of a subtree. Equations (39)–(40) combine those local costs with child messages, while (41)–(43) pass separator precision by observation and Schur complementation. Rounded counterparts are (44)–(48). Restricting every local observed subset to designated candidates simply removes choices; the proof’s rounding step retains the same observed subset, so every allowed optimal witness remains allowed. At the level of the covariance costs intended by Definition 18, replacing each local trace by a nonnegative weighted sum of its private-coordinate variances preserves relative-error comparisons: if $e^{-a}M \preceq \widehat{M} \preceq e^a M$ with $M \succ 0$, inversion gives the same factors for each covariance diagonal and their nonnegative weighted sum. The addition, observation, marginalization and rounding comparisons in Observation 33 and Lemmas 41–42 then give the same induction. Zero costs are compared by inequalities without division. This is an adaptation of the proof, not the wording of the stated theorem.

One printed local-cost formula needs care. Taking an inverse after extracting a private principal block generally differs from extracting that block after inverting the full unobserved bag precision. For example,

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad (M^{-1})_{11} = 3/5 \neq (M_{11})^{-1} = 1/2.$$

For the focused specialization we can avoid a nonzero cost in this disputed form. Retain the target ϑ in every separator, direct messages toward a singleton bag $\{\vartheta\}$ followed by an empty root, and give all earlier private variables weight zero. The only positive cost is then the target variance

in that singleton bag, whose separator is empty and where the two inverse/block orders coincide. Zero-weight local costs are set to zero without evaluating an unnecessary inverse. Information messages still use the stated Schur complements. Binary copies of bags retain ϑ until the singleton and preserve this property. Thus the focused construction does not depend on that erroneous local-cost identity.

B.2 Rational normalization and regularization

Take the scalar-observation class of Theorem 4.1, with one parameter ($p = 1$) and cardinality as the only constraint. Cardinalities $k < 0$ or $k > n$ are infeasible; for $k = 0$, return the empty schedule. If all sensitivities vanish, every feasible schedule is optimal. In the remaining case, assume $1 \leq k \leq n$ and at least one nonzero sensitivity f_t . Choose positive powers of two s, d_t with

$$s^2 \leq r_{\min} < 4s^2, \quad d_t^2 \leq r_t < 4d_t^2.$$

Define

$$U_t = X_t/s, \quad Z_t = Y_t/d_t, \quad c_t = s/d_t, \quad h = \max_t |f_t/d_t| > 0, \quad g_t = f_t/(d_t h).$$

Then $|c_t| \leq 2$, $|g_t| \leq 1$, some $|g_t| = 1$, and the normalized observation-noise variances lie in $[1, 4)$. The normalized parameter is $\vartheta = h\theta$; its data information equals original data information divided by h^2 , which preserves rankings and relative ratios. The latent covariance $K = \text{Cov}(U)$ satisfies

$$|K_{ij}| \leq 4B_0\rho_0^{|i-j|}, \quad \|K\|_2 \leq 4B_0(1 + \rho_0)/(1 - \rho_0).$$

For zero contraction the same formulas apply directly.

To allow zero original process variances, add independent variance ζ to each normalized initial/process innovation. Write A for the strict lower shift with entries a_t . Its norm is at most ρ_0 ; thus $T = (I - A)^{-1}$ satisfies

$$\frac{\zeta}{(1 + \rho_0)^2} I \preceq E := \zeta T T^\top \preceq \frac{\zeta}{(1 - \rho_0)^2} I.$$

The lower bound follows from $\|T^{-1}\|_2 \leq 1 + \rho_0$ and the upper one from the Neumann series. Hence $K_+ = K + E \succ 0$ has polynomially controlled spectrum. Let R' be the normalized observation-error covariance and $C = \text{diag}(c_t)$. Since $R' \succeq I$ and $\|C\|_2 \leq 2$,

$$R' \preceq R'_+ := R' + C E C^\top \preceq R' + \frac{4\zeta}{(1 - \rho_0)^2} I.$$

Choose $\tau = \varepsilon/4$ and $\zeta = \tau(1 - \rho_0)^2/4$. Every selected principal covariance obeys $R'_{SS} \preceq R'_{+,SS} \preceq (1 + \tau)R'_{SS}$, so for $I'(S) = g_S^\top (R'_{SS})^{-1} g_S$ and its regularized counterpart,

$$I'(S)/(1 + \tau) \leq I'_+(S) \leq I'(S). \tag{B.2}$$

All these changes are rational and have polynomial encoding length.

B.3 Constant width and conversion of the objective

Introduce an artificial independent target prior $\vartheta \sim N(0, 1)$ and the model

$$U \sim N(0, K_+), \quad Z_t = g_t \vartheta + c_t U_t + \nu_t, \quad 1 \leq \text{Var}(\nu_t) < 4.$$

Only the Z_t are candidates; only ϑ has positive variance weight. Then $v(S) = \text{Var}(\vartheta \mid Z_S) = 1/(1 + I'_+(S))$. The Gaussian precision graph has bags $\{\vartheta, U_t, U_{t+1}, Z_t\}$ for $t < n$, followed by $\{\vartheta, U_n, Z_n\}$ and the singleton target/empty root above. These cover the latent-chain edges and each observation-density triangle, and satisfy running intersection. Their width is at most three.

For conditioning, start from independent (ϑ, U, ν) with covariance $\text{diag}(1, K_+, D)$, $I \preceq D \prec 4I$. The shear to (ϑ, U, Z) and its inverse have norms at most $1 + \sqrt{n+4}$, because the added bottom block $[g, C]$ has norm at most $\sqrt{n+4}$. The joint covariance and inverse precision therefore have condition number bounded by

$$(1 + \sqrt{n+4})^4 \frac{\max\{4, 4B_0(1 + \rho_0)/(1 - \rho_0) + \zeta/(1 - \rho_0)^2\}}{\min\{1, \zeta/(1 + \rho_0)^2\}} = O_{\rho_0, B_0}(n^2/\varepsilon). \quad (\text{B.3})$$

In particular, normalization removes arbitrarily varying original noise and sensitivity scales from this bound.

At an index with $|g_t| = 1$, the regularized observation-error variance is at most $D_0 = 4 + 16B_0 + 4\zeta/(1 - \rho_0^2)$. A best nonempty schedule thus has $I'_{+,*} \geq 1/D_0$: start with that observation and use monotonicity to pad to cardinality k . Apply the focused predecessor with $\alpha = \varepsilon/[4(D_0 + 1)]$. Its variance ratio implies

$$I'_+(\hat{S}) \geq \frac{I'_{+,*} - \alpha}{1 + \alpha} \geq \frac{1 - \alpha D_0}{1 + \alpha} I'_{+,*} \geq (1 - \varepsilon/4) I'_{+,*}.$$

An at-most- k output can be padded to exactly k , since observing more variables cannot reduce information. Combining with (B.2) yields

$$I'(\hat{S}) \geq \frac{1 - \varepsilon/4}{1 + \varepsilon/4} \max_{|S|=k} I'(S) \geq (1 - \varepsilon) \max_{|S|=k} I'(S).$$

Multiplication by h^2 and addition of a common nonnegative scalar prior preserve this guarantee. Equations (B.1) and (B.3) make the predecessor's operation bound polynomial for this normalized class.

This establishes the mathematical reduction in the predecessor's spectral-rounding model; it does not supply a full rational implementation of that rounding net. Our explicit bit-complexity result, Theorem 4.1, is proved independently, and this remaining difference in numerical model is no basis for claiming a first scalar approximation scheme. The reduction uses cardinality alone and does not justify padding under spacing or arbitrary side constraints.

C Proof of the represented-matroid cover

We prove Theorem 4.6 using the owner-count variant of the exact profile construction. The Lean formalization of Theorem 4.6 covers this variant, including its execution from the original input and its polynomial bit-work bound.

If $q = \text{rank } A = 0$, the only base is empty and is returned, regardless of the prior. Otherwise select q independent rows of A and discard the redundant rows, preserving its column matroid. Apply Lemma 4.4, retaining its range and magnitude tests. For a rank- r trial, write E' for the retained elements and F for the distinct forced owners. Only original bases contained in E' and containing F may be returned. Such a base exists only if $F \subseteq E'$, A_F is independent, and $\text{rank } A_{E'} = q$. These conditions can be tested directly or through the exact support test below. A smaller base of a rank-deficient restriction is never accepted.

C.1 Forcing owners without contraction

Retain the original q rows, and zero columns outside E' ; call this matrix A_{ret} . Give each element the marker $b_e = \mathbf{1}\{e \in F\}$. Since a base contains each element at most once,

$$\sum_{e \in B} b_e = |F| \iff F \subseteq B. \quad (\text{C.1})$$

One extra integer profile coordinate therefore enforces all owners. This replaces rational contraction without changing the accepted bases. Write $q' = q - |F|$ for their common number of optional elements. If $q' = 0$, every accepted base is exactly F and has zero approximation error; the same support procedure may still be used.

C.2 Exact profile detection and witness recovery

Put $h = \eta/(rq)$ and $z_{e,ij} = \lfloor A_{e,ij}/h \rfloor$ for $i \leq j$. There are $d = r(r+1)/2$ information coordinates. The prior is unrounded. Forced elements may be rounded with the other retained elements because their common contribution cancels between accepted bases. Set $L_* = \lceil 4p/h \rceil$ and $w_{e,ij} = z_{e,ij} + L_* \in \{0, \dots, 2L_*\}$. Every original base has q elements, so subtracting $qL_*\mathbf{1}$ recovers its signed profile. Subtracting the shared forced profile leaves the optional profile of q' elements.

Over the rational field define

$$\begin{aligned} g(y, t) &= \det \left(A_{\text{ret}} \text{diag}_e (y^{w_e} t^{b_e}) A_{\text{ret}}^\top \right) \\ &= \sum_{\substack{B \text{ original base} \\ B \subseteq E'}} \det(A_B)^2 y^{\sum_{e \in B} w_e} t^{|B \cap F|}. \end{aligned} \quad (\text{C.2})$$

This is the Cauchy–Binet identity used by Bernstein et al. [7, Lemmas 4.3–4.4], with one additional integer coordinate. Expanding over q -column subsets gives zero for dependent subsets and the displayed squared minor for every base. Thus every coefficient is nonnegative and is positive exactly when its profile is attained. Retain only marker degree $|F|$, which enforces (C.1). Characteristic-zero arithmetic is essential: a positive coefficient may vanish modulo a prime. A zero residue alone cannot certify infeasibility.

The information degrees are at most $D_* = 2qL_*$ and the marker degree is at most $|F| \leq q$. Use the uniform grid $\{1, \dots, \overline{D} + 1\}^{d+1}$, where $\overline{D} = \max(D_*, q)$. Since $L_* \geq 1$ in these trials, $\overline{D} = D_*$. A sharper rectangular grid is also possible. Exact tensor Lagrange interpolation recovers the requested coefficient: for each univariate node j , form

$$\ell_j(X) = \prod_{\substack{1 \leq s \leq \overline{D}+1 \\ s \neq j}} \frac{X - s}{j - s},$$

and sum each grid value multiplied by the product of the corresponding coefficients of the ℓ_j . This is an explicit inverse of the tensor Vandermonde system. The coefficient-vector products construct the univariate polynomials with polynomial work, without enumerating all subsets of factors. The procedure is equivalent to the established exact interpolation method, whose original presentation uses a moment curve.

For every positive owner-complete profile, recover one original base by one-pass deletion. Start with the retained columns and delete a column exactly when the requested coefficient remains positive after its removal. Keep the original q rows in all tests: rank loss makes the polynomial zero. Never replace it by a smaller row basis. Feasibility of the target persists, and a failed deletion

cannot become feasible after further columns are removed. An extra surviving column outside one target-profile base would contradict its failed deletion. Thus at most m tests leave exactly an original base with that profile. No lifting is needed.

C.3 Coverage, zero information, and bit bounds

Fix an arbitrary target base B with information rank $r > 0$. Its maximum-volume trial from Lemma 4.4 retains all of B . The forced owners form a subset of B , and the retained set has original rank q . The marker-complete profile of B therefore has a positive coefficient. Let $\hat{B}$ be its recovered representative. If $q' = 0$, both bases equal F . Otherwise their optional signed profiles are equal after cancelling F , and each optional entry residual lies in $[0, q'h)$. The transformed information entries consequently differ by less than $q'h \leq qh$. Their symmetric difference has norm at most $rqh = \eta$. The forced identity floor and rectangular reconstruction give (4.3) and exact range equality. For information rank zero, require $Q_0 = 0$ and find an original-rank base using only zero-matrix elements. A nonzero prior excludes this branch.

There are at most $\binom{M}{r}$ rank- r trials. Fixing the marker at $|F|$ and subtracting the common forced profile leaves at most $R_* = (2q'L_* + 1)^d$ positive profiles. The extra interpolation coordinate therefore increases work but not the number of returned bases. In particular,

$$|\mathcal{C}| \leq 1 + \sum_{r=1}^p \binom{M}{r} [2q\lceil 4prq/\eta \rceil + 1]^{r(r+1)/2}.$$

The initial one permits the zero-information representative; $q' = 0$ contributes at most one base in its trial.

The variable matrix rank requires polynomial determinant evaluation, not a permutation expansion. The implementation clears denominators and uses a cached division-free Bird recurrence on the integer matrix [9]. It stores each preceding matrix once; an order- q determinant takes $O(q^4)$ ring operations. Bounds on the integer iterates and exact final division by the common denominator raised to q give polynomial bit work, even when q grows. Rational row selection and information normalization have polynomial bit bounds as well.

At a grid point the monomial has bit length $O((dL_* + 1)\log(\bar{D} + 1))$. The evaluated matrix, determinant, univariate Lagrange coefficients, tensor terms, and finite sums all have polynomial encoding lengths. The grid has polynomial size for fixed $d + 1$. Recomputing its values for each requested coefficient and each deletion test still gives polynomial work in input length and $1/\eta$ for fixed p . Basis output length is polynomial too. This completes the mathematical Turing-model argument without a floating zero test, independence-oracle conversion, or feasibility relaxation. The Lean cost model charges schoolbook arithmetic, finite scans, storage, and copies, including rejected trials and labels. It excludes construction of proof and cost-observer transcripts and makes no Lean wall-clock claim.

C.4 Exact witnesses for the representation restrictions

The squared-minor argument is specific to a single represented matroid. For two representations the mixed determinant contains products of different minors, whose signs may differ. With row matrices $A_1 = (1, 1)$ and $A_2 = (1, -1)$, and identical zero profiles, both singleton sets are common bases but $\det(A_1 A_2^\top) = 0$. Thus direct substitution loses feasible profiles. A separate randomized profile algorithm for represented matroid intersection is established by Bernstein et al. [8, Section 3]; element indeterminates and random substitution address that cancellation. Its guarantee is not

part of our deterministic theorem. Likewise,

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = -2$$

is nonzero over $\mathbb{Q}$ but zero over $\mathbb{F}_2$, so reinterpreting a finite-field representation can change the bases.

Projection does not remove attainability requirements either. The rank-one uniform matroid on two elements with scalar information 1 and 3 has projected information polytope $[1, 3]$. Its integer point 2 is not attained by any base. Integer optimization over that interval would therefore solve a different problem. Finally, the independence-oracle result of Bernstein et al. [7, Theorem 1.1] assumes a fixed number of distinct weight values. The number of our rounded labels grows with input size and $1/\eta$; substituting them in that oracle theorem does not retain a fixed polynomial exponent. These examples explain limitations of the proof routes, not hardness of every possible broader spectral-cover algorithm.

D Exact arithmetic for the support witnesses

For rational encoded data, rational elimination evaluates the selected covariance inverse and the conditional regressions used by the local and separator certificates. Gaussian innovation formulas are an efficient alternative, not a different information model. The exact checker also uses rational representations of its reference matrices N or $W = N^{-1}$, separator nuisance coefficients G , split and query witnesses, and scenario simplex weights. Their stated positivity, feasibility, and normalization conditions are checked exactly; a numerical proposal alone is not a witness. The rational upper enclosures $\hat{\delta} < 1$ of Section 5.1 replace analytic locality errors in all local scores, scenario by scenario when needed; the weight H below includes this rational inflation factor when applicable. These representation choices restrict the checker, not the real-valued theorems. All covariance matrices inverted must be SPD under the stated hypotheses. Symmetry and exact positive leading principal minors check an SPD reference; exact pivoted rational LDL^T or all principal minors can check a PSD split. A nonpositive pivot rejects an SPD claim. Floating-point eigenvalues alone certify neither assertion.

For $x \in \mathbb{Q}$, $x > 0$, choose an integer h with $x = 2^h y$, $1 \leq y < 2$. Put $u = (y - 1)/(y + 1) \in [0, 1/3)$, and for $m \geq 1$ define

$$s_m(u) = 2 \sum_{j=0}^{m-1} \frac{u^{2j+1}}{2j+1}, \quad 0 \leq \log y - s_m(u) \leq \frac{2u^{2m+1}}{(2m+1)(1-u^2)}. \quad (\text{D.1})$$

The series follows by integrating $2/(1-u^2)$; replacing the remaining odd denominators by $2m+1$ proves its geometric remainder bound. The same formula at $u = 1/3$ encloses $\log 2$. Combine the enclosures as $\log x = h \log 2 + \log y$, reversing the endpoints for the first term when $h < 0$. Thus determinants below one and negative log determinants cause no sign ambiguity. Exact positive rational determinants need no eigenvalue calculation. Refinement to any prescribed rational interval width terminates geometrically; the bit length of h and the requested interval width control the number of terms, and the arithmetic bit cost also depends on the rational input encoding. Integer floor and ceiling then yield outward decimal displays. A displayed efficiency percentage additionally needs an exponential enclosure if it is itself asserted as a rigorous bound.

D.1 Signed integer interval scores

For a scalar local innovation let

$$u = (F_t - \sum_j b_j F_{t-a_j})^T, \quad d > 0, \quad s = u^T H u / d,$$

where H is any symmetric rational weight matrix. In our logdet certificates $H \succeq 0$; allowing indefinite H makes the arithmetic contract explicit. Choose positive integer grids G_F, G, G_s . Enclose sensitivities on grid G_F^{-1} , and b_j, H_{ij} on grid G^{-1} , using exact floors and ceilings for signed rationals. For integer intervals, multiplication uses the minimum and maximum of all four endpoint products; subtraction uses $[a, b] - [c, d] = [a - d, b - c]$. A square interval has lower endpoint zero if it contains zero, and otherwise the smaller endpoint square; its upper endpoint is the larger endpoint square.

These rules enclose each component u_i by endpoints divided by GG_F . Compute diagonal terms $H_{ii}u_i^2$ by the square rule and off-diagonal terms $2H_{ij}u_iu_j$ for $i < j$ by interval multiplication. Let Q^+ be the integer upper endpoint after summation, so $u^T H u \leq Q^+ / (G^3 G_F^2)$. Repeated variables need not be independent; interval inclusion holds term by term and dependence can only widen the result. Rounded weight entries need not form a PSD matrix.

Require $d_\ell = \lfloor Gd \rfloor > 0$, and put $Q_+ = \max(0, Q^+)$. Then

$$s \leq \frac{Q_+}{G^2 G_F^2 d_\ell} \leq \frac{1}{G_s} \left\lceil \frac{G_s Q_+}{G^2 G_F^2 d_\ell} \right\rceil. \quad (\text{D.2})$$

For a nonnegative numerator, division by the positive lower variance gives the first inequality. For a negative numerator, the nonnegative right side already dominates it. The clamp is therefore necessary for this general indefinite-weight contract. A nonpositive variance floor causes rejection or grid refinement; it is never replaced by a positive number. The statistical innovation lower bound d_* used to establish δ and the arithmetic floor d_ℓ/G serve different purposes. Both must be valid for their respective conditional variances.

An exact integer path DP applied to these upper scores bounds the rational support price. If every reachable arc exceeds its direct rational ceiling by at most e integer units and paths have at most k scoring innovations, the additional price is at most ke/G_s . A separator pattern scored as a sum of scalar conditional innovations obeys the same bound with the total number of selected observations. A finite-state spacing restriction changes the feasible price, not this inclusion argument. For fixed positive variances and PSD weights, refining G_F, G shrinks the interval error. At a score-grid boundary one excess ceiling unit may persist at every finite interval grid; refining G_s too removes this final scale, so no universal negligible-error claim is needed. General block scores can instead use exact rational conditional inverses and an outward ceiling directly; the scalar formula above does not assume an unproved entrywise bound on a block inverse.

E Structured evaluation of the established dense relaxation

Let $K_{ij} = P\rho^{|i-j|}$, $R = K + rI$, $P \geq 0$, $r > 0$, $|\rho| < 1$, and choose an admissible scalar split $a > 0$. For $P > 0$ and $n \geq 2$, put $c = P(1 - \rho^2) > 0$. The scaled latent precision M has endpoint diagonals 1, interior diagonals $1 + \rho^2$, and adjacent off-diagonals $-\rho$, so $K^{-1} = M/c$. Defining $\beta_i = a(1 - z_i) + rz_i > 0$ gives the exact tridiagonal identities

$$[M + c \text{diag}(z_i/\beta_i)]U = MF, \quad V_i = (a/\beta_i)U_i, \quad J_D = J_0 + F^T \text{diag}(z_i/\beta_i)U. \quad (\text{E.1})$$

To derive them, multiply $(I + (K + (r - a)I)Z)V = F$ by K^{-1} , set $U = \text{diag}(\beta_i/a)V$, and collect terms. The coefficient matrix is SPD. These are instances of established fast latent-precision plus diagonal-noise algebra [50]. Tridiagonal solves require $O(np)$ field operations, but information formation and all gradient quadratics bring the complete value/gradient cost to $O(np^2 + p^3)$, with $O(np + p^2 + n)$ storage. Field-operation counts are distinct from rational denominator growth.

An exact split check follows from the congruence

$$R - aI = M^{-1/2}[cI + (r - a)M]M^{-1/2}. \quad (\text{E.2})$$

Strict positive pivots in rational tridiagonal elimination of $cI + (r - a)M$ prove $a < \lambda_{\min}(R)$. Conversely, its first nonpositive pivot, with all earlier pivots positive, proves it is not SPD by Sylvester's criterion, and hence $a \geq \lambda_{\min}(R)$. Elimination stops at that pivot: no division by a zero or negative pivot is needed. This is the scalar all-split witness in (5.9). A PSD boundary split can alternatively be checked by a complete PSD test. For $n = 1$, use variance $P + r$ directly; the endpoint formula M/c does not apply. For $P = 0$ or $\rho = 0$, use the corresponding diagonal covariance directly.

E.1 Covariance innovations and the reverse adjoint

Precision coordinates can be ill-conditioned when $|\rho|$ approaches one, even if the observation nugget makes R moderately conditioned. The following exact covariance formulas avoid the identified precision solve. Start $P_1^- = P$, $m_1^- = 0$, where each m_i^- is a row with p entries. For $i = 1, \dots, n$ set

$$\begin{aligned} e_i &= F_i - m_i^-, & q_i &= z_i P_i^- + \beta_i, & w_i &= z_i / q_i, \\ m_i &= m_i^- + P_i^- w_i e_i, & P_i^+ &= P_i^- \beta_i / q_i, \\ m_{i+1}^- &= \rho m_i, & P_{i+1}^- &= \rho^2 P_i^+ + P(1 - \rho^2). \end{aligned} \quad (\text{E.3})$$

The final unused prediction can be omitted. Then

$$J_D(z) = J_0 + \sum_i w_i e_i^T e_i. \quad (\text{E.4})$$

On $T = \{i : z_i > 0\}$ these are ordinary Kalman innovations for covariance $K_{TT} + \text{diag}(\beta_i/z_i)$ and deterministic observation rows F_T . This is the covariance in (5.3). At $z_i = 0$ the update and information are zero, with no infinite variance formed. Classical Kalman filtering and RTS smoothing are described by Särkkä and Svensson [52, Theorems 6.6 and 12.2]; the present equations apply their linear algebra to the design sensitivity right-hand sides.

All rows of V are needed in the selection gradient, including those at zero weights. Initialize a row $b_{n+1} = 0$ and sweep backward:

$$V_i = \frac{a}{q_i}(e_i - \rho P_i^- b_{i+1}), \quad b_i = w_i e_i + \rho \frac{\beta_i}{q_i} b_{i+1}. \quad (\text{E.5})$$

For one right-hand side consider $Q(F) = \frac{1}{2} \sum_i w_i e_i^2$ with the variances and weights fixed. The forward mean map in (E.3) is $m_{i+1}^- = \rho[(1 - P_i^- w_i)m_i^- + P_i^- w_i F_i]$. Its negative adjoint with respect to m_i^- is $b_i = w_i e_i + \rho(1 - P_i^- w_i)b_{i+1}$. Since $1 - P_i^- w_i = \beta_i/q_i$, $\partial Q/\partial F_i = w_i(e_i - \rho P_i^- b_{i+1})$. By (5.3) this derivative is row i of ZV . Cancel z_i/a when $z_i > 0$ to prove the first identity in (E.5); continuity of the resolvent and the positive denominators proves it also at $z_i = 0$. Finally apply (5.4). This is an ordinary reverse derivative of the innovation quadratic, not a new filter or relaxation.

In exact arithmetic, $a/q_i \leq 1$ for an admissible scalar split. The conditional variance $P_i^- + r$ arises by predicting the original observation from previous observations with additional independent

nonnegative virtual noise. For every predictor coefficient vector c_0 , the original residual variance is at least $\lambda_{\min}(R)(1 + \|c_0\|^2)$; added noise cannot reduce it. Minimization gives $P_i^- + r \geq \lambda_{\min}(R) \geq a$, and therefore $q_i = a(1 - z_i) + (P_i^- + r)z_i \geq a$. Also $|\rho\beta_i/q_i| \leq |\rho|$. These bounds explain why the reverse sweep avoids one amplification mechanism; they are not a complete floating-point error analysis.

Two implementation failures have different causes. The algebraically correct tridiagonal precision formulation can lose accuracy near unit correlation because it works in a poorly conditioned latent precision. A first covariance implementation instead computed $V_i = a(F_i - m_i^{\text{smooth}})/\beta_i$ from RTS smoothed means. That identity is exact, but a large latent-to-nugget ratio can cause catastrophic cancellation in $F_i - m_i^{\text{smooth}}$ before scaling. The reverse sweep (E.5) avoids that particular subtraction and uses the bounded factor a/q_i . Remaining small-residual cancellation, ill-conditioned parameter information, and finite floating-point range can still cause error. Neither successful numerical evaluation nor observed agreement replaces the rational certificate.

The boundary cases follow directly. When all $z_i = 0$, $J_D = J_0$ and $V = F$, so the gradient generally is nonzero. Negative ρ changes the signs of mean transitions and adjoints but not variance positivity. A split $a > r$ is valid whenever $a \leq \lambda_{\min}(R)$; positivity of β_i does not require $a \leq r$. In floating point the process variance is evaluated as $P[(1 - \rho)(1 + \rho)]$ to avoid an unnecessary intermediate overflow for negative ρ . Exact certificates are unaffected by expression order.

F Robust certificate witnesses and normalization limits

Consider two scalar scenarios with prior one and two feasible paths whose information pairs are $(4, 1)$ and $(1, 4)$. This is an additive two-arc graph with increments $(3, 0)$ and $(0, 3)$. Both individual optima are four and each feasible standardized efficiency is $1/4$. At $N_1 = N_2 = 5/2$ and equal scenario weights, the common linear support price is $3/5$, whereas the average of the separate support maxima is $6/5$. The shared certificate is $\log(5/8)$, a strict improvement of $3/5$ over that separate-price bound. It still exceeds the true optimum $\log(1/4)$: the equal mixture has information $(5/2, 5/2)$ and must satisfy every such tangent upper bound. Thus zero memory error does not remove a common-mixture integrality gap.

The extra factor in Corollary 5.6 cannot be omitted for an arbitrary maximizing tie choice in the stated same-set recipe. Let $a = 3/4$, $p = 1$, and let the returned set of two-scenario information pairs be

$$\mathcal{C} = \{(3/4, 1/10), (1/10, 1), (3/8, 3/8), (9/32, 3/8)\}.$$

Add the feasible pairs $(1/2, 1/2)$ and $(1, 1/10)$ to form $\mathcal{F}$. Every pair in $\mathcal{F}$ has a representative in $\mathcal{C}$ between simultaneous factors $3/4$ and $5/4$: use itself for retained pairs, $(3/8, 3/8)$ for $(1/2, 1/2)$, and $(3/4, 1/10)$ for $(1, 1/10)$. All pairs are realizable on an additive choice graph with common positive prior $1/20$ in each scenario and nonnegative scalar increments. The true scenario maxima are $(1, 1)$ and the set maxima are $(3/4, 1)$. The last two retained pairs tie for best set-normalized minimum, $3/8$. Choosing $(9/32, 3/8)$ yields true efficiency $9/32$, while the true robust optimum is $1/2$. Their ratio is $9/16 = a^2$. This example limits that comparison procedure; it is not a lower bound against all possible algorithms for robust normalization.

G Changing the latent representation and stationary bridges

G.1 The cross-representation Schur identity

We prove (5.22) algebraically so that no Gaussian prior cross term is implicit. Fix $A \subseteq B$ and put $C = B \setminus A$. Write $U = X_A$, $V = X_C$. The anchor prior can be expressed as

$$V = LU + \xi, \quad L = K_{CA}K_A^{-1}, \quad \Sigma = K_{CC} - K_{CA}K_A^{-1}K_{AC} \succ 0,$$

where ξ is independent of U . Empty blocks have their usual omitted interpretation. Partition the observation loading given (U, V) as $H_B = [H_U, H_V]$. For the selected rows and the B -conditional covariance $D = (D_B)_{SS}$, the joint quadratic, including the parameter prior, is

$$\begin{aligned} & \theta^T J_0 \theta + U^T K_A^{-1} U + (V - LU)^T \Sigma^{-1} (V - LU) \\ & + (F_S \theta + H_{U,S} U + H_{V,S} V)^T D^{-1} (F_S \theta + H_{U,S} U + H_{V,S} V). \end{aligned} \quad (\text{G.1})$$

Its coefficient matrix in (θ, U, V) is precisely $M_B(S)$. In particular the nuisance prior is

$$K_B^{-1} = \begin{pmatrix} K_A^{-1} + L^T \Sigma^{-1} L & -L^T \Sigma^{-1} \\ -\Sigma^{-1} L & \Sigma^{-1} \end{pmatrix},$$

which is generally not block diagonal between retained and removed anchors. This positive observation quadratic is the quadratic part of the residual likelihood in (θ, U, V) : both mean terms enter its residual with the same minus sign.

Substitute $V = LU + \xi$. For fixed (θ, U) this is a bijection of the minimized coordinates, so it leaves their Schur complement unchanged. The retained observation loading becomes $\tilde{H} = H_U + H_V L = H_A$, by iterated Gaussian conditioning (or directly by multiplying covariance blocks). Minimizing the remaining terms over ξ gives the quadratic

$$\theta^T J_0 \theta + U^T K_A^{-1} U + (F_S \theta + H_{A,S} U)^T [D + H_{V,S} \Sigma H_{V,S}^T]^{-1} (F_S \theta + H_{A,S} U).$$

Indeed, completing the square in ξ gives the inverse in brackets by Woodbury's identity. Conditional covariance decomposition gives $D + H_{V,S} \Sigma H_{V,S}^T = (D_A)_{SS}$. The resulting coefficient matrix is $M_A(S)$, proving (5.22) for every selected set, including the empty set. This proof needs only $K \succ 0$ and positive observation noise; Markov structure is used for cheap block pattern enumeration, not the identity itself.

For completeness, if $M \preceq N$ are SPD with the same coordinate split, then for every parameter vector x ,

$$x^T \mathcal{S}_\theta(M) x = \min_y [x; y]^T M [x; y] \leq \min_y [x; y]^T N [x; y] = x^T \mathcal{S}_\theta(N) x.$$

The minimum of the average is at least the average of the minima, proving Schur concavity in the same way. Eliminating two nuisance groups in either order is the same joint minimum over their vectors. These facts justify every matrix-order step in Theorem 5.8.

The hierarchy can be strict even when the empty-anchor hull has no integer gap. Take $n = 2$, $p = 1$, $k = 1$, $J_0 = 1$, $F = (1, -1)^T$, $K = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$, and $r = 1$. Both true singleton informations equal $3/2$, so $U_\emptyset = \log(3/2)$. For all anchors the symmetric concave scalar-split relaxation attains its maximum at equal weights, with information $9/5$; thus $U_{\{1,2\}} = \log(9/5) > U_\emptyset$. The equal mixture with one anchor has information $75/44$; this value illustrates an intermediate representation, and we do not claim that this asymmetric level is optimized at equal weights.

G.2 Singleton patterns recover the actual diagonal split

Suppose $D_A = D$ is diagonal. Set $L_0 = H_A K_A^{1/2}$, so $R = D + L_0 L_0^T$. At a distribution over schedules the augmented data matrix depends only on $Z = \text{diag}(z_i/D_{ii})$:

$$\bar{M}_A = \text{diag}(J_0, K_A^{-1}) + [F, H_A]^T Z [F, H_A].$$

Eliminating the nuisance variables gives

$$J_0 + F^T \{Z - Z L_0 (I + L_0^T Z L_0)^{-1} L_0^T Z\} F = J_0 + F^T Z (I + (R - D)Z)^{-1} F.$$

The inverse identity follows by multiplying by $I + (R - D)Z$ and is valid at zero weights and with rank-deficient $R - D$. Hence the objective is the virtual-noise extension for this exact D . For cardinality alone, the marginals of mixtures over size- k subsets fill $\mathcal{Z}_k$; more generally they fill the actual indicator hull. Replacing that hull by a larger polytope gives a further relaxation, not an equality of feasible sets. For no anchors the same formulas have no nuisance columns and apply when R itself is diagonal.

G.3 Signed stationary bridge formulas

For $K_{ij} = P\rho^{|i-j|}$ with $P > 0$ and $|\rho| < 1$, two neighboring anchors $a < c$ give, at an interior time i , the nonzero loadings

$$H_{i,a} = \rho^{i-a} \frac{1 - \rho^{2(c-i)}}{1 - \rho^{2(c-a)}}, \quad H_{i,c} = \rho^{c-i} \frac{1 - \rho^{2(i-a)}}{1 - \rho^{2(c-a)}}. \quad (\text{G.2})$$

At an anchor its loading is the corresponding unit vector. Before the first or after the last anchor the only loading is the appropriate power of ρ . For $a < i \leq j < c$ the conditional latent covariance is

$$\text{Cov}(X_i, X_j \mid X_A) = P\rho^{j-i} \frac{(1 - \rho^{2(i-a)})(1 - \rho^{2(c-j)})}{1 - \rho^{2(c-a)}}. \quad (\text{G.3})$$

With only a left anchor, omit the right numerator factor and denominator; with only a right anchor, omit the left factor and denominator. With no anchors use $P\rho^{j-i}$. Rows and columns at an anchor have zero latent conditional covariance, and r is added to every observation diagonal. These identities follow by inverting the two-by-two anchor covariance and subtracting $K_{:A} K_A^{-1} K_{A:}$; the Markov property makes all other anchor loadings zero and distinct residual blocks independent. They retain signs for negative ρ . Rational input yields rational expressions.

One can price a block pattern by scalar filtering rather than repeatedly inverting its selected bridge covariance. Write V_t for its unconditional conditional-latent variance, obtained by setting $i = j = t$ in (G.3) or its endpoint versions. For successive selected nonanchor times $s < t$ in a block with right anchor c , set

$$A_{ts} = \rho^{t-s} \frac{1 - \rho^{2(c-t)}}{1 - \rho^{2(c-s)}}; \quad (\text{G.4})$$

with no right anchor set $A_{ts} = \rho^{t-s}$. The denominator is positive because $s < c$. Dividing the conditional cross covariance by $V_s > 0$ proves this transition, and its process variance is $V_t - A_{ts}^2 V_s \geq 0$ by conditional covariance positivity. Starting the first selected time at prediction variance V_t , propagate $P_t^- = V_t - A_{ts}^2 V_s + A_{ts}^2 P_s^+$ and update with observation-noise variance r . Filter the deterministic rows $F_t + H_{A,t} G$ in parallel. At a selected anchor the residual state is deterministically zero; its information contribution is the row outer product divided by r , and the

latent filter is reset. This explicit reset also handles partitions assigning an anchor observation to a block that continues on its right. Thus no formula divides by its zero residual state variance.

For ordered anchors $a_1 < \dots < a_m$, let $c_j = \rho^{a_j - a_{j-1}}$. The anchor-prior trace in (5.20) has the sparse form

$$\text{tr}(WG^TK_A^{-1}G) = \frac{G_1WG_1^T}{P} + \sum_{j=2}^m \frac{(G_j - c_jG_{j-1})W(G_j - c_jG_{j-1})^T}{P(1 - c_j^2)}. \quad (\text{G.5})$$

This is the ordinary Markov density factorization. Hence the exact support calculation needs neither a dense anchor-prior inverse nor a long-window covariance approximation. Enumerating 2^b block patterns and handling m nuisance coordinates still have real computational costs.

H Computational models, implementation, and retained evidence

H.1 Kinetic sensitivities and packet spectra

For (6.1), put $d = k_2 - k_1$ and $u = e^{-k_1t} - e^{-k_2t}$. In the declared log coordinates the sensitivity row is

$$\left(B(t), \quad A_0k_1 \left\{ \frac{k_2u}{d^2} - \frac{k_1te^{-k_1t}}{d} \right\}, \quad A_0k_2 \left\{ -\frac{k_1u}{d^2} + \frac{k_1te^{-k_2t}}{d} \right\} \right). \quad (\text{H.1})$$

The tested rates are separated; no near-equal-rate numerical stability claim is made for this expression. An independently assembled linear mass-balance system for (A, B, C) has generator

$$G = \begin{pmatrix} -k_1 & 0 & 0 \\ k_1 & -k_2 & 0 \\ 0 & k_2 & 0 \end{pmatrix}.$$

For each log parameter θ_j , its sensitivity obeys $\dot{s}_j = Gs_j + (\partial_{\theta_j}G)x$. Initial conditions are $x(0) = (A_0, 0, 0)^T$, $s_{\log A_0}(0) = x(0)$, and zero for the rate sensitivities. We integrate this augmented twelve-dimensional linear system with DOP853 (relative tolerance $2 \cdot 10^{-13}$, absolute tolerance $2 \cdot 10^{-14}$) and separately evaluate its matrix exponential at 70 digits. This comparison uses the balances, not numerical differentiation of the same closed form. The supplement additionally records complex-step checks, mass conservation, and the rate-swap identity. All large-instance certificates continue to refer to the saved rational decimal arrays.

For the complete-packet spectra, let channel coordinates be equally spaced on $[0, 1]$. The three profiles are $h_m(\lambda) = \exp\{-(\lambda - c_m)^2/(2w_m^2)\}$ with $(c_1, c_2, c_3) = (.15, .50, .85)$ and $(w_1, w_2, w_3) = (.16, .19, .16)$. The mean is $h_1A(t) + h_2B(t) + h_3C(t)$, with unit path length, $C = A_0 - A - B$, $A_0 = 1$, $k_1 = .7$, $k_2 = .2$, and only $\log k_1$ unknown. Times are $.5, 1, \dots, 12$. The latent recursion starts in its stationary covariance I_d , with process covariance $I_d - AA^T$ and independent observation covariance I_d . Its complete observation covariance has diagonal blocks $2I_d$, lower blocks A^{t-s} and transposed upper blocks. These are normalized synthetic spectra, without measured-noise or calibration claims.

H.2 Numerical proposals and exact checks

The practical finite-history logdet solver optimizes convex combinations of complete path information matrices, using a cardinality-layered support oracle and fully corrective numerical mixture steps. It retains true-covariance integer incumbents separately. A support tangent bounds the entire path hull, while the locality correction bounds the true selected information. For fixed weighted trace, a

single support price suffices. Minimum spacing is encoded in feasibility state as well as used in the covariance bound.

The dense comparator uses the selected-covariance extension (5.3). Numerical continuous proposals use SLSQP; tangent support prices over an exact-cardinality cube use the k largest gradient coordinates (block sums for complete packets). Robust versions use common visits, an epigraph, and nonnegative scenario weights before a common price. The early mixed-integer comparisons add up to 200 root outer-approximation rounds and Gurobi branching, within the recorded per-invocation time allowance. The archive retains the actual optimizer options and tolerances, completed iteration counts, cap outcomes, selected schedules and tangent points. A numerical success flag is never a premise of a rational certificate.

The repaired covariance oracle computes virtual-noise information by scalar filtering and its derivative by the reverse recursions in Appendix E. The archival precision-space prototype remains available to reconstruct historical proposals and its documented instability; it is not used to assert robust near-unit-correlation numerical evaluation. The diagonal-split proposal uses SLSQP eigenvalue constraints and a separate linear dual SDP. The unsuccessful kinetic proposal and the synthetic `optimal_inaccurate` dual status are preserved. Exact factor repair supplies a valid PSD dual independently of either status. Optional proposal reproduction uses CVXPY 1.9.2 and Clarabel 0.11.1; verification of its saved rational dual requires neither package.

Ordinary scalar log certificates use 10^8 reference and score grids and rational logarithm enclosures with explicit remainder. Spacing and partial certificates use coefficient grid 10^{12} and feature grid 10^{18} . Robust certificates use exact simplex weights and 10^{12} final log grids; all individual normalizer certificates are recomputed. Separator certificates use grid 10^8 for both the reference matrix and nuisance witness, 10^{12} for bridge regression coefficients, 10^{18} for features, 10^8 for scores, and 10^{12} for final logarithms. Their exact feasibility checks include cardinality, covariance promises, positive reference matrices, positive variance floors and declared pattern caps. The supplement saves all fractions, not just decimal summaries.

The fresh nested-anchor study uses SLSQP tolerance 10^{-12} and at most 1000 iterations, starting from the uniform mixture over all 56 schedules. Proposed nonnegative weights are floored on a 10^9 grid and divided by their integer sum; this creates an exactly feasible mixture. Its exact Schur information supplies $W = J^{-1}$ and its exact optimizing nuisance matrix supplies G . The support maximum is evaluated over every actual schedule, so the certified mixture gap is independent of SLSQP convergence. The dense fractional visits are rounded and adjusted on a 10^8 grid to retain exact cardinality. Every stored split is checked by rational leading principal minors. The fresh block experiment uses exhaustive exact pricing, with no numerical optimization. All indices in these two small fixtures are zero-based; the temporal kinetic times in the main experiment are one-based grid multiples.

H.3 What the archive retains and what each result establishes

The archive is a frozen record of the investigation, not an additional set of unproved manuscript claims. Earlier $L = 6$ scalar studies, the small 12- and 24-candidate certificates, and the initial $L = 8$ run are retained as precursors; the six-case exact scalar table and four kinetic certificates provide the main unconstrained evidence. All fixed-split dense and all-scalar comparisons retain exact common-model checks. The all-diagonal kinetic witness is the stronger single-case separation; the unsuccessful synthetic fixed-point witness is a negative result.

The first spacing certificate, its integer-interval replay, and its refined pair-bound certificate are separate artifacts; their improved guarantees do not imply a change in incumbent. The full-packet numerical studies retain both dimensions, their identical greedy schedules, tighter dense bounds,

workspace estimates, and tiny exhaustive diagnostics. The fresh block certificate is stored separately and does not replace their numerical bounds. Partial-observation records retain the original weaker bounds and preliminary $L = 6$ cases; the four normalized $L = 8$ certificates establish the reported trace guarantees.

The robust archive retains the original hull, greedy, nominal, dense rounding, independent exchange-polishing, and exact-certification records, including all scenario values and individual-reference costs. Earlier bounds are not retroactively assigned to later incumbents. In the fixed-physical study, primary timeouts, the separately budgeted $L = 12$ diagnostic, and the refined numerical transfer are separate records. Separator block-size sweeps retain smaller blocks, five exact certificates, nonnested partitions, generation and certification costs, and the missed .01 target.

Supplement contents, validation and reproduction commands are listed in the data and code availability statement (Section 7).

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